



## **Module Manual**

Master of Science (M.Sc.)

# **Mechanical Engineering and Management**

Cohort: Winter Term 2025

Updated: 8th May 2025



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## Program description

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### Content

Nowadays engineers work not only as designers or as problem solvers in technical issues, but also fill management positions and have to make strategic and operative decisions. In addition to profound and specialized knowledge in diverse engineering fields, engineers also need a basic understanding in economics and business studies. Graduates, who already bring along both, specialized knowledge in engineering as well as a basic understanding of economic sciences, have excellent prospects in the labor market.

The international master study course "Mechanical Engineering and Management" gives students with a bachelor's degree in mechanical engineering or similar the opportunity to build up an individual profile within two specializations.

In the first specialization students gain basic knowledge in management, business administration, accounting as well as in specialized management topics, such as corporate management, human resources or logistics.

For the second specialization students can choose between three main topics: Materials, Mechatronics, or Product Development and Production. Because of the material behavior and its great impact on product design and manufacturing, the Materials specialization represents a bridge between natural science and engineering science. The Mechatronics specialization represents an interdisciplinary field between mechanics, electronics and computer science. The last specialization, Product Development and Production, includes the computation as well as the manufacturing of products. Therefore not only the structure of the master study course is interdisciplinary, but also its specializations.

### Career prospects

The international master study course "Mechanical Engineering and Management" prepares graduates for a wide range of job profiles in international operating companies and in service providers, such as consulting. They are able to work as a facilitator between technical and business sectors and to take leading positions as technical and executive managers with budget and personnel responsibilities. The program is designed to be diverse and allows graduates to work in a variety of different industrial sectors (especially in mechanical engineering) and with different products and services. Graduates may decide for direct entry into companies or to take up academic careers, e.g. Ph.D. studies, in universities or other research institutions.

### Learning target

#### Knowledge

- Graduates can demonstrate specialised interdisciplinary knowledge in the fields of Robotics, Computer Aided Design and Computation, as well as in the Structure and Properties of Fibre-Polymer Composites, based on their broad theoretical and methodological foundation.
- In addition to a fundamental understanding of business-related topics, they also acquire in-depth knowledge in various subjects enabling them to analyse relationships within marketing, intercultural communication or project management contexts, and can describe various methods and current research directions in these areas.
- They can explain in detail the principles, methods and application areas of the chosen mechanical engineering specialisations.
- Graduates are aware of the social, economic and ecological implications of their discipline and their actions and are able to analyse and evaluate them. Graduates with a non-German mother tongue can communicate in German in simple routine situations involving a simple and direct exchange of information on familiar and common matters.
- Graduates will be able to present the state of the art in technology and research and provide an overview of applications in industry and research.

#### Skills

For all specializations:

- Graduates are able to use the interdisciplinary knowledge they have acquired, to solve complex problems. They can recognise implications in the interplay between business and technology and mediate between the relevant functional areas.
- Graduates are able to convert their theoretical knowledge into practice, analyse business management issues in complex corporate situations and select advanced methods and procedures from materials science, mechatronics or from the available methods for calculation and manufacturing, and apply them to solve complex problems.
- They are in a position to investigate or assess future technologies, materials and methods as well as scientific developments, and are able to conduct independent research (qualification for a doctorate).

#### Specialisation in Management

- Graduates are able to evaluate various business metrics and make decisions based on them.
- They can use a variety of methods in marketing and communications, marketing, management of business, human resources, technology or manufacturing, enterprise financing and planning as well as quantitative techniques in economics to solve problems in their respective business fields.

#### Specialisation in Materials

- Graduates become aware of new fields of application for various materials and are able to make an application-specific selection of the material, taking into account function, cost and quality.
- They can calculate various material properties and make design decisions based on them.

#### Specialisation in Mechatronics

- Graduates can solve systematically and methodically mechatronic and constructive tasks.
- They can apply their knowledge of new development methods as well as automation and simulation to investigate real systems, evaluate and analyse the results, and select and apply appropriate solution strategies based on these results.

#### Specialisation in Product Development and Production

- Graduates can evaluate and select various production and manufacturing processes in the contexts of geometry generation, error control and work cost-effectiveness.
- They can design state-of-the-art products and calculate and simulate them using numerical methods.

### Social Competence

- Graduates can clearly communicate their approach and results in writing and orally in English.
- They will be able to communicate in English about advanced content and problems of two chosen specialisations and especially at their interdisciplinary interfaces with professionals and laypersons. They can respond appropriately to inquiries, additions and comments.

## Module Manual M.Sc. "Mechanical Engineering and Management"

- Graduates can work in groups. They can define, distribute, and integrate subtasks, make time agreements and engage in social interaction. They have the ability and willingness to take-on leadership responsibilities.

### Independence

- Graduates are able to expand and deepen their knowledge in technical, economic and social topics as well as to acquire necessary information and put it into the context of their knowledge.
- They can systematically reflect on the non-technical impacts of engineering work and evaluate their actions in a socio-economic context.
- They can realistically assess their existing competencies and weaknesses, address deficiencies independently and make meaningful improvements.
- They can independently and self-motivated explore research areas and find or define new research questions (lifelong researching).

### Program structure

The course is designed modular and is based on the university-wide standardized course structure with uniform module sizes (multiples of six credit points (CP)). The course combines the engineering and management disciplines and allows the deepening in two of four specializations. The students can broadly personalize their studies due to high number and variety of elective courses.

In the common core skills, students take the following modules:

- Computer Aided Design and Computation (6 CP)
- Fibre-polymer-composites (6 CP)
- Robotics (6 CP)
- Management and complementary technical elective courses or an internship can be chosen (12 CP)
- Complementary courses business and management (catalog) (6 CP)
- Complementary nontechnical elective courses (catalog) (6 CP), of that 4 CP are intended for German classes

Students specialize by selecting two of the following areas, each covering 18 credit points. Students have to choose the Management specialization. Solely students of the Northern Institute of Technology have to choose two engineering specializations:

- Management (18 CP)
- Materials (18 CP)
- Mechatronics (18 CP)
- Product Development and Production (18 CP)

Within each area of specialization students can choose within a catalogue of modules (each 6 CP).

Students write also a master thesis and one additional scientific project work.

- Research Project (12 CP)
- Master thesis (30 CP)

## Core Qualification

The core qualification provides the basic fundamentals for the four specializations and also includes a catalogue of nontechnical elective complementary courses. For all three engineering specializations (Materials, Mechatronics, Product Development and Production) a compulsory module is included. As preparation for the Management specialization students choose three lecturers from the Business and Management catalogue and can also choose up to two more management related modules. Alternatively technical complementary courses or an internship can be chosen here. In total two modules have to be chosen.

| Module M0524: Non-technical Courses for Master     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <b>Module Responsible</b>                          | Dagmar Richter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Admission Requirements</b>                      | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Recommended Previous Knowledge</b>              | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Educational Objectives</b>                      | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Professional Competence</b><br><i>Knowledge</i> | <p><b>The Nontechnical Academic Programms (NTA)</b></p> <p>imparts skills that, in view of the TUHH's training profile, professional engineering studies require but are not able to cover fully. Self-reliance, self-management, collaboration and professional and personnel management competences. The department implements these training objectives in its <b>teaching architecture</b>, in its <b>teaching and learning arrangements</b>, in <b>teaching areas</b> and by means of teaching offerings in which students can qualify by opting for <b>specific competences</b> and a <b>competence level</b> at the Bachelor's or Master's level. The teaching offerings are pooled in two different catalogues for nontechnical complementary courses.</p> <p><b>The Learning Architecture</b></p> <p>consists of a cross-disciplinary study offering. The centrally designed teaching offering ensures that courses in the nontechnical academic programms follow the specific profiling of TUHH degree courses.</p> <p>The learning architecture demands and trains independent educational planning as regards the individual development of competences. It also provides orientation knowledge in the form of "profiles".</p> <p>The subjects that can be studied in parallel throughout the student's entire study program - if need be, it can be studied in one to two semesters. In view of the adaptation problems that individuals commonly face in their first semesters after making the transition from school to university and in order to encourage individually planned semesters abroad, there is no obligation to study these subjects in one or two specific semesters during the course of studies.</p> <p><b>Teaching and Learning Arrangements</b></p> <p>provide for students, separated into B.Sc. and M.Sc., to learn with and from each other across semesters. The challenge of dealing with interdisciplinarity and a variety of stages of learning in courses are part of the learning architecture and are deliberately encouraged in specific courses.</p> <p><b>Fields of Teaching</b></p> <p>are based on research findings from the academic disciplines cultural studies, social studies, arts, historical studies, communication studies, migration studies and sustainability research, and from engineering didactics. In addition, from the winter semester 2014/15 students on all Bachelor's courses will have the opportunity to learn about business management and start-ups in a goal-oriented way.</p> <p>The fields of teaching are augmented by soft skills offers and a foreign language offer. Here, the focus is on encouraging goal-oriented communication skills, e.g. the skills required by outgoing engineers in international and intercultural situations.</p> <p><b>The Competence Level</b></p> <p>of the courses offered in this area is different as regards the basic training objective in the Bachelor's and Master's fields. These differences are reflected in the practical examples used, in content topics that refer to different professional application contexts, and in the higher scientific and theoretical level of abstraction in the B.Sc.</p> <p>This is also reflected in the different quality of soft skills, which relate to the different team positions and different group leadership functions of Bachelor's and Master's graduates in their future working life.</p> <p><b>Specialized Competence (Knowledge)</b></p> <p>Students can</p> <ul style="list-style-type: none"> <li>• explain specialized areas in context of the relevant non-technical disciplines,</li> <li>• outline basic theories, categories, terminology, models, concepts or artistic techniques in the disciplines represented in the learning area,</li> <li>• different specialist disciplines relate to their own discipline and differentiate it as well as make connections,</li> <li>• sketch the basic outlines of how scientific disciplines, paradigms, models, instruments, methods and forms of representation in the specialized sciences are subject to individual and socio-cultural interpretation and historicity,</li> <li>• Can communicate in a foreign language in a manner appropriate to the subject.</li> </ul> |
| <i>Skills</i>                                      | <p><b>Professional Competence (Skills)</b></p> <p>In selected sub-areas students can</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Personal Competence</b><br/> <i>Social Competence</i></p> <p><i>Autonomy</i></p> | <ul style="list-style-type: none"> <li>• apply basic and specific methods of the said scientific disciplines,</li> <li>• aquestion a specific technical phenomena, models, theories from the viewpoint of another, aforementioned specialist discipline,</li> <li>• to handle simple and advanced questions in aforementioned scientific disciplines in a sucessful manner,</li> <li>• justify their decisions on forms of organization and application in practical questions in contexts that go beyond the technical relationship to the subject.</li> </ul>                                                                     |
|                                                                                        | <p><b>Personal Competences (Social Skills)</b></p> <p>Students will be able</p> <ul style="list-style-type: none"> <li>• to learn to collaborate in different manner,</li> <li>• to present and analyze problems in the abovementioned fields in a partner or group situation in a manner appropriate to the addressees,</li> <li>• to express themselves competently, in a culturally appropriate and gender-sensitive manner in the language of the country (as far as this study-focus would be chosen),</li> <li>• to explain nontechnical items to auditorium with technical background knowledge.</li> </ul>                  |
|                                                                                        | <p><b>Personal Competences (Self-reliance)</b></p> <p>Students are able in selected areas</p> <ul style="list-style-type: none"> <li>• to reflect on their own profession and professionalism in the context of real-life fields of application</li> <li>• to organize themselves and their own learning processes</li> <li>• to reflect and decide questions in front of a broad education background</li> <li>• to communicate a nontechnical item in a competent way in written form or verbally</li> <li>• to organize themselves as an entrepreneurial subject country (as far as this study-focus would be chosen)</li> </ul> |
| <b>Workload in Hours</b>                                                               | Depends on choice of courses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Credit points</b>                                                                   | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## Courses

**Information regarding lectures and courses can be found in the corresponding module handbook published separately.**

## Module M0809: Computer Aided Design and Computation

### Courses

| Title                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Typ                        | Hrs/wk | CP |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------|----|
| Computer Aided Design and Computation (L0525)                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Lecture                    | 2      | 3  |
| Computer Aided Design and Computation (L0527)                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Recitation Section (small) | 2      | 3  |
| <b>Module Responsible</b>                                                                                                                                | Prof. Claus Emmelmann                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                            |        |    |
| <b>Admission Requirements</b>                                                                                                                            | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                            |        |    |
| <b>Recommended Previous Knowledge</b>                                                                                                                    | <ul style="list-style-type: none"> <li>- Mechanical parts and basic operations of manufacturing techniques</li> <li>- Basic knowledge in mathematics, physics, and statics</li> <li>- Mechanics I (statics, mechanics of materials) and mechanics II (hydrostatics, kinematics, dynamics)</li> <li>- Mathematics I, II, III (in particular differential equations)</li> </ul>                                                                                                                                                                                           |                            |        |    |
| <b>Educational Objectives</b>                                                                                                                            | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                            |        |    |
| <b>Professional Competence</b><br><i>Knowledge</i><br><br><i>Skills</i><br><br><b>Personal Competence</b><br><i>Social Competence</i><br><i>Autonomy</i> | <ul style="list-style-type: none"> <li>- Understanding of the capabilities and limitations of 3D-CAD-Systems, PDM systems, and computer aided simulation Tools</li> <li>- General knowledge of the finite element method in combination with a basic theoretical and methodology basis</li> <li>- Basic understanding of the structural optimizations potential and fields of application</li> <li>- Hands-on practice with an exemplary 3D-CAD-system to demonstrate basic modeling techniques as well as interfaces for concurrent finite element analysis</li> </ul> |                            |        |    |
| <b>Workload in Hours</b>                                                                                                                                 | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                            |        |    |
| <b>Credit points</b>                                                                                                                                     | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                            |        |    |
| <b>Course achievement</b>                                                                                                                                | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                            |        |    |
| <b>Examination</b>                                                                                                                                       | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                            |        |    |
| <b>Examination duration and scale</b>                                                                                                                    | 120 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                            |        |    |
| <b>Assignment for the Following Curricula</b>                                                                                                            | Mechanical Engineering and Management: Core Qualification: Compulsory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                            |        |    |



| Course L0525: Computer Aided Design and Computation |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                          | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Hrs/wk</b>                                       | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>CP</b>                                           | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Workload in Hours</b>                            | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Lecturer</b>                                     | Prof. Claus Emmelmann, Dr. Konrad Schneider, Prof. Nikola Bursac                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Language</b>                                     | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Cycle</b>                                        | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Content</b>                                      | <p><b>Part 1: System generation development (Prof. Dr.-Ing. N. Bursac)</b></p> <ul style="list-style-type: none"> <li>• Introduction to integrated product development</li> <li>• 3D-CAD-systems and CAD-interfaces</li> <li>• Introduction to PDM-systems</li> <li>• Additional computer aided engineering/simulation tools (FEA, DMU, VR)</li> </ul> <p><b>Part 2: Introduction to the Finite Element Method (Dr.-Ing. K. Schneider)</b></p> <ul style="list-style-type: none"> <li>• General overview on the finite element method</li> <li>• Displacement method</li> <li>• Isoparametric elements</li> <li>• Numerical integration</li> <li>• Applications</li> <li>• Programming of elements (Matlab, hands-on sessions)</li> </ul> <p><b>Part 3: Structural Optimization Methods (Prof. Dr.-Ing. C. Emmelmann)</b></p> <ul style="list-style-type: none"> <li>• Introduction to structural optimization theory</li> <li>• Fields of application for structural optimization and commercial software tools</li> </ul> <p>This module relies heavily on the interconnection of theory and the application of commercial software systems via live demonstrations as well as hands-on sessions in a PC-pool.</p> |
| <b>Literature</b>                                   | <p>Lee, K.: Principles of CAD / CAM / CAE Systems, Addison Wesley</p> <p>Bathe, K.-J.: Finite element procedures, Prentice Hall</p> <p>Christensen, P.W.; Klarbring, A.: An introduction to structural optimization; Springer</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Course L0527: Computer Aided Design and Computation |                                                                  |
|-----------------------------------------------------|------------------------------------------------------------------|
| <b>Typ</b>                                          | Recitation Section (small)                                       |
| <b>Hrs/wk</b>                                       | 2                                                                |
| <b>CP</b>                                           | 3                                                                |
| <b>Workload in Hours</b>                            | Independent Study Time 62, Study Time in Lecture 28              |
| <b>Lecturer</b>                                     | Prof. Claus Emmelmann, Dr. Konrad Schneider, Prof. Nikola Bursac |
| <b>Language</b>                                     | EN                                                               |
| <b>Cycle</b>                                        | WiSe                                                             |
| <b>Content</b>                                      | See interlocking course                                          |
| <b>Literature</b>                                   | See interlocking course                                          |

| Module M0563: Robotics                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |              |                                        |                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------------------------|---------------------------------------------------------------------------------------------|
| Courses                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |              |                                        |                                                                                             |
| Title                                                                                                                                                | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              | Hrs/wk                                 | CP                                                                                          |
| Robotics: Modelling and Control (L0168)                                                                                                              | Integrated Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              | 4                                      | 4                                                                                           |
| Robotics: Modelling and Control (L1305)                                                                                                              | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |              | 2                                      | 2                                                                                           |
| <b>Module Responsible</b>                                                                                                                            | Dr. Martin Gomse                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              |                                        |                                                                                             |
| <b>Admission Requirements</b>                                                                                                                        | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |              |                                        |                                                                                             |
| <b>Recommended Previous Knowledge</b>                                                                                                                | Fundamentals of electrical engineering<br>Broad knowledge of mechanics<br>Fundamentals of control theory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              |                                        |                                                                                             |
| <b>Educational Objectives</b>                                                                                                                        | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |              |                                        |                                                                                             |
| <b>Professional Competence</b><br><i>Knowledge</i><br><i>Skills</i><br><br><b>Personal Competence</b><br><i>Social Competence</i><br><i>Autonomy</i> | Students are able to describe fundamental properties of robots and solution approaches for multiple problems in robotics.<br>Students are able to derive and solve equations of motion for various manipulators.<br><br>Students can generate trajectories in various coordinate systems.<br><br>Students can design linear and partially nonlinear controllers for robotic manipulators.<br><br>Students are able to work goal-oriented in small mixed groups.<br>Students are able to recognize and improve knowledge deficits independently.<br><br>With instructor assistance, students are able to evaluate their own knowledge level and define a further course of study.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              |                                        |                                                                                             |
| <b>Workload in Hours</b>                                                                                                                             | Independent Study Time 96, Study Time in Lecture 84                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |                                        |                                                                                             |
| <b>Credit points</b>                                                                                                                                 | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              |                                        |                                                                                             |
| <b>Course achievement</b>                                                                                                                            | <b>Compulsory</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Bonus</b> | <b>Form</b>                            | <b>Description</b>                                                                          |
|                                                                                                                                                      | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None         | Subject theoretical and practical work | Teilnahme an PBL-Einheiten sowie Erreichen des Gesamtziels und der jeweiligen Session-Ziele |
| <b>Examination</b>                                                                                                                                   | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |                                        |                                                                                             |
| <b>Examination duration and scale</b>                                                                                                                | 120 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |                                        |                                                                                             |
| <b>Assignment for the Following Curricula</b>                                                                                                        | Aircraft Systems Engineering: Core Qualification: Elective Compulsory<br>International Management and Engineering: Specialisation II. Product Development and Production: Elective Compulsory<br>International Management and Engineering: Specialisation II. Mechatronics: Elective Compulsory<br>Aeronautics: Core Qualification: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory<br>Mechanical Engineering and Management: Core Qualification: Compulsory<br>Mechatronics: Core Qualification: Compulsory<br>Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Production: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Materials: Elective Compulsory<br>Theoretical Mechanical Engineering: Specialisation Product Development and Production: Elective Compulsory<br>Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory |              |                                        |                                                                                             |

| Course L0168: Robotics: Modelling and Control |                                                                                                                                                                                                                                 |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                    | Integrated Lecture                                                                                                                                                                                                              |
| <b>Hrs/wk</b>                                 | 4                                                                                                                                                                                                                               |
| <b>CP</b>                                     | 4                                                                                                                                                                                                                               |
| <b>Workload in Hours</b>                      | Independent Study Time 64, Study Time in Lecture 56                                                                                                                                                                             |
| <b>Lecturer</b>                               | Dr. Martin Gomse                                                                                                                                                                                                                |
| <b>Language</b>                               | EN                                                                                                                                                                                                                              |
| <b>Cycle</b>                                  | WiSe                                                                                                                                                                                                                            |
| <b>Content</b>                                | Fundamental kinematics of rigid body systems<br><br>Newton-Euler equations for manipulators<br><br>Trajectory generation<br><br>Linear and nonlinear control of robots                                                          |
| <b>Literature</b>                             | Craig, John J.: Introduction to Robotics Mechanics and Control, Third Edition, Prentice Hall. ISBN 0201-54361-3<br><br>Spong, Mark W.; Hutchinson, Seth; Vidyasagar, M. : Robot Modeling and Control. WILEY. ISBN 0-471-64990-2 |

| Course L1305: Robotics: Modelling and Control |                                                     |
|-----------------------------------------------|-----------------------------------------------------|
| <b>Typ</b>                                    | Project-/problem-based Learning                     |
| <b>Hrs/wk</b>                                 | 2                                                   |
| <b>CP</b>                                     | 2                                                   |
| <b>Workload in Hours</b>                      | Independent Study Time 32, Study Time in Lecture 28 |
| <b>Lecturer</b>                               | Dr. Martin Gomse                                    |
| <b>Language</b>                               | EN                                                  |
| <b>Cycle</b>                                  | WiSe                                                |
| <b>Content</b>                                | See interlocking course                             |
| <b>Literature</b>                             | See interlocking course                             |

| Module M0523: Business & Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Module Responsible</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Prof. Matthias Meyer                                                                 |
| <b>Admission Requirements</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Successful completion of the modul "Foundations of Management"                       |
| <b>Recommended Previous Knowledge</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                                                                 |
| <b>Educational Objectives</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | After taking part successfully, students have reached the following learning results |
| <b>Professional Competence</b><br><i>Knowledge</i> <ul style="list-style-type: none"> <li>• Students are able to find their way around selected special areas of management within the scope of business management.</li> <li>• Students are able to explain basic theories, categories, and models in selected special areas of business management.</li> <li>• Students are able to interrelate technical and management knowledge.</li> </ul><br><i>Skills</i> <ul style="list-style-type: none"> <li>• Students are able to apply basic methods in selected areas of business management.</li> <li>• Students are able to explain and give reasons for decision proposals on practical issues in areas of business management.</li> </ul><br><b>Personal Competence</b><br><i>Social Competence</i> <ul style="list-style-type: none"> <li>• Students are able to communicate in small interdisciplinary groups and to jointly develop solutions for complex problems</li> </ul><br><i>Autonomy</i> <ul style="list-style-type: none"> <li>• Students are capable of acquiring necessary knowledge independently by means of research and preparation of material.</li> </ul> |                                                                                      |
| <b>Workload in Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Depends on choice of courses                                                         |
| <b>Credit points</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 6                                                                                    |
| Courses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                      |
| Information regarding lectures and courses can be found in the corresponding module handbook published separately.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                      |

**Module M1282: Selected Topics of Mechanical Engineering and Management (Alternative A: 12 CP)**

**Courses**

| <b>Title</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Type</b>                                                                                | <b>Hrs/wk</b> | <b>CP</b> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------|-----------|
| Fatigue & Damage Tolerance (L0310)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Lecture                                                                                    | 2             | 3         |
| International Law for Engineers (L1750)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Seminar                                                                                    | 2             | 2         |
| International Law for Engineers (L1749)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Lecture                                                                                    | 2             | 2         |
| Lightweight Design Practical Course (L1258)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Project-/problem-based Learning                                                            | 3             | 3         |
| NaPe: Sustainable Product Development - Lecture Series of the Scientific Society for Product Development (WiGeP) (L3340)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Lecture                                                                                    | 2             | 3         |
| Sustainable Industrial Production (L2863)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Lecture                                                                                    | 2             | 3         |
| Productivity Management (L0928)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Project-/problem-based Learning                                                            | 2             | 2         |
| Productivity Management (L0931)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Recitation Section (small)                                                                 | 1             | 1         |
| SMS: Smart Machine Systems (L3341)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Lecture                                                                                    | 2             | 3         |
| Reliability in Engineering Dynamics (L2994)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Lecture                                                                                    | 2             | 2         |
| Reliability in Engineering Dynamics (L2995)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Recitation Section (small)                                                                 | 1             | 2         |
| <b>Module Responsible</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Prof. Nikola Bursac                                                                        |               |           |
| <b>Admission Requirements</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | None                                                                                       |               |           |
| <b>Recommended Previous Knowledge</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | see lecture description                                                                    |               |           |
| <b>Educational Objectives</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | After taking part successfully, students have reached the following learning results       |               |           |
| <b>Professional Competence</b><br><i>Knowledge</i> <ul style="list-style-type: none"> <li>Students are able to express their extended knowledge and discuss the connection of different special fields or application areas of Materials, Mechatronics and Product Development and Production</li> <li>Students are qualified to connect different special fields with each other</li> </ul> <i>Skills</i> <ul style="list-style-type: none"> <li>Students can apply specialized solution strategies and new scientific methods in selected areas</li> <li>Students are able to transfer learned skills to new and unknown problems and can develop own solution approaches</li> </ul> <b>Personal Competence</b><br><i>Social Competence</i><br><i>Autonomy</i> | Students are able to develop their knowledge and skills by autonomous election of courses. |               |           |
| <b>Workload in Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Depends on choice of courses                                                               |               |           |
| <b>Credit points</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 12                                                                                         |               |           |
| <b>Assignment for the Following Curricula</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Mechanical Engineering and Management: Core Qualification: Elective Compulsory             |               |           |

**Course L0310: Fatigue & Damage Tolerance**

|                                       |                                                                                                                                                                                                      |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Type</b>                           | Lecture                                                                                                                                                                                              |
| <b>Hrs/wk</b>                         | 2                                                                                                                                                                                                    |
| <b>CP</b>                             | 3                                                                                                                                                                                                    |
| <b>Workload in Hours</b>              | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                  |
| <b>Examination Form</b>               | Mündliche Prüfung                                                                                                                                                                                    |
| <b>Examination duration and scale</b> | 45 min                                                                                                                                                                                               |
| <b>Lecturer</b>                       | Dr. Martin Flamm                                                                                                                                                                                     |
| <b>Language</b>                       | EN                                                                                                                                                                                                   |
| <b>Cycle</b>                          | WiSe                                                                                                                                                                                                 |
| <b>Content</b>                        | Design principles, fatigue strength, crack initiation and crack growth, damage calculation, counting methods, methods to improve fatigue strength, environmental influences                          |
| <b>Literature</b>                     | Jaap Schijve, Fatigue of Structures and Materials. Kluwer Academic Publisher, Dordrecht, 2001 E. Haibach. Betriebsfestigkeit Verfahren und Daten zur Bauteilberechnung. VDI-Verlag, Düsseldorf, 1989 |

| Course L1750: International Law for Engineers |                                                                                                                                                                                  |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                    | Seminar                                                                                                                                                                          |
| <b>Hrs/wk</b>                                 | 2                                                                                                                                                                                |
| <b>CP</b>                                     | 2                                                                                                                                                                                |
| <b>Workload in Hours</b>                      | Independent Study Time 32, Study Time in Lecture 28                                                                                                                              |
| <b>Examination Form</b>                       | Schriftliche Ausarbeitung                                                                                                                                                        |
| <b>Examination duration and scale</b>         | 10-20 Seiten                                                                                                                                                                     |
| <b>Lecturer</b>                               | Markus A. Meyer-Chory                                                                                                                                                            |
| <b>Language</b>                               | EN                                                                                                                                                                               |
| <b>Cycle</b>                                  | SoSe                                                                                                                                                                             |
| <b>Content</b>                                | <ul style="list-style-type: none"> <li>basics and selected legal aspects of international Engineers work - i.e. on contracts, construction, labor, patents, insurance</li> </ul> |
| <b>Literature</b>                             | As per Stud.IP                                                                                                                                                                   |

| Course L1749: International Law for Engineers |                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                    | Lecture                                                                                                                                                                                                                                                                                                                                    |
| <b>Hrs/wk</b>                                 | 2                                                                                                                                                                                                                                                                                                                                          |
| <b>CP</b>                                     | 2                                                                                                                                                                                                                                                                                                                                          |
| <b>Workload in Hours</b>                      | Independent Study Time 32, Study Time in Lecture 28                                                                                                                                                                                                                                                                                        |
| <b>Examination Form</b>                       | Klausur                                                                                                                                                                                                                                                                                                                                    |
| <b>Examination duration and scale</b>         | 90 Minuten                                                                                                                                                                                                                                                                                                                                 |
| <b>Lecturer</b>                               | Markus A. Meyer-Chory                                                                                                                                                                                                                                                                                                                      |
| <b>Language</b>                               | EN                                                                                                                                                                                                                                                                                                                                         |
| <b>Cycle</b>                                  | WiSe                                                                                                                                                                                                                                                                                                                                       |
| <b>Content</b>                                | <ul style="list-style-type: none"> <li>basics and selected legal aspects of international Engineers work and international laws, such as civil/common law, questions of jurisdiction and courts as well as arbitration and enforcement of titles, etc.</li> <li>also laws on contracts, construction, labor, patents, companies</li> </ul> |
| <b>Literature</b>                             | As per Stud.IP.                                                                                                                                                                                                                                                                                                                            |

| Course L1258: Lightweight Design Practical Course |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                        | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Hrs/wk</b>                                     | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>CP</b>                                         | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Workload in Hours</b>                          | Independent Study Time 48, Study Time in Lecture 42                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Examination Form</b>                           | Mündliche Prüfung                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Examination duration and scale</b>             | 30 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Lecturer</b>                                   | Prof. Dieter Krause                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Language</b>                                   | DE/EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Cycle</b>                                      | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Content</b>                                    | <p>Development of a sandwich structure made of fibre reinforced plastics</p> <ul style="list-style-type: none"> <li>• getting familiar with fibre reinforced plastics as well as lightweight design</li> <li>• Design of a sandwich structure made of fibre reinforced plastics using finite element analysis (FEA)</li> <li>• Determination of material properties based on sample tests</li> <li>• manufacturing of the structure in the composite lab</li> <li>• Testing of the developed structure</li> <li>• Concept presentation</li> <li>• Self-organised teamwork</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Literature</b>                                 | <ul style="list-style-type: none"> <li>• Schürmann, H., „Konstruieren mit Faser-Kunststoff-Verbunden“, Springer, Berlin, 2005.</li> <li>• Puck, A., „Festigkeitsanalyse von Faser-Matrix-Laminaten“, Hanser, München, Wien, 1996.</li> <li>• R&amp;G, „Handbuch Faserverbundwerkstoffe“, Waldenbuch, 2009.</li> <li>• VDI 2014 „Entwicklung von Bauteilen aus Faser-Kunststoff-Verbund“</li> <li>• Ehrenstein, G. W., „Faserverbundkunststoffe“, Hanser, München, 2006.</li> <li>• Klein, B., „Leichtbau-Konstruktion“, Vieweg &amp; Sohn, Braunschweig, 1989.</li> <li>• Wiedemann, J., „Leichtbau Band 1: Elemente“, Springer, Berlin, Heidelberg, 1986.</li> <li>• Wiedemann, J., „Leichtbau Band 2: Konstruktion“, Springer, Berlin, Heidelberg, 1986.</li> <li>• Backmann, B.F., „Composite Structures, Design, Safety and Innovation“, Oxford (UK), Elsevier, 2005.</li> <li>• Krause, D., „Leichtbau“, In: Handbuch Konstruktion, Hrsg.: Rieg, F., Steinhilper, R., München, Carl Hanser Verlag, 2012.</li> <li>• Schulte, K., Fiedler, B., „Structure and Properties of Composite Materials“, Hamburg, TUHH - TuTech Innovation GmbH, 2005.</li> </ul> |

| Course L3340: NaPe: Sustainable Product Development - Lecture Series of the Scientific Society for Product Development (WiGeP) |                                                     |
|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Typ</b>                                                                                                                     | Lecture                                             |
| <b>Hrs/wk</b>                                                                                                                  | 2                                                   |
| <b>CP</b>                                                                                                                      | 3                                                   |
| <b>Workload in Hours</b>                                                                                                       | Independent Study Time 62, Study Time in Lecture 28 |
| <b>Examination Form</b>                                                                                                        | Klausur                                             |
| <b>Examination duration and scale</b>                                                                                          | 120 min                                             |
| <b>Lecturer</b>                                                                                                                | Prof. Nikola Bursac, Prof. Dieter Krause            |
| <b>Language</b>                                                                                                                | DE                                                  |
| <b>Cycle</b>                                                                                                                   | SoSe                                                |
| <b>Content</b>                                                                                                                 |                                                     |
| <b>Literature</b>                                                                                                              |                                                     |

| Course L2863: Sustainable Industrial Production |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                      | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Hrs/wk</b>                                   | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>CP</b>                                       | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Workload in Hours</b>                        | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Examination Form</b>                         | Klausur                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Examination duration and scale</b>           | 60 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Lecturer</b>                                 | Dr. Simon Markus Kothe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Language</b>                                 | DE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Cycle</b>                                    | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Content</b>                                  | <p>Industrial production deals with the manufacture of physical products to satisfy human needs using various manufacturing processes that change the form and physical properties of raw materials. Manufacturing is a central driver of economic development and has a major impact on the well-being of humanity. However, the scale of current manufacturing activities results in enormous global energy and material demands that are harmful to both the environment and people. Historically, industrial activities were mostly oriented towards economic constraints, while social and environmental consequences were only hardly considered. As a result, today's global consumption rates of many resources and associated emissions often exceed the natural regeneration rate of our planet. In this respect, current industrial production can mostly be described as unsustainable. This is emphasized each year by the Earth Overshoot Day, which marks the day when humanity's ecological footprint exceeds the Earth's annual regenerative capacity.</p> <p>This lecture aims to provide the motivation, analytical methods as well as approaches for sustainable industrial production and to clarify the influence of the production phase in relation to the raw material, use and recycling phases in the entire life cycle of products. For this, the following topics will be highlighted:</p> <ul style="list-style-type: none"> <li>- Motivation for sustainable production, the 17 Sustainable Development Goals (SDGs) of the UN and their relevance for tomorrow's manufacturing;</li> <li>- raw material vs. production phase vs. use phase vs. recycling/end-of-life phase: importance of the production phase for the environmental impact of manufactured products;</li> <li>- Typical energy- and resource-intensive processes in industrial production and innovative approaches to increase energy and resource efficiency;</li> <li>- Methodology for optimizing the energy and resource efficiency of industrial manufacturing chains with the three steps of modeling (1), evaluating (2) and improving (3);</li> <li>- Resource efficiency of industrial manufacturing value chains and its assessment using life cycle analysis (LCA);</li> <li>- Exercise: LCA analysis of a manufacturing process (thermoplastic joining of an aircraft fuselage segment) as part of a product life cycle assessment.</li> </ul> |
| <b>Literature</b>                               | <p>Literatur:</p> <ul style="list-style-type: none"> <li>- Stefan Alexander (2020): Resource efficiency in manufacturing value chains. Cham: Springer International Publishing.</li> <li>- Hauschild, Michael Z.; Rosenbaum, Ralph K.; Olsen, Stig Irving (Hg.) (2018): Life Cycle Assessment. Theory and Practice. Cham: Springer International Publishing.</li> <li>- Kishita, Yusuke; Matsumoto, Mitsutaka; Inoue, Masato; Fukushige, Shinichi (2021): EcoDesign and sustainability. Singapore: Springer.</li> <li>- Schebek, Liselotte; Herrmann, Christoph; Cerdas, Felipe (2019): Progress in Life Cycle Assessment. Cham: Springer International Publishing.</li> <li>- Thiede, Sebastian; Hermann, Christoph (2019): Eco-factories of the future. Cham: Springer Nature Switzerland AG.</li> <li>- Vorlesungsskript.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |



| Course L0928: Productivity Management |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                            | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Hrs/wk</b>                         | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>CP</b>                             | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Workload in Hours</b>              | Independent Study Time 32, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Examination Form</b>               | Klausur                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Examination duration and scale</b> | 90 Minuten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Lecturer</b>                       | Prof. Hermann Lödding, Christopher Mundt                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Language</b>                       | DE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Cycle</b>                          | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Content</b>                        | <ul style="list-style-type: none"> <li>• Principles of productivity management</li> <li>• Shop floor management and standardisation</li> <li>• Takt analysis and design of manual operations</li> <li>• Maintenance Principles</li> <li>• Total Productive Maintenance (TPM)</li> <li>• Optimisation of set-up operations</li> <li>• Analysis of interlinked production systems</li> </ul>                                                                                                            |
| <b>Literature</b>                     | <p>Bokranz, R.; Landau, K.: Produktivitätsmanagement von Arbeitssystemen. Schäffer-Poeschel, Stuttgart, 2006.</p> <p>Takeda, H.: Das synchrone Produktionssystem: Just-in-Time für das ganze Unternehmen. 5. Aufl., mi-Wirtschaftsbuch, FinanzBuch Verlag, München, 2006.</p> <p>Nakajima, S.: Management der Produktionseinrichtungen (Total Productive Maintenance). Campus Verlag, New York, 1995.</p> <p>Shingo, S.: A Revolution in Manufacturing: The SMED System. Productivity, Inc., 1985</p> |

| Course L0931: Productivity Management |                                                     |
|---------------------------------------|-----------------------------------------------------|
| <b>Typ</b>                            | Recitation Section (small)                          |
| <b>Hrs/wk</b>                         | 1                                                   |
| <b>CP</b>                             | 1                                                   |
| <b>Workload in Hours</b>              | Independent Study Time 16, Study Time in Lecture 14 |
| <b>Examination Form</b>               | Klausur                                             |
| <b>Examination duration and scale</b> | 90 Minuten                                          |
| <b>Lecturer</b>                       | Prof. Hermann Lödding                               |
| <b>Language</b>                       | DE                                                  |
| <b>Cycle</b>                          | SoSe                                                |
| <b>Content</b>                        | See interlocking course                             |
| <b>Literature</b>                     | See interlocking course                             |

| Course L3341: SMS: Smart Machine Systems |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                               | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Hrs/wk</b>                            | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>CP</b>                                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Workload in Hours</b>                 | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Examination Form</b>                  | Klausur                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Examination duration and scale</b>    | 60 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Lecturer</b>                          | Prof. Nikola Bursac                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Language</b>                          | DE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Cycle</b>                             | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Content</b>                           | <p><b>Knowledge:</b> Students acquire a comprehensive understanding of how data can be methodically used in product development to optimize complex development processes, support informed decision-making, and design innovative products. They understand the fundamental principles and methods by which data can be effectively utilized in the development process of technical products or systems. They learn to identify the strategic potentials of data-driven approaches, assess specific challenges such as data quality, security, or availability, and strategically leverage the opportunities of data-driven decisions. Furthermore, students develop the ability to critically analyze data-driven decision-making processes in product development and adapt them to specific application scenarios. A special emphasis is placed on not considering data-driven insights in isolation but integrating them interdisciplinarily to enable innovative and holistic solutions. With this knowledge, they are able to sustainably enhance value creation through the targeted use of data in product development and successfully develop data-driven strategies in a practical context.</p> <p><b>Skills:</b> Students can independently apply methods for using data in the development process of technical products and systems and use the insights gained to optimize products. They are capable of independently implementing methods for utilizing operational data to support data-based decisions in the development process and derive well-founded action options. Additionally, they master the independent execution of data analyses and the critical interpretation of results. They can handle challenges related to data management and are able to evaluate and effectively manage the opportunities and risks of data-driven approaches. Students are also able to specifically use insights from data analyses to support decision-making in development processes and advance data-driven innovation strategies.</p> <p><b>Personal Competencies:</b></p> <p><b>Social Competence:</b><br/>Students are capable of collaborating in interdisciplinary teams to develop sustainable solutions for real product development challenges based on operational data.</p> <p><b>Autonomy:</b><br/>Students can independently identify problems in the field of data-driven product development and develop appropriate measures to solve them.</p> |
| <b>Literature</b>                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Course L2994: Reliability in Engineering Dynamics |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                        | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Hrs/wk</b>                                     | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>CP</b>                                         | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Workload in Hours</b>                          | Independent Study Time 32, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Examination Form</b>                           | Klausur                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Examination duration and scale</b>             | 90 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Lecturer</b>                                   | Dr. Eric Groß, Prof. Benedikt Kriegesmann                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Language</b>                                   | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Cycle</b>                                      | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Content</b>                                    | <p>Method for calculation and testing of reliability of dynamic machine systems</p> <p>Modeling</p> <p>System identification</p> <p>Simulation</p> <p>Processing of measurement data</p> <p>Damage accumulation</p> <p>Test planning and execution</p>                                                                                                                                                                                                                                |
| <b>Literature</b>                                 | <p>Bertsche, B.: Reliability in Automotive and Mechanical Engineering. Springer, 2008. ISBN: 978-3-540-33969-4</p> <p>Inman, Daniel J.: Engineering Vibration. Prentice Hall, 3rd Ed., 2007. ISBN-13: 978-0132281737</p> <p>Dresig, H., Holzweißig, F.: Maschinendynamik, Springer Verlag, 9. Auflage, 2009. ISBN 3540876936.</p> <p>VDA (Hg.): Zuverlässigkeitssicherung bei Automobilherstellern und Lieferanten. Band 3 Teil 2, 3. überarbeitete Auflage, 2004. ISSN 0943-9412</p> |

| Course L2995: Reliability in Engineering Dynamics |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                        | Recitation Section (small)                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Hrs/wk</b>                                     | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>CP</b>                                         | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Workload in Hours</b>                          | Independent Study Time 46, Study Time in Lecture 14                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Examination Form</b>                           | Klausur                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Examination duration and scale</b>             | 90 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Lecturer</b>                                   | Prof. Benedikt Kriegesmann, Dr. Eric Groß                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Language</b>                                   | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Cycle</b>                                      | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Content</b>                                    | <p>Method for calculation and testing of reliability of dynamic machine systems</p> <p>Modeling</p> <p>System identification</p> <p>Simulation</p> <p>Processing of measurement data</p> <p>Damage accumulation</p> <p>Test planning and execution</p>                                                                                                                                                                                                                                |
| <b>Literature</b>                                 | <p>Bertsche, B.: Reliability in Automotive and Mechanical Engineering. Springer, 2008. ISBN: 978-3-540-33969-4</p> <p>Inman, Daniel J.: Engineering Vibration. Prentice Hall, 3rd Ed., 2007. ISBN-13: 978-0132281737</p> <p>Dresig, H., Holzweißig, F.: Maschinendynamik, Springer Verlag, 9. Auflage, 2009. ISBN 3540876936.</p> <p>VDA (Hg.): Zuverlässigkeitssicherung bei Automobilherstellern und Lieferanten. Band 3 Teil 2, 3. überarbeitete Auflage, 2004. ISSN 0943-9412</p> |

**Module M1438: Selected Topics of Mechanical Engineering and Management (Alternative B: 6 CP)**

**Courses**

| <b>Title</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Type</b>                                                                                | <b>Hrs/wk</b> | <b>CP</b> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------|-----------|
| Fatigue & Damage Tolerance (L0310)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Lecture                                                                                    | 2             | 3         |
| International Law for Engineers (L1749)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Lecture                                                                                    | 2             | 2         |
| International Law for Engineers (L1750)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Seminar                                                                                    | 2             | 2         |
| Lightweight Design Practical Course (L1258)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Project-/problem-based Learning                                                            | 3             | 3         |
| NaPe: Sustainable Product Development - Lecture Series of the Scientific Society for Product Development (WiGeP) (L3340)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Lecture                                                                                    | 2             | 3         |
| Sustainable Industrial Production (L2863)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Lecture                                                                                    | 2             | 3         |
| Productivity Management (L0928)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Project-/problem-based Learning                                                            | 2             | 2         |
| Productivity Management (L0931)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Recitation Section (small)                                                                 | 1             | 1         |
| SMS: Smart Machine Systems (L3341)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Lecture                                                                                    | 2             | 3         |
| Reliability in Engineering Dynamics (L2994)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Lecture                                                                                    | 2             | 2         |
| Reliability in Engineering Dynamics (L2995)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Recitation Section (small)                                                                 | 1             | 2         |
| <b>Module Responsible</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Prof. Nikola Bursac                                                                        |               |           |
| <b>Admission Requirements</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | None                                                                                       |               |           |
| <b>Recommended Previous Knowledge</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | see lecture description                                                                    |               |           |
| <b>Educational Objectives</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | After taking part successfully, students have reached the following learning results       |               |           |
| <b>Professional Competence</b><br><i>Knowledge</i> <ul style="list-style-type: none"> <li>Students are able to express their extended knowledge and discuss the connection of different special fields or application areas of Materials, Mechatronics and Product Development and Production</li> <li>Students are qualified to connect different special fields with each other</li> </ul> <i>Skills</i> <ul style="list-style-type: none"> <li>Students can apply specialized solution strategies and new scientific methods in selected areas</li> <li>Students are able to transfer learned skills to new and unknown problems and can develop own solution approaches</li> </ul> <b>Personal Competence</b><br><i>Social Competence</i><br><i>Autonomy</i> | Students are able to develop their knowledge and skills by autonomous election of courses. |               |           |
| <b>Workload in Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Depends on choice of courses                                                               |               |           |
| <b>Credit points</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 6                                                                                          |               |           |
| <b>Assignment for the Following Curricula</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Mechanical Engineering and Management: Core Qualification: Elective Compulsory             |               |           |

**Course L0310: Fatigue & Damage Tolerance**

|                                       |                                                                                                                                                                                                      |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Type</b>                           | Lecture                                                                                                                                                                                              |
| <b>Hrs/wk</b>                         | 2                                                                                                                                                                                                    |
| <b>CP</b>                             | 3                                                                                                                                                                                                    |
| <b>Workload in Hours</b>              | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                  |
| <b>Examination Form</b>               | Mündliche Prüfung                                                                                                                                                                                    |
| <b>Examination duration and scale</b> | 45 min                                                                                                                                                                                               |
| <b>Lecturer</b>                       | Dr. Martin Flamm                                                                                                                                                                                     |
| <b>Language</b>                       | EN                                                                                                                                                                                                   |
| <b>Cycle</b>                          | WiSe                                                                                                                                                                                                 |
| <b>Content</b>                        | Design principles, fatigue strength, crack initiation and crack growth, damage calculation, counting methods, methods to improve fatigue strength, environmental influences                          |
| <b>Literature</b>                     | Jaap Schijve, Fatigue of Structures and Materials. Kluwer Academic Publisher, Dordrecht, 2001 E. Haibach. Betriebsfestigkeit Verfahren und Daten zur Bauteilberechnung. VDI-Verlag, Düsseldorf, 1989 |

| Course L1749: International Law for Engineers |                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                    | Lecture                                                                                                                                                                                                                                                                                                                                    |
| <b>Hrs/wk</b>                                 | 2                                                                                                                                                                                                                                                                                                                                          |
| <b>CP</b>                                     | 2                                                                                                                                                                                                                                                                                                                                          |
| <b>Workload in Hours</b>                      | Independent Study Time 32, Study Time in Lecture 28                                                                                                                                                                                                                                                                                        |
| <b>Examination Form</b>                       | Klausur                                                                                                                                                                                                                                                                                                                                    |
| <b>Examination duration and scale</b>         | 90 Minuten                                                                                                                                                                                                                                                                                                                                 |
| <b>Lecturer</b>                               | Markus A. Meyer-Chory                                                                                                                                                                                                                                                                                                                      |
| <b>Language</b>                               | EN                                                                                                                                                                                                                                                                                                                                         |
| <b>Cycle</b>                                  | WiSe                                                                                                                                                                                                                                                                                                                                       |
| <b>Content</b>                                | <ul style="list-style-type: none"> <li>basics and selected legal aspects of international Engineers work and international laws, such as civil/common law, questions of jurisdiction and courts as well as arbitration and enforcement of titles, etc.</li> <li>also laws on contracts, construction, labor, patents, companies</li> </ul> |
| <b>Literature</b>                             | As per Stud.IP.                                                                                                                                                                                                                                                                                                                            |

| Course L1750: International Law for Engineers |                                                                                                                                                                                  |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                    | Seminar                                                                                                                                                                          |
| <b>Hrs/wk</b>                                 | 2                                                                                                                                                                                |
| <b>CP</b>                                     | 2                                                                                                                                                                                |
| <b>Workload in Hours</b>                      | Independent Study Time 32, Study Time in Lecture 28                                                                                                                              |
| <b>Examination Form</b>                       | Schriftliche Ausarbeitung                                                                                                                                                        |
| <b>Examination duration and scale</b>         | 10-20 Seiten                                                                                                                                                                     |
| <b>Lecturer</b>                               | Markus A. Meyer-Chory                                                                                                                                                            |
| <b>Language</b>                               | EN                                                                                                                                                                               |
| <b>Cycle</b>                                  | SoSe                                                                                                                                                                             |
| <b>Content</b>                                | <ul style="list-style-type: none"> <li>basics and selected legal aspects of international Engineers work - i.e. on contracts, construction, labor, patents, insurance</li> </ul> |
| <b>Literature</b>                             | As per Stud.IP                                                                                                                                                                   |

| Course L1258: Lightweight Design Practical Course |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                        | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Hrs/wk</b>                                     | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>CP</b>                                         | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Workload in Hours</b>                          | Independent Study Time 48, Study Time in Lecture 42                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Examination Form</b>                           | Mündliche Prüfung                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Examination duration and scale</b>             | 30 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Lecturer</b>                                   | Prof. Dieter Krause                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Language</b>                                   | DE/EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Cycle</b>                                      | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Content</b>                                    | <p>Development of a sandwich structure made of fibre reinforced plastics</p> <ul style="list-style-type: none"> <li>• getting familiar with fibre reinforced plastics as well as lightweight design</li> <li>• Design of a sandwich structure made of fibre reinforced plastics using finite element analysis (FEA)</li> <li>• Determination of material properties based on sample tests</li> <li>• manufacturing of the structure in the composite lab</li> <li>• Testing of the developed structure</li> <li>• Concept presentation</li> <li>• Self-organised teamwork</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Literature</b>                                 | <ul style="list-style-type: none"> <li>• Schürmann, H., „Konstruieren mit Faser-Kunststoff-Verbunden“, Springer, Berlin, 2005.</li> <li>• Puck, A., „Festigkeitsanalyse von Faser-Matrix-Laminaten“, Hanser, München, Wien, 1996.</li> <li>• R&amp;G, „Handbuch Faserverbundwerkstoffe“, Waldenbuch, 2009.</li> <li>• VDI 2014 „Entwicklung von Bauteilen aus Faser-Kunststoff-Verbund“</li> <li>• Ehrenstein, G. W., „Faserverbundkunststoffe“, Hanser, München, 2006.</li> <li>• Klein, B., „Leichtbau-Konstruktion“, Vieweg &amp; Sohn, Braunschweig, 1989.</li> <li>• Wiedemann, J., „Leichtbau Band 1: Elemente“, Springer, Berlin, Heidelberg, 1986.</li> <li>• Wiedemann, J., „Leichtbau Band 2: Konstruktion“, Springer, Berlin, Heidelberg, 1986.</li> <li>• Backmann, B.F., „Composite Structures, Design, Safety and Innovation“, Oxford (UK), Elsevier, 2005.</li> <li>• Krause, D., „Leichtbau“, In: Handbuch Konstruktion, Hrsg.: Rieg, F., Steinhilper, R., München, Carl Hanser Verlag, 2012.</li> <li>• Schulte, K., Fiedler, B., „Structure and Properties of Composite Materials“, Hamburg, TUHH - TuTech Innovation GmbH, 2005.</li> </ul> |

| Course L3340: NaPe: Sustainable Product Development - Lecture Series of the Scientific Society for Product Development (WiGeP) |                                                     |
|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Typ</b>                                                                                                                     | Lecture                                             |
| <b>Hrs/wk</b>                                                                                                                  | 2                                                   |
| <b>CP</b>                                                                                                                      | 3                                                   |
| <b>Workload in Hours</b>                                                                                                       | Independent Study Time 62, Study Time in Lecture 28 |
| <b>Examination Form</b>                                                                                                        | Klausur                                             |
| <b>Examination duration and scale</b>                                                                                          | 120 min                                             |
| <b>Lecturer</b>                                                                                                                | Prof. Nikola Bursac, Prof. Dieter Krause            |
| <b>Language</b>                                                                                                                | DE                                                  |
| <b>Cycle</b>                                                                                                                   | SoSe                                                |
| <b>Content</b>                                                                                                                 |                                                     |
| <b>Literature</b>                                                                                                              |                                                     |

| Course L2863: Sustainable Industrial Production |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                      | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Hrs/wk</b>                                   | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>CP</b>                                       | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Workload in Hours</b>                        | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Examination Form</b>                         | Klausur                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Examination duration and scale</b>           | 60 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Lecturer</b>                                 | Dr. Simon Markus Kothe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Language</b>                                 | DE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Cycle</b>                                    | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Content</b>                                  | <p>Industrial production deals with the manufacture of physical products to satisfy human needs using various manufacturing processes that change the form and physical properties of raw materials. Manufacturing is a central driver of economic development and has a major impact on the well-being of humanity. However, the scale of current manufacturing activities results in enormous global energy and material demands that are harmful to both the environment and people. Historically, industrial activities were mostly oriented towards economic constraints, while social and environmental consequences were only hardly considered. As a result, today's global consumption rates of many resources and associated emissions often exceed the natural regeneration rate of our planet. In this respect, current industrial production can mostly be described as unsustainable. This is emphasized each year by the Earth Overshoot Day, which marks the day when humanity's ecological footprint exceeds the Earth's annual regenerative capacity.</p> <p>This lecture aims to provide the motivation, analytical methods as well as approaches for sustainable industrial production and to clarify the influence of the production phase in relation to the raw material, use and recycling phases in the entire life cycle of products. For this, the following topics will be highlighted:</p> <ul style="list-style-type: none"> <li>- Motivation for sustainable production, the 17 Sustainable Development Goals (SDGs) of the UN and their relevance for tomorrow's manufacturing;</li> <li>- raw material vs. production phase vs. use phase vs. recycling/end-of-life phase: importance of the production phase for the environmental impact of manufactured products;</li> <li>- Typical energy- and resource-intensive processes in industrial production and innovative approaches to increase energy and resource efficiency;</li> <li>- Methodology for optimizing the energy and resource efficiency of industrial manufacturing chains with the three steps of modeling (1), evaluating (2) and improving (3);</li> <li>- Resource efficiency of industrial manufacturing value chains and its assessment using life cycle analysis (LCA);</li> <li>- Exercise: LCA analysis of a manufacturing process (thermoplastic joining of an aircraft fuselage segment) as part of a product life cycle assessment.</li> </ul> |
| <b>Literature</b>                               | <p>Literatur:</p> <ul style="list-style-type: none"> <li>- Stefan Alexander (2020): Resource efficiency in manufacturing value chains. Cham: Springer International Publishing.</li> <li>- Hauschild, Michael Z.; Rosenbaum, Ralph K.; Olsen, Stig Irving (Hg.) (2018): Life Cycle Assessment. Theory and Practice. Cham: Springer International Publishing.</li> <li>- Kishita, Yusuke; Matsumoto, Mitsutaka; Inoue, Masato; Fukushige, Shinichi (2021): EcoDesign and sustainability. Singapore: Springer.</li> <li>- Schebek, Liselotte; Herrmann, Christoph; Cerdas, Felipe (2019): Progress in Life Cycle Assessment. Cham: Springer International Publishing.</li> <li>- Thiede, Sebastian; Hermann, Christoph (2019): Eco-factories of the future. Cham: Springer Nature Switzerland AG.</li> <li>- Vorlesungsskript.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Course L0928: Productivity Management |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                            | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Hrs/wk</b>                         | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>CP</b>                             | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Workload in Hours</b>              | Independent Study Time 32, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Examination Form</b>               | Klausur                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Examination duration and scale</b> | 90 Minuten                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Lecturer</b>                       | Prof. Hermann Lödding, Christopher Mundt                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Language</b>                       | DE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Cycle</b>                          | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Content</b>                        | <ul style="list-style-type: none"> <li>• Principles of productivity management</li> <li>• Shop floor management and standardisation</li> <li>• Takt analysis and design of manual operations</li> <li>• Maintenance Principles</li> <li>• Total Productive Maintenance (TPM)</li> <li>• Optimisation of set-up operations</li> <li>• Analysis of interlinked production systems</li> </ul>                                                                                                            |
| <b>Literature</b>                     | <p>Bokranz, R.; Landau, K.: Produktivitätsmanagement von Arbeitssystemen. Schäffer-Poeschel, Stuttgart, 2006.</p> <p>Takeda, H.: Das synchrone Produktionssystem: Just-in-Time für das ganze Unternehmen. 5. Aufl., mi-Wirtschaftsbuch, FinanzBuch Verlag, München, 2006.</p> <p>Nakajima, S.: Management der Produktionseinrichtungen (Total Productive Maintenance). Campus Verlag, New York, 1995.</p> <p>Shingo, S.: A Revolution in Manufacturing: The SMED System. Productivity, Inc., 1985</p> |

| Course L0931: Productivity Management |                                                     |
|---------------------------------------|-----------------------------------------------------|
| <b>Typ</b>                            | Recitation Section (small)                          |
| <b>Hrs/wk</b>                         | 1                                                   |
| <b>CP</b>                             | 1                                                   |
| <b>Workload in Hours</b>              | Independent Study Time 16, Study Time in Lecture 14 |
| <b>Examination Form</b>               | Klausur                                             |
| <b>Examination duration and scale</b> | 90 Minuten                                          |
| <b>Lecturer</b>                       | Prof. Hermann Lödding                               |
| <b>Language</b>                       | DE                                                  |
| <b>Cycle</b>                          | SoSe                                                |
| <b>Content</b>                        | See interlocking course                             |
| <b>Literature</b>                     | See interlocking course                             |



| Course L3341: SMS: Smart Machine Systems |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                               | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Hrs/wk</b>                            | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>CP</b>                                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Workload in Hours</b>                 | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Examination Form</b>                  | Klausur                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Examination duration and scale</b>    | 60 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Lecturer</b>                          | Prof. Nikola Bursac                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Language</b>                          | DE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Cycle</b>                             | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Content</b>                           | <p><b>Knowledge:</b> Students acquire a comprehensive understanding of how data can be methodically used in product development to optimize complex development processes, support informed decision-making, and design innovative products. They understand the fundamental principles and methods by which data can be effectively utilized in the development process of technical products or systems. They learn to identify the strategic potentials of data-driven approaches, assess specific challenges such as data quality, security, or availability, and strategically leverage the opportunities of data-driven decisions. Furthermore, students develop the ability to critically analyze data-driven decision-making processes in product development and adapt them to specific application scenarios. A special emphasis is placed on not considering data-driven insights in isolation but integrating them interdisciplinarily to enable innovative and holistic solutions. With this knowledge, they are able to sustainably enhance value creation through the targeted use of data in product development and successfully develop data-driven strategies in a practical context.</p> <p><b>Skills:</b> Students can independently apply methods for using data in the development process of technical products and systems and use the insights gained to optimize products. They are capable of independently implementing methods for utilizing operational data to support data-based decisions in the development process and derive well-founded action options. Additionally, they master the independent execution of data analyses and the critical interpretation of results. They can handle challenges related to data management and are able to evaluate and effectively manage the opportunities and risks of data-driven approaches. Students are also able to specifically use insights from data analyses to support decision-making in development processes and advance data-driven innovation strategies.</p> <p><b>Personal Competencies:</b></p> <p><b>Social Competence:</b><br/>Students are capable of collaborating in interdisciplinary teams to develop sustainable solutions for real product development challenges based on operational data.</p> <p><b>Autonomy:</b><br/>Students can independently identify problems in the field of data-driven product development and develop appropriate measures to solve them.</p> |
| <b>Literature</b>                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Course L2994: Reliability in Engineering Dynamics |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                        | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Hrs/wk</b>                                     | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>CP</b>                                         | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Workload in Hours</b>                          | Independent Study Time 32, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Examination Form</b>                           | Klausur                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Examination duration and scale</b>             | 90 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Lecturer</b>                                   | Dr. Eric Groß, Prof. Benedikt Kriegesmann                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Language</b>                                   | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Cycle</b>                                      | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Content</b>                                    | <p>Method for calculation and testing of reliability of dynamic machine systems</p> <p>Modeling</p> <p>System identification</p> <p>Simulation</p> <p>Processing of measurement data</p> <p>Damage accumulation</p> <p>Test planning and execution</p>                                                                                                                                                                                                                                |
| <b>Literature</b>                                 | <p>Bertsche, B.: Reliability in Automotive and Mechanical Engineering. Springer, 2008. ISBN: 978-3-540-33969-4</p> <p>Inman, Daniel J.: Engineering Vibration. Prentice Hall, 3rd Ed., 2007. ISBN-13: 978-0132281737</p> <p>Dresig, H., Holzweißig, F.: Maschinendynamik, Springer Verlag, 9. Auflage, 2009. ISBN 3540876936.</p> <p>VDA (Hg.): Zuverlässigkeitssicherung bei Automobilherstellern und Lieferanten. Band 3 Teil 2, 3. überarbeitete Auflage, 2004. ISSN 0943-9412</p> |

| Course L2995: Reliability in Engineering Dynamics |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                        | Recitation Section (small)                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Hrs/wk</b>                                     | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>CP</b>                                         | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Workload in Hours</b>                          | Independent Study Time 46, Study Time in Lecture 14                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Examination Form</b>                           | Klausur                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Examination duration and scale</b>             | 90 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Lecturer</b>                                   | Prof. Benedikt Kriegesmann, Dr. Eric Groß                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Language</b>                                   | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Cycle</b>                                      | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Content</b>                                    | <p>Method for calculation and testing of reliability of dynamic machine systems</p> <p>Modeling</p> <p>System identification</p> <p>Simulation</p> <p>Processing of measurement data</p> <p>Damage accumulation</p> <p>Test planning and execution</p>                                                                                                                                                                                                                                |
| <b>Literature</b>                                 | <p>Bertsche, B.: Reliability in Automotive and Mechanical Engineering. Springer, 2008. ISBN: 978-3-540-33969-4</p> <p>Inman, Daniel J.: Engineering Vibration. Prentice Hall, 3rd Ed., 2007. ISBN-13: 978-0132281737</p> <p>Dresig, H., Holzweißig, F.: Maschinendynamik, Springer Verlag, 9. Auflage, 2009. ISBN 3540876936.</p> <p>VDA (Hg.): Zuverlässigkeitssicherung bei Automobilherstellern und Lieferanten. Band 3 Teil 2, 3. überarbeitete Auflage, 2004. ISSN 0943-9412</p> |

| Module M1919: Sustainable operation of technical assets        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |        |    |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------|----|
| Courses                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |        |    |
| Title                                                          | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  | Hrs/wk | CP |
| Fundamentals of Maintenance, Repair and Overhaul (MRO) (L3160) | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  | 3      | 4  |
| Fundamentals of Maintenance, Repair and Overhaul (MRO) (L3161) | Recitation Section (large)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  | 1      | 2  |
| <b>Module Responsible</b>                                      | Prof. Gerko Wende                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |        |    |
| <b>Admission Requirements</b>                                  | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |        |    |
| <b>Recommended Previous Knowledge</b>                          | We recommend knowledge in the areas of general engineering sciences, aeronautics and aircraft systems engineering. Technical fields like mechanical engineering, mechatronics and production engineering will be introduced into the relevant aeronautical content.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |        |    |
| <b>Educational Objectives</b>                                  | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |        |    |
| <b>Professional Competence</b>                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |        |    |
| <i>Knowledge</i>                                               | The students are able to describe fundamental correlations for the sustainable operation of technical assets and to identify solution approaches for complex optimization problems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |        |    |
| <i>Skills</i>                                                  | The students are enabled to apply the general engineering capabilities of the individual course towards the optimization of the sustainability in operation of technical assets. The resulting competencies will open an entry into positions in the development, production and technical operation of sustainable products in the mobility and engineering industries.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |        |    |
| <b>Personal Competence</b>                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |        |    |
| <i>Social Competence</i>                                       | The students are able to work in mixed groups with a clear focus on the approached solutions by respecting the complex environment of multiple stakeholders.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |        |    |
| <i>Autonomy</i>                                                | The students are enabled to find solutions for optimization problems and to take required decision for the assessment of determining factors independently.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |        |    |
| <b>Workload in Hours</b>                                       | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |        |    |
| <b>Credit points</b>                                           | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |        |    |
| <b>Course achievement</b>                                      | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |        |    |
| <b>Examination</b>                                             | Oral exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |        |    |
| <b>Examination duration and scale</b>                          | 30 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |        |    |
| <b>Assignment for the Following Curricula</b>                  | Aeronautics: Core Qualification: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory<br>Mechanical Engineering and Management: Core Qualification: Elective Compulsory<br>Mechatronics: Core Qualification: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Production: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Materials: Elective Compulsory<br>Theoretical Mechanical Engineering: Specialisation Product Development and Production: Elective Compulsory<br>Theoretical Mechanical Engineering: Specialisation Aircraft Systems Engineering: Elective Compulsory |  |        |    |

| Course L3160: Fundamentals of Maintenance, Repair and Overhaul (MRO) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                                           | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Hrs/wk</b>                                                        | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>CP</b>                                                            | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Workload in Hours</b>                                             | Independent Study Time 78, Study Time in Lecture 42                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Lecturer</b>                                                      | Prof. Gerko Wende                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Language</b>                                                      | DE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Cycle</b>                                                         | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Content</b>                                                       | <p>Fundamentals for the sustainable operation of technical assets by means of maintenance, repair and overhaul (MRO):</p> <ul style="list-style-type: none"> <li>• Life cycle analytics</li> <li>• Material circularity and service products</li> <li>• Rules and regulations</li> <li>• Processes and production methods</li> <li>• Tools and technologies</li> <li>• Data handling and usage</li> <li>• Design for maintenance</li> <li>• Self-healing technical systems</li> </ul>                                                                                                                                                                                                                                   |
| <b>Literature</b>                                                    | <ul style="list-style-type: none"> <li>• The Greenhouse Gas Protocol, World Resources Institute and World Business Council for Sustainable Development, 2004, ISBN 1-56973-568-9</li> <li>• The Environmental Performance Index 2022, Ranking country performance on sustainability issues, Yale Center for Environmental Law &amp; Policy, Yale University</li> <li>• Business model innovation in the aerospace industry: Strategic options for maintenance, repair and overhaul firms, Oliver Wirths, Dissertation, Universität zu Köln, 2019</li> <li>• Industrielles Luftfahrtmanagement, Technik und Organisation luftfahrttechnischer Betriebe, Martin Hinsch, Springer, 2019, ISBN 978-3-662-58803-1</li> </ul> |

| Course L3161: Fundamentals of Maintenance, Repair and Overhaul (MRO) |                                                     |
|----------------------------------------------------------------------|-----------------------------------------------------|
| <b>Typ</b>                                                           | Recitation Section (large)                          |
| <b>Hrs/wk</b>                                                        | 1                                                   |
| <b>CP</b>                                                            | 2                                                   |
| <b>Workload in Hours</b>                                             | Independent Study Time 46, Study Time in Lecture 14 |
| <b>Lecturer</b>                                                      | Prof. Gerko Wende                                   |
| <b>Language</b>                                                      | DE                                                  |
| <b>Cycle</b>                                                         | WiSe                                                |
| <b>Content</b>                                                       | See interlocking course                             |
| <b>Literature</b>                                                    | See interlocking course                             |

| Module M2079: GSD - Generational Sheet-Metal Development |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |              |                    |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|--------------------|
| Courses                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |              |                    |
| Title                                                    | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              | Hrs/wk       | CP                 |
| GSD - Generational Sheet-Metal Development (L3064)       | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              | 4            | 6                  |
| <b>Module Responsible</b>                                | Prof. Nikola Bursac                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |              |                    |
| <b>Admission Requirements</b>                            | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              |              |                    |
| <b>Recommended Previous Knowledge</b>                    | No specific prior knowledge is required.<br>Experience in engineering design and basic manufacturing techniques is advantageous.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |              |              |                    |
| <b>Educational Objectives</b>                            | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              |              |                    |
| <b>Professional Competence</b>                           | <p><i>Knowledge</i> The students gain a comprehensive understanding of product development processes. Through practical examples and the teaching of methodological knowledge, they learn how to directly apply these concepts in development projects, allowing them to gain valuable hands-on experience. They understand the fundamental principles and methods of system generation development and practice these within the scope of their own product development project.</p> <p>A particular focus is placed on design for manufacturability in sheet metal construction, the development process, and problem-solving strategies. Students learn how to apply design principles to develop innovative and practical solutions that are both functional and feasible. Additionally, CAD tools are used to optimize the preparation of their ideas for manufacturing.</p> <p>The course combines theoretical knowledge with creativity and practical experience. This enables students to actively participate in product development and further enhance their skills.</p> <p><i>Skills</i> Students acquire the ability to:</p> <ul style="list-style-type: none"> <li>• <b>Learn and apply methods</b> for problem-solving within the development context</li> <li>• <b>Analyze and evaluate functional principles</b></li> <li>• <b>Identify and incorporate customer requirements</b></li> <li>• <b>Understand, deepen, and implement system generation development</b></li> <li>• <b>Develop designs optimized for manufacturability</b></li> <li>• <b>Use CAD tools for design tasks</b></li> <li>• <b>Leverage creative ideas in product development</b></li> </ul> |              |              |                    |
| <b>Personal Competence</b>                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |              |                    |
| <i>Social Competence</i>                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |              |                    |
| <i>Autonomy</i>                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |              |                    |
| <i>Autonomy</i>                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |              |                    |
| <b>Workload in Hours</b>                                 | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              |              |                    |
| <b>Credit points</b>                                     | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |              |              |                    |
| <b>Course achievement</b>                                | <b>Compulsory</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Bonus</b> | <b>Form</b>  | <b>Description</b> |
|                                                          | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None         | Presentation | GSD Konstruktion   |
| <b>Examination</b>                                       | Oral exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |              |              |                    |
| <b>Examination duration and scale</b>                    | 40 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |              |              |                    |
| <b>Assignment for the Following Curricula</b>            | Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory<br>Mechanical Engineering and Management: Core Qualification: Elective Compulsory<br>Mechatronics: Core Qualification: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Materials: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Production: Elective Compulsory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |              |                    |

| Course L3064: GSD - Generational Sheet-Metal Development |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                               | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Hrs/wk</b>                                            | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>CP</b>                                                | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Workload in Hours</b>                                 | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Lecturer</b>                                          | Prof. Nikola Bursac                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Language</b>                                          | DE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Cycle</b>                                             | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Content</b>                                           | <p>Experience in mechanical engineering design and the fundamentals of manufacturing engineering</p> <p>After successful completion of the course, students will be able to explain development projects using the theory of product generation engineering and explain design rules for sheet metal development.</p> <p>After successful completion of the course, students will be able to apply the theory of product generation engineering to development tasks and develop sheet-metal products suitable for production.</p> <p>After successful completion of the course, students will be able to develop a product in a team and to compete against other teams.</p> <p>After successful completion of the course, students will be able to independently access knowledge required for sheet metal development.</p>                                                                                                                                                         |
| <b>Literature</b>                                        | <p>[1] Albers, A., Rapp, S., Spadinger, M., Richter, T., Birk, C., Marthaler, M., Heimicke, J., Kurtz, V., Wessels, H. (2019). Das Referenzsystem im Modell der PGE - Produktgenerationsentwicklung: Vorschlag einer generalisierten Beschreibung von Referenzprodukten und ihrer Wechselbeziehungen, ICED 2019</p> <p>[2] Albers et al. (2017). PGE - Produktgenerationsentwicklung am Beispiel des Zweimassenschwungrads, Forschung im Ingenieurwesen 81(1), 13-31, 2017</p> <p>[3] Albers, A., Heimicke, J., et. al. (2018). Product Profiles: Modelling customer benefits as a foundation to bring inventions to innovations. Procedia CIRP. (70), 253-258</p> <p>[4] Stachowiak, H., (1973). Allgemeine Modelltheorie. ISBN 3-211-81106-0</p> <p>[5] Rophol, G., (2009). Systemtheorie der Technik. ISBN: 978-3-86644-374-7</p> <p>[6] Herstatt, C., &amp; Verworn, B. (2007). Management der frühen Innovationsphasen: Grundlagen - Methoden - Neue Ansätze. Gabler Verlag.</p> |

| Module M1292: Marketing and Communication           |                                                                                                                                                                |                            |        |    |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------|----|
| Courses                                             |                                                                                                                                                                |                            |        |    |
| Title                                               |                                                                                                                                                                | Typ                        | Hrs/wk | CP |
| Business-to-Business Marketing (L0762)              |                                                                                                                                                                | Lecture                    | 2      | 2  |
| Business-to-Business Marketing Case Studies (L1760) |                                                                                                                                                                | Recitation Section (small) | 2      | 2  |
| Intercultural Management and Communication (L0846)  |                                                                                                                                                                | Lecture                    | 2      | 2  |
| Module Responsible                                  | Prof. Christian Lüthje                                                                                                                                         |                            |        |    |
| Admission Requirements                              | None                                                                                                                                                           |                            |        |    |
| Recommended Previous Knowledge                      | No specific knowledge required. Bachelor-level knowledge in business administration with some insights into marketing and international management is helpful. |                            |        |    |
| Educational Objectives                              | After taking part successfully, students have reached the following learning results                                                                           |                            |        |    |
| Professional Competence                             |                                                                                                                                                                |                            |        |    |
| <i>Knowledge</i>                                    |                                                                                                                                                                |                            |        |    |
| <i>Skills</i>                                       |                                                                                                                                                                |                            |        |    |
| <i>Social Competence</i>                            |                                                                                                                                                                |                            |        |    |
| Personal Competence                                 |                                                                                                                                                                |                            |        |    |
| <i>Social Competence</i>                            |                                                                                                                                                                |                            |        |    |
| <i>Autonomy</i>                                     |                                                                                                                                                                |                            |        |    |
| Workload in Hours                                   | Independent Study Time 96, Study Time in Lecture 84                                                                                                            |                            |        |    |
| Credit points                                       | 6                                                                                                                                                              |                            |        |    |
| Course achievement                                  | None                                                                                                                                                           |                            |        |    |
| Examination                                         | Subject theoretical and practical work                                                                                                                         |                            |        |    |
| Examination duration and scale                      | Written elaboration, excercises, presentation, oral participation, written exam                                                                                |                            |        |    |
| Assignment for the Following Curricula              | Global Innovation Management: Core Qualification: Compulsory<br>Mechanical Engineering and Management: Core Qualification: Elective Compulsory                 |                            |        |    |

| Course L0762: Business-to-Business Marketing |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                   | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Hrs/wk</b>                                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>CP</b>                                    | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Workload in Hours</b>                     | Independent Study Time 32, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Lecturer</b>                              | Prof. Christian Lüthje                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Language</b>                              | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Cycle</b>                                 | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Content</b>                               | <p>Business-to-business (B2B) markets play an important role in most economies. At the same time, B2B markets differ strongly from consumer goods markets. For example, companies' buying decisions follow different rules than those of consuming individuals. Consequently, marketing mix decisions in B2B markets need to follow the specific circumstances in such markets. The aim of this lecture is to enable students to understand the specifics of marketing in B2B markets.</p> <p>Why this course is essential:</p> <p>B2B marketing students learn which strategic marketing decisions may be most appropriate in industrial markets. Following that, the lecture will focus more on different options to design marketing mix elements - Pricing, Communication and Distribution - in B2B markets. We extend the student's basic knowhow in marketing and focus on the specific requirements in B2B markets.</p> <p>Content:</p> <p>The following key topics will be explored throughout the course:</p> <ul style="list-style-type: none"> <li>• The importance, specific characteristics and developments of B2B markets today.</li> <li>• Organizational buying behavior and the corporate buying process.</li> <li>• B2B marketing strategies regarding modes and time of market entry with focus on innovative industrial products.</li> <li>• Types of project-related cooperation in the B2B project business.</li> <li>• Specific operational marketing methods in communication (success factors of fairs and exhibitions, importance of public relations for B2B markets); pricing (measuring willingness-to-pay via auctions; value-based pricing in industrial markets, bidding models and auctioning); distribution and channel strategies for B2B markets.</li> <li>• Marketing in complex value chains: Solving the problem of direct customers' unwillingness to adopt innovative products by directly addressing indirect customers.</li> </ul> <p>What you will learn and get:</p> <p>By the end of this course students will develop a thorough understanding of:</p> <ul style="list-style-type: none"> <li>• How organizations and firms buy.</li> <li>• How marketing can be performed in complex value chains.</li> <li>• Promising market and competitive strategies in B2B markets.</li> <li>• Modes of cooperation in B2B markets.</li> <li>• Marketing-mix decisions in B2B marketing (communication, pricing, distribution).</li> </ul> |
| <b>Literature</b>                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |



| Course L1760: Business-to-Business Marketing Case Studies |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                                | Recitation Section (small)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Hrs/wk</b>                                             | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>CP</b>                                                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Workload in Hours</b>                                  | Independent Study Time 32, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Lecturer</b>                                           | Prof. Christian Lühje                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Language</b>                                           | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Cycle</b>                                              | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Content</b>                                            | <p>This course aims at deepening and applying the subjects taught in the lecture Business-to-Business Marketing (B2B). Students work on case studies in teams comprising 5-6 people. Each of these cases is related to a specific topic that has been tackled in the previous lectures of this module. The cases can comprise scientific studies or specific company examples (e.g. how company X built up a new salesforce; how company Y designed a successful communication campaign for other countries, how research study Z contributes to the understanding of intercultural differences). The student teams receive material (e.g. scientific articles, press articles) and work with this material to complete presentation documents. The results will be illustrated and discussed in a short presentation.</p> <p>Why this course is essential:</p> <p>The cases will enable the student teams to analyze problems, to discuss theoretical frameworks and scientific results, to evaluate decisions made in companies and/or to develop own ideas for solutions.</p> <p>Content:</p> <p>The course is divided into two sessions where a selection of case studies is presented and discussed to challenge student's understanding of the topics and allow them to elaborate their own solutions.</p> <p>What you will learn and get:</p> <p>By the end of this course students will develop a thorough understanding of the specificities of B2B marketing:</p> <ul style="list-style-type: none"> <li>• Market potential and forecast strategies.</li> <li>• Distribution channels.</li> <li>• Pricing strategy.</li> <li>• Market segmentation.</li> </ul> |
| <b>Literature</b>                                         | <p>Die Materialien werden jedes Semester neu zusammengestellt, um die ausgewählten Fälle aktuell zu halten.</p> <p>Will be newly compiled each semester to keep the cases up-to-date and fresh.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Course L0846: Intercultural Management and Communication |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                               | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Hrs/wk</b>                                            | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>CP</b>                                                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Workload in Hours</b>                                 | Independent Study Time 32, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Lecturer</b>                                          | Prof. Christian Lühje                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Language</b>                                          | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Cycle</b>                                             | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Content</b>                                           | <p>Globalization of business processes and the revolution in information and communication technologies (ICT) have resulted in distributed workflows across geographic boundaries. These developments have led to the creation of (virtual) multi-cultural, multi-ethnic teams with diverse cultural backgrounds. Such diversity generally has a positive impact on creativity and innovativeness, as many empirical studies confirm. Nevertheless, varying cultural practices, communication styles, and contextual sensibilities have the potential to disturb or even disrupt collaborative work processes, if left unmanaged. This course focuses on inter-cultural management from both, theoretical as well as practical, points of view to provide a solid fundament to students enabling them to operate successfully in cross-cultural settings. Case studies and guest lecture(s) will be used to provide added practical relevance to the course. In addition, where practicable, student assignments will be used to foster autonomous learning. Some of the main topics covered in this course include:</p> <ul style="list-style-type: none"> <li>• Understanding "culture" and its impact on human interaction</li> <li>• Verbal and non-verbal communication</li> <li>• High and low context communication</li> <li>• Role of formality and non-formality in communication</li> <li>• Varying interpretations of symbols, rituals &amp; gestures</li> <li>• Managing diversity in domestic settings</li> </ul> |
| <b>Literature</b>                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Module M1285: Internship MEM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                      |        |    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------|----|
| Courses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                      |        |    |
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Typ                                                                                  | Hrs/wk | CP |
| <b>Module Responsible</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NN                                                                                   |        |    |
| <b>Admission Requirements</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                                                                                 |        |    |
| <b>Recommended Previous Knowledge</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Basic knowledge of German language                                                   |        |    |
| <b>Educational Objectives</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | After taking part successfully, students have reached the following learning results |        |    |
| <b>Professional Competence</b><br><i>Knowledge</i> <ul style="list-style-type: none"> <li>• Students are able to describe business structures and processes</li> <li>• They can summarise and present the contents of the project(s) they worked on during the internship</li> </ul><br><i>Skills</i> <ul style="list-style-type: none"> <li>• Students are able to transfer knowledge and methods learned from the project on other applications</li> <li>• They are able to plan their work and their procedure</li> <li>• During their project, they can make decisions, justify them and based upon these they can draw conclusions on future work</li> </ul><br><b>Personal Competence</b><br><i>Social Competence</i> <ul style="list-style-type: none"> <li>• Students know and understand social structures of companies and are able to integrate themselves into these</li> <li>• They can discuss their work with colleagues and respond adequately to critique</li> <li>• They can work in teams, undertake tasks and comply with the time schedule</li> </ul><br><i>Autonomy</i> <ul style="list-style-type: none"> <li>• Students know their interests, strengths and weaknesses. Based on this, they can find a suitable position for an internship, apply for it and explain their competences to others.</li> </ul> |                                                                                      |        |    |
| <b>Workload in Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Independent Study Time 180, Study Time in Lecture 0                                  |        |    |
| <b>Credit points</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 6                                                                                    |        |    |
| <b>Course achievement</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | None                                                                                 |        |    |
| <b>Examination</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Written elaboration (accord. to Internship Regulations)                              |        |    |
| <b>Examination duration and scale</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | see internship guidelines                                                            |        |    |
| <b>Assignment for the Following Curricula</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Mechanical Engineering and Management: Core Qualification: Elective Compulsory       |        |    |

## Module M1343: Structure and properties of fibre-polymer-composites

### Courses

| Title                                                        | Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Hrs/wk | CP |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|
| Structure and properties of fibre-polymer-composites (L1894) | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2      | 3  |
| Structure and properties of fibre-polymer-composites (L2614) | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2      | 2  |
| Structure and properties of fibre-polymer-composites (L2613) | Recitation Section (large)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1      | 1  |
| <b>Module Responsible</b>                                    | Prof. Bodo Fiedler                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |    |
| <b>Admission Requirements</b>                                | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Recommended Previous Knowledge</b>                        | Basics: chemistry / physics / materials science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |    |
| <b>Educational Objectives</b>                                | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Professional Competence</b><br><i>Knowledge</i>           | <p>Students can use the knowledge of fiber-reinforced composites (FRP) and its constituents to play (fiber / matrix) and define the necessary testing and analysis.</p> <p>They can explain the complex relationships structure-property relationship and the interactions of chemical structure of the polymers, their processing with the different fiber types, including to explain neighboring contexts (e.g. sustainability, environmental protection).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| <i>Skills</i>                                                | <p>Students are capable of</p> <ul style="list-style-type: none"> <li>• using standardized calculation methods in a given context to mechanical properties (modulus, strength) to calculate and evaluate the different materials.</li> <li>• approximate sizing using the network theory of the structural elements implement and evaluate.</li> <li>• selecting appropriate solutions for mechanical recycling problems and sizing example stiffness, corrosion resistance.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |    |
| <b>Personal Competence</b><br><i>Social Competence</i>       | <p>Students can</p> <ul style="list-style-type: none"> <li>• arrive at funded work results in heterogenous groups and document them.</li> <li>• provide appropriate feedback and handle feedback on their own performance constructively.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <i>Autonomy</i>                                              | <p>Students are able to</p> <ul style="list-style-type: none"> <li>- assess their own strengths and weaknesses.</li> <li>- assess their own state of learning in specific terms and to define further work steps on this basis.</li> <li>- assess possible consequences of their professional activity.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |    |
| <b>Workload in Hours</b>                                     | Independent Study Time 110, Study Time in Lecture 70                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Credit points</b>                                         | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| <b>Course achievement</b>                                    | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Examination</b>                                           | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |    |
| <b>Examination duration and scale</b>                        | 90 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| <b>Assignment for the Following Curricula</b>                | <p>Aircraft Systems Engineering: Core Qualification: Elective Compulsory</p> <p>International Management and Engineering: Specialisation II. Product Development and Production: Elective Compulsory</p> <p>Aeronautics: Core Qualification: Elective Compulsory</p> <p>Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory</p> <p>Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory</p> <p>Mechanical Engineering - Product Development and Production: Specialisation Materials: Compulsory</p> <p>Materials Science and Engineering: Specialisation Engineering Materials: Elective Compulsory</p> <p>Mechanical Engineering and Management: Core Qualification: Compulsory</p> <p>Biomedical Engineering: Specialisation Implants and Endoprotheses: Elective Compulsory</p> <p>Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory</p> <p>Product Development, Materials and Production: Specialisation Production: Elective Compulsory</p> <p>Product Development, Materials and Production: Specialisation Materials: Compulsory</p> <p>Renewable Energies: Specialisation Bioenergy Systems: Elective Compulsory</p> <p>Renewable Energies: Specialisation Wind Energy Systems: Elective Compulsory</p> <p>Renewable Energies: Specialisation Solar Energy Systems: Elective Compulsory</p> <p>Theoretical Mechanical Engineering: Specialisation Materials Science: Elective Compulsory</p> |        |    |

| Course L1894: Structure and properties of fibre-polymer-composites |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                                         | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Hrs/wk</b>                                                      | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>CP</b>                                                          | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Workload in Hours</b>                                           | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Lecturer</b>                                                    | Prof. Bodo Fiedler                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Language</b>                                                    | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Cycle</b>                                                       | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Content</b>                                                     | <ul style="list-style-type: none"> <li>- Microstructure and properties of the matrix and reinforcing materials and their interaction</li> <li>- Development of composite materials</li> <li>- Mechanical and physical properties</li> <li>- Mechanics of Composite Materials</li> <li>- Laminate theory</li> <li>- Test methods</li> <li>- Non destructive testing</li> <li>- Failure mechanisms</li> <li>- Theoretical models for the prediction of properties</li> <li>- Application</li> </ul> |
| <b>Literature</b>                                                  | Hall, Clyne: Introduction to Composite materials, Cambridge University Press<br>Daniel, Ishai: Engineering Mechanics of Composites Materials, Oxford University Press<br>Mallick: Fibre-Reinforced Composites, Marcel Dekker, New York                                                                                                                                                                                                                                                            |

| Course L2614: Structure and properties of fibre-polymer-composites |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                                         | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Hrs/wk</b>                                                      | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>CP</b>                                                          | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Workload in Hours</b>                                           | Independent Study Time 32, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Lecturer</b>                                                    | Prof. Bodo Fiedler                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Language</b>                                                    | DE/EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Cycle</b>                                                       | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Content</b>                                                     | <p>The students receive the assignment in the form of a material design for test bodies made of fibre composites. Technical and normative requirements are listed in the assignment, all other required information comes from the lectures and exercises or the respective documents (electronically and in conversation).</p> <p>The procedure is specified in a milestone plan and enables the students to plan subtasks and thus work continuously. At the end of the project, different test specimens were tested in tensile or bending tests.</p> <p>In the individual project meetings, the conception (discussion of requirements and risks) is scrutinised. The calculations are analysed, the production methods are evaluated and determined. Materials are selected and the test specimens are manufactured according to standards. The quality and mechanical properties are checked and classified. At the end, a final report is prepared and the results are presented to all participants in the form of a presentation and discussed.</p> <p>Translated with <a href="https://www.DeepL.com/Translator">www.DeepL.com/Translator</a> (free version)</p> |
| <b>Literature</b>                                                  | Hall, Clyne: Introduction to Composite materials, Cambridge University Press<br>Daniel, Ishai: Engineering Mechanics of Composites Materials, Oxford University Press<br>Mallick: Fibre-Reinforced Composites, Marcel Dekker, New York                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Course L2613: Structure and properties of fibre-polymer-composites |                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                                         | Recitation Section (large)                                                                                                                                                                                                             |
| <b>Hrs/wk</b>                                                      | 1                                                                                                                                                                                                                                      |
| <b>CP</b>                                                          | 1                                                                                                                                                                                                                                      |
| <b>Workload in Hours</b>                                           | Independent Study Time 16, Study Time in Lecture 14                                                                                                                                                                                    |
| <b>Lecturer</b>                                                    | Prof. Bodo Fiedler                                                                                                                                                                                                                     |
| <b>Language</b>                                                    | EN                                                                                                                                                                                                                                     |
| <b>Cycle</b>                                                       | SoSe                                                                                                                                                                                                                                   |
| <b>Content</b>                                                     | <p>The contents of the lecture are repeated and deepened using practical examples.</p> <p>Calculations are carried out together or individually, and the results are discussed critically.</p>                                         |
| <b>Literature</b>                                                  | Hall, Clyne: Introduction to Composite materials, Cambridge University Press<br>Daniel, Ishai: Engineering Mechanics of Composites Materials, Oxford University Press<br>Mallick: Fibre-Reinforced Composites, Marcel Dekker, New York |

| Module M0805: Technical Acoustics I (Acoustic Waves, Noise Protection, Psycho Acoustics ) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                  |    |  |
|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--|
| Courses                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                  |    |  |
| Title                                                                                     | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Hrs/wk                                                                                                                                                                                                           | CP |  |
| Technical Acoustics I (Acoustic Waves, Noise Protection, Psycho Acoustics ) (L0516)       | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2                                                                                                                                                                                                                | 3  |  |
| Technical Acoustics I (Acoustic Waves, Noise Protection, Psycho Acoustics ) (L0518)       | Recitation Section (large)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2                                                                                                                                                                                                                | 3  |  |
| Module Responsible                                                                        | Dr. Alessandro Bassetti                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                  |    |  |
| Admission Requirements                                                                    | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                  |    |  |
| Recommended Previous Knowledge                                                            | Mechanics I (Statics, Mechanics of Materials) and Mechanics II (Hydrostatics, Kinematics, Dynamics)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                  |    |  |
|                                                                                           | Mathematics I, II, III (in particular differential equations)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                  |    |  |
| Educational Objectives                                                                    | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                  |    |  |
| Professional Competence                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                  |    |  |
|                                                                                           | Knowledge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | The students possess an in-depth knowledge in acoustics regarding acoustic waves, noise protection, and psycho acoustics and are able to give an overview of the corresponding theoretical and methodical basis. |    |  |
|                                                                                           | Skills                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | The students are capable to handle engineering problems in acoustics by theory-based application of the demanding methodologies and measurement procedures treated within the module.                            |    |  |
|                                                                                           | Personal Competence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                  |    |  |
| Social Competence                                                                         | Students can work in small groups on specific problems to arrive at joint solutions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                  |    |  |
| Autonomy                                                                                  | The students are able to independently solve challenging acoustical problems in the areas treated within the module. Possible conflicting issues and limitations can be identified and the results are critically scrutinized.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                  |    |  |
| Workload in Hours                                                                         | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                  |    |  |
| Credit points                                                                             | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                  |    |  |
| Course achievement                                                                        | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                  |    |  |
| Examination                                                                               | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                  |    |  |
| Examination duration and scale                                                            | 90 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                  |    |  |
| Assignment for the Following Curricula                                                    | Computational Engineering: Core Qualification: Elective Compulsory<br>Aircraft Systems Engineering: Core Qualification: Elective Compulsory<br>International Management and Engineering: Specialisation II. Aviation Systems: Elective Compulsory<br>Aeronautics: Core Qualification: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory<br>Mechanical Engineering and Management: Core Qualification: Elective Compulsory<br>Mechatronics: Core Qualification: Elective Compulsory<br>Product Development, Materials and Production: Core Qualification: Elective Compulsory<br>Technomathematics: Specialisation III. Engineering Science: Elective Compulsory<br>Theoretical Mechanical Engineering: Specialisation Product Development and Production: Elective Compulsory<br>Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory |                                                                                                                                                                                                                  |    |  |

| Course L0516: Technical Acoustics I (Acoustic Waves, Noise Protection, Psycho Acoustics ) |                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                                                                | Lecture                                                                                                                                                                                                                                                                                                                                                              |
| <b>Hrs/wk</b>                                                                             | 2                                                                                                                                                                                                                                                                                                                                                                    |
| <b>CP</b>                                                                                 | 3                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Workload in Hours</b>                                                                  | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                  |
| <b>Lecturer</b>                                                                           | Dr. Alessandro Bassetti                                                                                                                                                                                                                                                                                                                                              |
| <b>Language</b>                                                                           | EN                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Cycle</b>                                                                              | SoSe                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Content</b>                                                                            | <ul style="list-style-type: none"> <li>- Introduction and Motivation</li> <li>- Acoustic quantities</li> <li>- Acoustic waves</li> <li>- Sound sources, sound radiation</li> <li>- Sound energy and intensity</li> <li>- Sound propagation</li> <li>- Signal processing</li> <li>- Psycho acoustics</li> <li>- Noise</li> <li>- Measurements in acoustics</li> </ul> |
| <b>Literature</b>                                                                         | Cremer, L.; Heckl, M. (1996): Körperschall. Springer Verlag, Berlin<br>Veit, I. (1988): Technische Akustik. Vogel-Buchverlag, Würzburg<br>Veit, I. (1988): Flüssigkeitsschall. Vogel-Buchverlag, Würzburg                                                                                                                                                            |

| Course L0518: Technical Acoustics I (Acoustic Waves, Noise Protection, Psycho Acoustics ) |                                                     |
|-------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Typ</b>                                                                                | Recitation Section (large)                          |
| <b>Hrs/wk</b>                                                                             | 2                                                   |
| <b>CP</b>                                                                                 | 3                                                   |
| <b>Workload in Hours</b>                                                                  | Independent Study Time 62, Study Time in Lecture 28 |
| <b>Lecturer</b>                                                                           | Dr. Alessandro Bassetti                             |
| <b>Language</b>                                                                           | EN                                                  |
| <b>Cycle</b>                                                                              | SoSe                                                |
| <b>Content</b>                                                                            | See interlocking course                             |
| <b>Literature</b>                                                                         | See interlocking course                             |

## Module M1283: Research Project IMPMEM

### Courses

| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Typ                                                                                  | Hrs/wk | CP |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------|----|
| <b>Module Responsible</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Dozenten des Studiengangs                                                            |        |    |
| <b>Admission Requirements</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | None                                                                                 |        |    |
| <b>Recommended Previous Knowledge</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Subjects of the Master program and the chosen specialisation.                        |        |    |
| <b>Educational Objectives</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | After taking part successfully, students have reached the following learning results |        |    |
| <b>Professional Competence</b><br><i>Knowledge</i> <ul style="list-style-type: none"> <li>Students can explain the project as well as their autonomously gained knowledge and relate it to current issues of their field of study.</li> <li>They can explain the basic scientific methods they have worked with.</li> </ul> <i>Skills</i> <p>The students are able to autonomously solve a limited scientific task under the guidance of an experienced researcher. They can justify and explain their approach for problem solving; they can draw conclusions from their results, and then can find new ways and methods for their work. Students are capable of comparing and assessing alternative approaches with their own with regard to given criteria.</p> <b>Personal Competence</b><br><i>Social Competence</i> <p>The students are able to condense the relevance and the structure of the project work, the work procedure and the sub-problems for the presentation and discussion in front of a bigger group. They can lead the discussion and give a feedback on the project to their peers and supervisors.</p> <i>Autonomy</i> <p>The students are capable of independently planning and documenting the work steps and procedures while considering the given deadlines. This includes the ability to accurately procure the newest scientific information. Furthermore, they can obtain feedback from experts with regard to the progress of the work, and to accomplish results on the state of the art in science and technology.</p> |                                                                                      |        |    |
| <b>Workload in Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Independent Study Time 360, Study Time in Lecture 0                                  |        |    |
| <b>Credit points</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 12                                                                                   |        |    |
| <b>Course achievement</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                                                                                 |        |    |
| <b>Examination</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Study work                                                                           |        |    |
| <b>Examination duration and scale</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | see FSPO                                                                             |        |    |
| <b>Assignment for the Following Curricula</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Mechanical Engineering and Management: Core Qualification: Compulsory                |        |    |

| Module M0806: Technical Acoustics II (Room Acoustics, Computational Methods) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |        |    |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------|----|
| Courses                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |        |    |
| Title                                                                        | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  | Hrs/wk | CP |
| Technical Acoustics II (Room Acoustics, Computational Methods) (L0519)       | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  | 2      | 3  |
| Technical Acoustics II (Room Acoustics, Computational Methods) (L0521)       | Recitation Section (large)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  | 2      | 3  |
| <b>Module Responsible</b>                                                    | Dr. Alessandro Bassetti                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |        |    |
| <b>Admission Requirements</b>                                                | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |        |    |
| <b>Recommended Previous Knowledge</b>                                        | <p>Technical Acoustics I (Acoustic Waves, Noise Protection, Psycho Acoustics)</p> <p>Mechanics I (Statics, Mechanics of Materials) and Mechanics II (Hydrostatics, Kinematics, Dynamics)</p> <p>Mathematics I, II, III (in particular differential equations)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |        |    |
| <b>Educational Objectives</b>                                                | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |        |    |
| <b>Professional Competence</b>                                               | <p><i>Knowledge</i> The students possess an in-depth knowledge in acoustics regarding room acoustics and computational methods and are able to give an overview of the corresponding theoretical and methodical basis.</p> <p><i>Skills</i> The students are capable to handle engineering problems in acoustics by theory-based application of the demanding computational methods and procedures treated within the module.</p> <p><i>Personal Competence</i></p> <p><i>Social Competence</i> Students can work in small groups on specific problems to arrive at joint solutions.</p> <p><i>Autonomy</i> The students are able to independently solve challenging acoustical problems in the areas treated within the module. Possible conflicting issues and limitations can be identified and the results are critically scrutinized.</p> |  |        |    |
|                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |        |    |
|                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |        |    |
|                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |        |    |
| <b>Workload in Hours</b>                                                     | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |        |    |
| <b>Credit points</b>                                                         | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |        |    |
| <b>Course achievement</b>                                                    | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |        |    |
| <b>Examination</b>                                                           | Oral exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |        |    |
| <b>Examination duration and scale</b>                                        | 20 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |        |    |
| <b>Assignment for the Following Curricula</b>                                | <p>Computational Engineering: Core Qualification: Elective Compulsory</p> <p>Aircraft Systems Engineering: Core Qualification: Elective Compulsory</p> <p>Aeronautics: Core Qualification: Elective Compulsory</p> <p>Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory</p> <p>Mechanical Engineering and Management: Core Qualification: Elective Compulsory</p> <p>Mechatronics: Core Qualification: Elective Compulsory</p> <p>Product Development, Materials and Production: Core Qualification: Elective Compulsory</p> <p>Theoretical Mechanical Engineering: Specialisation Product Development and Production: Elective Compulsory</p> <p>Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory</p>                                         |  |        |    |

| Course L0519: Technical Acoustics II (Room Acoustics, Computational Methods) |                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                                                   | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Hrs/wk</b>                                                                | 2                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>CP</b>                                                                    | 3                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Workload in Hours</b>                                                     | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Lecturer</b>                                                              | Dr. Alessandro Bassetti                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Language</b>                                                              | EN                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Cycle</b>                                                                 | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Content</b>                                                               | <ul style="list-style-type: none"> <li>- Room acoustics</li> <li>- Sound absorber</li> <li>- Standard computations</li> <li>- Statistical Energy Approaches</li> <li>- Finite Element Methods</li> <li>- Boundary Element Methods</li> <li>- Geometrical acoustics</li> <li>- Special formulations</li> <li>- Practical applications</li> <li>- Hands-on Sessions: Programming of elements (Matlab)</li> </ul>               |
| <b>Literature</b>                                                            | <p>Cremer, L.; Heckl, M. (1996): Körperschall. Springer Verlag, Berlin</p> <p>Veit, I. (1988): Technische Akustik. Vogel-Buchverlag, Würzburg</p> <p>Veit, I. (1988): Flüssigkeitsschall. Vogel-Buchverlag, Würzburg</p> <p>Gaul, L.; Fiedler, Ch. (1997): Methode der Randelemente in Statik und Dynamik. Vieweg, Braunschweig, Wiesbaden</p> <p>Bathe, K.-J. (2000): Finite-Elemente-Methoden. Springer Verlag, Berlin</p> |



| Course L0521: Technical Acoustics II (Room Acoustics, Computational Methods) |                                                     |
|------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Typ</b>                                                                   | Recitation Section (large)                          |
| <b>Hrs/wk</b>                                                                | 2                                                   |
| <b>CP</b>                                                                    | 3                                                   |
| <b>Workload in Hours</b>                                                     | Independent Study Time 62, Study Time in Lecture 28 |
| <b>Lecturer</b>                                                              | Dr. Alessandro Bassetti                             |
| <b>Language</b>                                                              | EN                                                  |
| <b>Cycle</b>                                                                 | WiSe                                                |
| <b>Content</b>                                                               | See interlocking course                             |
| <b>Literature</b>                                                            | See interlocking course                             |

| Module M2102: Mechanics of Composites                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|
| Courses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Hrs/wk | CP |
| Mechanics of Composites (L3377)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Integrated Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 4      | 6  |
| <b>Module Responsible</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Prof. Benedikt Kriegesmann                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| <b>Admission Requirements</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |    |
| <b>Recommended Previous Knowledge</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>• Courses <ul style="list-style-type: none"> <li>◦ Engineering mechanics I &amp; II</li> <li>◦ Fundamentals of materials science</li> </ul> </li> <li>• General prior knowledge <ul style="list-style-type: none"> <li>◦ Basic understanding of structural mechanics</li> <li>◦ Basic programming knowledge</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |    |
| <b>Educational Objectives</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |    |
| <b>Professional Competence</b><br><i>Knowledge</i> <ul style="list-style-type: none"> <li>• Classical laminate theory</li> <li>• Rules of mixture</li> <li>• Failure mechanisms and criteria of composites</li> <li>• Boundary value problems of isotropic and anisotropic shells</li> <li>• Stability of composite structures</li> <li>• Optimization of laminated composites</li> <li>• Modelling composites in FEM</li> <li>• Numerical multiscale analysis of composites</li> <li>• Progressive failure analysis</li> </ul> <i>Skills</i> <ul style="list-style-type: none"> <li>• Calculation of fiber composite structures</li> <li>• Verification of the strength of fiber composite structures</li> <li>• Determination of the deformation of fiber composite structures</li> <li>• Determination of the buckling load of thin-walled structures</li> </ul> <b>Personal Competence</b><br><i>Social Competence</i> <p>Students learn in teams to divide a given task into subtasks, to divide subtasks and to combine partial results into an overall result.</p> <i>Autonomy</i> <p>Students learn to independently analyze, simulate and optimize fiber composite structures.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| <b>Workload in Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |    |
| <b>Credit points</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |    |
| <b>Course achievement</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |    |
| <b>Examination</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Presentation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |    |
| <b>Examination duration and scale</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 30 minutes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| <b>Assignment for the Following Curricula</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Computational Engineering: Core Qualification: Elective Compulsory<br>Aeronautics: Core Qualification: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory<br>Mechanical Engineering and Management: Core Qualification: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Production: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Materials: Elective Compulsory<br>Theoretical Mechanical Engineering: Core Qualification: Elective Compulsory |        |    |

| Course L3377: Mechanics of Composites |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                            | Integrated Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Hrs/wk</b>                         | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>CP</b>                             | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Workload in Hours</b>              | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Lecturer</b>                       | Prof. Benedikt Kriegesmann                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Language</b>                       | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Cycle</b>                          | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Content</b>                        | <ul style="list-style-type: none"> <li>• Classical laminate theory</li> <li>• Rules of mixture</li> <li>• Failure mechanisms and criteria of composites</li> <li>• Boundary value problems of isotropic and anisotropic shells</li> <li>• Stability of composite structures</li> <li>• Optimization of laminated composites</li> <li>• Modelling composites in FEM</li> <li>• Numerical multiscale analysis of composites</li> <li>• Progressive failure analysis</li> </ul>                |
| <b>Literature</b>                     | <ul style="list-style-type: none"> <li>• Jones, R.M., „Mechanics of Composite Materials“, Scripta Book Co., Washington, current edition.</li> <li>• Mittelstedt, C., Becker, W., „Strukturmechanik ebener Lamine“, aktuelle Auflage.</li> <li>• Wiedemann, J., „Leichtbau Band 1: Elemente“, Springer, Berlin, Heidelberg, , aktuelle Auflage.</li> <li>• Reddy, J.N., „Mechanics of Composite Laminated Plates and Shells“, CRC Publishing, Boca Raton et al., current edition.</li> </ul> |

| Module M2174: Product Planning                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              |             |                    |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------|--------------------|
| Courses                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              |             |                    |
| Title                                         | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |              | Hrs/wk      | CP                 |
| Product Planning (L0851)                      | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              | 3           | 3                  |
| Product Planning Seminar (L0853)              | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              | 2           | 3                  |
| <b>Module Responsible</b>                     | Prof. Tim Schweisfurth                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |             |                    |
| <b>Admission Requirements</b>                 | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |              |             |                    |
| <b>Recommended Previous Knowledge</b>         | Good basic-knowledge of Business Administration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              |             |                    |
| <b>Educational Objectives</b>                 | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |              |             |                    |
| <b>Professional Competence</b>                | <p><i>Knowledge</i> Students will gain insights into:</p> <ul style="list-style-type: none"> <li>• Product Planning <ul style="list-style-type: none"> <li>◦ Process</li> <li>◦ Methods</li> </ul> </li> <li>• Design thinking <ul style="list-style-type: none"> <li>◦ Process</li> <li>◦ Methods</li> <li>◦ User integration</li> </ul> </li> </ul> <p><i>Skills</i> Students will gain deep insights into:</p> <ul style="list-style-type: none"> <li>• Product Planning <ul style="list-style-type: none"> <li>◦ Process-related aspects</li> <li>◦ Organisational-related aspects</li> <li>◦ Human-Ressource related aspects</li> <li>◦ Working-tools, methods and instruments</li> <li>◦</li> </ul> </li> </ul> <p><b>Personal Competence</b></p> <p><i>Social Competence</i></p> <ul style="list-style-type: none"> <li>• Interact within a team</li> <li>• Raise awareness for globabl issues</li> </ul> <p><i>Autonomy</i></p> <ul style="list-style-type: none"> <li>• Gain access to knowledge sources</li> <li>• Interpret complex cases</li> <li>• Develop presentation skills</li> </ul> |              |             |                    |
|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              |             |                    |
|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              |             |                    |
|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              |             |                    |
| <b>Workload in Hours</b>                      | Independent Study Time 110, Study Time in Lecture 70                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |              |             |                    |
| <b>Credit points</b>                          | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |             |                    |
| <b>Course achievement</b>                     | <b>Compulsory</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Bonus</b> | <b>Form</b> | <b>Description</b> |
|                                               | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None         | Midterm     | mini tests         |
| <b>Examination</b>                            | Subject theoretical and practical work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |             |                    |
| <b>Examination duration and scale</b>         | several contributions spread over the semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |              |             |                    |
| <b>Assignment for the Following Curricula</b> | International Management and Engineering: Specialisation I. Electives Management: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory<br>Mechanical Engineering and Management: Core Qualification: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Production: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Materials: Elective Compulsory<br>Theoretical Mechanical Engineering: Specialisation Product Development and Production: Elective Compulsory                                                                                                                                  |              |             |                    |

| Course L0851: Product Planning |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                     | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Hrs/wk</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>CP</b>                      | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Workload in Hours</b>       | Independent Study Time 48, Study Time in Lecture 42                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Lecturer</b>                | Prof. Tim Schweisfurth                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Language</b>                | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Cycle</b>                   | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Content</b>                 | <p>Product Planning Process</p> <p>This integrated lecture is designed to understand major issues, activities and tools in the context of systematic product planning, a key activity for managing the front-end of innovation, i.e.:</p> <ul style="list-style-type: none"> <li>• Systematic scanning of markets for innovation opportunities</li> <li>• Understanding strengths/weakness and specific core competences of a firm as platforms for innovation</li> <li>• Exploring relevant sources for innovation (customers, suppliers, Lead Users, etc.)</li> <li>• Developing ideas for radical innovation, relying on the creativeness of employees, using techniques to stimulate creativity and creating a stimulating environment</li> <li>• Transferring ideas for innovation into feasible concepts which have a high market attractively</li> </ul> <p>Voluntary presentations in the third hour (articles / case studies)</p> <ul style="list-style-type: none"> <li>- Guest lectures by researchers</li> <li>- Lecture on Sustainability with frequent reference to current research</li> <li>- Permanent reference to current research</li> </ul> <p>Examination:</p> <p>In addition to the written exam at the end of the module, students have to attend the PBL-exercises and prepare presentations in groups in order to pass the module. Additionally, students have the opportunity to present research papers on a voluntary base. With these presentations it is possible to gain a bonus of max. 20% for the exam. However, the bonus is only valid if the exam is passed without the bonus.</p> |
| <b>Literature</b>              | Ulrich, K./Eppinger, S.: Product Design and Development, 2nd. Edition, McGraw-Hill 2010                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Course L0853: Product Planning Seminar |                                                                                                                           |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                             | Project-/problem-based Learning                                                                                           |
| <b>Hrs/wk</b>                          | 2                                                                                                                         |
| <b>CP</b>                              | 3                                                                                                                         |
| <b>Workload in Hours</b>               | Independent Study Time 62, Study Time in Lecture 28                                                                       |
| <b>Lecturer</b>                        | Prof. Tim Schweisfurth                                                                                                    |
| <b>Language</b>                        | EN                                                                                                                        |
| <b>Cycle</b>                           | WiSe                                                                                                                      |
| <b>Content</b>                         | Seminar is integrative part of the Module Product Planning (for content see lecture) and can not be chosen independantly. |
| <b>Literature</b>                      | See lecture information "Product Planning".                                                                               |

## Specialization Management

Graduates of the Management specialization learn to use their knowledge in management and business topics for the planning of production processes and projects. Furthermore they have extended knowledge in special topics, such as human resources, entrepreneurship or logistics. Graduates are able to evaluate the necessary business and financial key figures and to make decisions based on these. They are able to put their theoretical knowledge into practice and to analyze complex questions in business administration. They learn diverse methods and techniques of management and business administration and are able to use them successfully for different tasks.

Students have to choose the Management specialization. Solely students of the Northern Institute of Technology have to choose two engineering specializations.

| Module M0814: Technology Management                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |    |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|
| Courses                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |    |
| Title                                                  | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Hrs/wk | CP |
| Technology Management (L0849)                          | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3      | 3  |
| Technology Management Seminar (L0850)                  | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2      | 3  |
| <b>Module Responsible</b>                              | Prof. Cornelius Herstatt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |    |
| <b>Admission Requirements</b>                          | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |    |
| <b>Recommended Previous Knowledge</b>                  | Bachelor knowledge in business management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |    |
| <b>Educational Objectives</b>                          | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |    |
| <b>Professional Competence</b><br><i>Knowledge</i>     | Students will gain deep insights into: <ul style="list-style-type: none"> <li>• International R&amp;D-Management</li> <li>• Technology Timing Strategies               <ul style="list-style-type: none"> <li>◦ Technology Strategies and Lifecycle Management (I/II)</li> <li>◦ Technology Intelligence and Planning</li> </ul> </li> <li>• Technology Portfolio Management               <ul style="list-style-type: none"> <li>◦ Technology Portfolio Methodology</li> <li>◦ Technology Acquisition and Exploitation</li> <li>◦ IP Management</li> </ul> </li> <li>• Organizing Technology Development               <ul style="list-style-type: none"> <li>◦ Technology Organization &amp; Management</li> <li>◦ Technology Funding &amp; Controlling</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                    |        |    |
| <b>Personal Competence</b><br><i>Social Competence</i> | The course aims to: <ul style="list-style-type: none"> <li>• Develop an understanding of the importance of Technology Management - on a national as well as international level</li> <li>• Equip students with an understanding of important elements of Technology Management (strategic, operational, organizational and process-related aspects)</li> <li>• Foster a strategic orientation to problem-solving within the innovation process as well as Technology Management and its importance for corporate strategy</li> <li>• Clarify activities of Technology Management (e.g. technology sourcing, maintenance and exploitation)</li> <li>• Strengthen essential communication skills and a basic understanding of managerial, organizational and financial issues concerning Technology-, Innovation- and R&amp;D-management. Further topics to be discussed include:               <ul style="list-style-type: none"> <li>• Basic concepts, models and tools, relevant to the management of technology, R&amp;D and innovation</li> <li>• Innovation as a process (steps, activities and results)</li> </ul> </li> </ul> |        |    |
| <b>Personal Competence</b><br><i>Autonomy</i>          | <ul style="list-style-type: none"> <li>• Interact within a team</li> <li>• Raise awareness for global issues</li> <li>• Gain access to knowledge sources</li> <li>• Discuss recent research debates in the context of Technology and Innovation Management</li> <li>• Develop presentation skills</li> <li>• Discussion of international cases in R&amp;D-Management</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |    |
| <b>Workload in Hours</b>                               | Independent Study Time 110, Study Time in Lecture 70                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |    |
| <b>Credit points</b>                                   | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |    |
| <b>Course achievement</b>                              | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |    |
| <b>Examination</b>                                     | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |    |
| <b>Examination duration and scale</b>                  | 90 minutes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |    |
| <b>Assignment for the Following Curricula</b>          | Global Innovation Management: Core Qualification: Compulsory<br>International Management and Engineering: Specialisation I. Electives Management: Elective Compulsory<br>Mechanical Engineering and Management: Specialisation Management: Elective Compulsory<br>Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory<br>Biomedical Engineering: Specialisation Implants and Endoprostheses: Elective Compulsory<br>Biomedical Engineering: Specialisation Medical Technology and Control Theory: Elective Compulsory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |    |

**Course L0849: Technology Management**

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>               | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Hrs/wk</b>            | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>CP</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Workload in Hours</b> | Independent Study Time 48, Study Time in Lecture 42                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Lecturer</b>          | Prof. Cornelius Herstatt, Dr. Vytaute Dlugoborskyte                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Language</b>          | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Cycle</b>             | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Content</b>           | <p>The role of technology for the competitive advantage of the firm and industries; Basic concepts, models and tools for the management of technology; managerial decision making regarding the identification, selection and protection of technology (make or buy, keep or sell, current and future technologies). Theories, practical examples (cases), lectures, interactive sessions and group study.</p> <p>This lecture is part of the Module Technology Management and can not separately choosen.</p> |
| <b>Literature</b>        | Leiblein, M./Ziedonis, A.: Technology Strategy and Inoovation Management, Elgar Research Collection, Northhampton (MA) 2011                                                                                                                                                                                                                                                                                                                                                                                    |

**Course L0850: Technology Management Seminar**

|                          |                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>               | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                |
| <b>Hrs/wk</b>            | 2                                                                                                                                                                                                                                                                                                                                                                              |
| <b>CP</b>                | 3                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Workload in Hours</b> | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                            |
| <b>Lecturer</b>          | Prof. Cornelius Herstatt                                                                                                                                                                                                                                                                                                                                                       |
| <b>Language</b>          | EN                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Cycle</b>             | WiSe                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Content</b>           | <p>Beside the written exam at the end of the module, students have to give one presentation (RE) on a research paper and two presentations as part of a group discussion (GD) in the seminar in order to pass. With these presentations it is possible to gain a bonus of max. 20% for the exam. However, the bonus is only valid if the exam is passed without the bonus.</p> |
| <b>Literature</b>        | see lecture Technology Management.                                                                                                                                                                                                                                                                                                                                             |

## Module M0978: Sustainable Mobility of Goods and Logistics Systems

### Courses

| Title                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Type                            | Hrs/wk                      | CP                 |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----------------------------|--------------------|
| International Logistics and Transport Systems (L1168)     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Project-/problem-based Learning | 3                           | 4                  |
| Sustainable Mobility of Goods, Logistics, Traffic (L1165) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Lecture                         | 2                           | 2                  |
| <b>Module Responsible</b>                                 | Prof. Heike Flämig                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                 |                             |                    |
| <b>Admission Requirements</b>                             | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                 |                             |                    |
| <b>Recommended Previous Knowledge</b>                     | <ul style="list-style-type: none"> <li>• Introduction to Logistics and Mobility</li> <li>• Foundations of Management</li> <li>• Legal Foundations of Transportation and Logistics</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                 |                             |                    |
| <b>Educational Objectives</b>                             | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                 |                             |                    |
| <b>Professional Competence</b><br><i>Knowledge</i>        | <p>Students are able to...</p> <ul style="list-style-type: none"> <li>• give definitions of system theory, (international) transport chains and logistics in the context of supply chain management</li> <li>• explain trends and strategies for mobility of goods and logistics</li> <li>• describe elements of integrated and multi-modal transport chains and their advantages and disadvantages</li> <li>• deduce impacts of management decisions on logistics system and traffic system and explain how stakeholders influence them</li> <li>• explain the correlations between economy and logistics systems, mobility of goods, space-time-structures and the traffic system as well as ecology and politics</li> </ul> |                                 |                             |                    |
| <i>Skills</i>                                             | <p>Students are able to...</p> <ul style="list-style-type: none"> <li>• Design intermodal transport chains and logistic concepts</li> <li>• apply the commodity chain theory and case study analysis</li> <li>• evaluate different international transport chains</li> <li>• cope with differences in cultures that influence international transport chains</li> </ul>                                                                                                                                                                                                                                                                                                                                                        |                                 |                             |                    |
| <b>Personal Competence</b><br><i>Social Competence</i>    | <p>Students are able to...</p> <ul style="list-style-type: none"> <li>• develop a feeling of social responsibility for their future jobs</li> <li>• give constructive feedback to others about their presentation skills</li> <li>• plan and execute teamwork tasks</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                 |                             |                    |
| <i>Autonomy</i>                                           | Students are able to improve presentation skills by feedback of others                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                 |                             |                    |
| <b>Workload in Hours</b>                                  | Independent Study Time 110, Study Time in Lecture 70                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                 |                             |                    |
| <b>Credit points</b>                                      | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                 |                             |                    |
| <b>Course achievement</b>                                 | <b>Compulsory</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Bonus</b>                    | <b>Form</b>                 | <b>Description</b> |
|                                                           | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | None                            | Participation in excursions |                    |
|                                                           | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 5 %                             | Exercises                   | einschl. Referat   |
| <b>Examination</b>                                        | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                 |                             |                    |
| <b>Examination duration and scale</b>                     | written exam (60 minutes), exercises in groups (min. 80% attendance), one-day excursion with short presentations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |                             |                    |
| <b>Assignment for the Following Curricula</b>             | International Management and Engineering: Specialisation II. Logistics: Elective Compulsory<br>Logistics, Infrastructure and Mobility: Specialisation Production and Logistics: Elective Compulsory<br>Logistics, Infrastructure and Mobility: Specialisation Infrastructure and Mobility: Elective Compulsory<br>Mechanical Engineering and Management: Specialisation Management: Elective Compulsory                                                                                                                                                                                                                                                                                                                        |                                 |                             |                    |



| Course L1168: International Logistics and Transport Systems |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                                  | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Hrs/wk</b>                                               | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>CP</b>                                                   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Workload in Hours</b>                                    | Independent Study Time 78, Study Time in Lecture 42                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Lecturer</b>                                             | Prof. Heike Flämig                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Language</b>                                             | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Cycle</b>                                                | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Content</b>                                              | The problem-oriented-learning lecture consists of case studies and complex problems concerning the systemic characteristics of different modes of transport as well as the organization and realization of transport chains. Students get to know specific issues from practice of logistics and mobility of goods and work out recommendations for solutions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Literature</b>                                           | <p>David, P.; Stewart, R. (2010): International Logistics: The Management of International Trade Operations, 3rd Edition, Mason.</p> <p>EcoTransIT (2023): Environmental Methodology and Data - Update 2023. Commissioned by EcoTransIT World Initiative (EWI). Version 3.1 (15.05.2023). URL: <a href="https://www.ecotransit.org/wp-content/uploads/20230612_Methodology_Report_Update_2023.pdf">https://www.ecotransit.org/wp-content/uploads/20230612_Methodology_Report_Update_2023.pdf</a> (last access: 21.12.2023)</p> <p>Flämig, H.; Sjöstedt, L.; Hertel, C. (2002): Multimodal Transport: An Integrated Element for Last-Mile-Solutions? Proceedings, part 1; International Congress on Freight Transport Automation and Multimodality: Organisational and Technological Innovations. Delft, Suppl.</p> <p>Schieck, A. (2009): Internationale Logistik: Objekte, Prozesse und Infrastrukturen grenzüberschreitender Güterströme, München.</p> |

| Course L1165: Sustainable Mobility of Goods, Logistics, Traffic |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                                      | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Hrs/wk</b>                                                   | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>CP</b>                                                       | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Workload in Hours</b>                                        | Independent Study Time 32, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Lecturer</b>                                                 | Prof. Heike Flämig                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Language</b>                                                 | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Cycle</b>                                                    | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Content</b>                                                  | <p>The intention of this lecture is to provide a general system analysis-based overview of how transport chains emerge and how they can be developed in a sustainable way. The respective advantages and disadvantages of different international transportation chains of goods are to be pointed out from a micro- and a macroeconomic point of view. The effects on the traffic system as well as the ecological and social consequences of a spatial division of economic activities are to be discussed.</p> <p>The overview of current international transport chains is carried out on the basis of concrete material- and appendant information flows. Established transport chains and some of their individual elements are to become transparent to the students by a number of practical examples.</p> <ol style="list-style-type: none"> <li>1. A conceptual systems model</li> <li>2. Elements of integrated and multi-modal transport chains</li> <li>3. interaction of transport and traffic, demand and supply on different layers of the transport system</li> <li>4. Global Issues in Supply Chain Management</li> <li>5. Global Players and networks</li> <li>6. Logistics and corporate social responsibility (CSR)</li> <li>7. Methods and data for assessment of international transport chains</li> <li>8. Influence of cultural aspects on international transport chains</li> <li>9. New solutions using different focuses of the transport and logistics system</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Literature</b>                                               | <p>Biebig, P.; Althof, W.; Wagener, N. (2008): Seeverkehrswirtschaft: De Gruyter Oldenbourg.</p> <p>Bowersox, D.J.; Closs D. J.; Cooper M. B.; Bowerbox, J. C. (2007): Supply Chain Logistics Management. 2nd Edition. McGraw-Hill/Irwin, NY.</p> <p>Büker, T. (2018): Maps and facts on european interoperability issues.</p> <p>Bundesministerium für Verkehr und digitale Infrastruktur (2020): Verkehr in Zahlen 2020/2021. 49. Jahrgang</p> <p>Chopra, S.; Meindl, P. (2016): Supply chain management (3. ed., Pearson internat. ed.). Upper Saddle River, NJ: Pearson Prentice Hall.</p> <p>Domschke et al. (2018) Betriebliche Standortplanung. In: Horst Tempelmeier (Hg.): Planung logistischer Systeme. Berlin, Heidelberg: Springer Berlin Heidelberg, S. 1-27.</p> <p>European Commission (2020): EU Transport Figures. Statistical Pocketbook 2020.</p> <p>Flämig, H. (2015): Logistik und Nachhaltigkeit. In: Heidbrink, Ludger; Meyer, Nora; Reidel, Johannes; Schmidt, Imke (Hrsg.): Corporate Social Responsibility in der Logistikbranche- Anforderungen an eine nachhaltige Unternehmensführung. Berlin: E. Schmidt, 2015. S. 25-44.</p> <p>Flämig, H.; Sjöstedt, L.; Hertel, C. (2002): Multimodal Transport: An Integrated Element for Last-Mile-Solutions? Proceedings, part 1; International Congress on Freight Transport Automation and Multimodality: Organisational and Technological Innovations. Delft, Suppl.</p> <p>Lumsden, Kenth R. (2020): Fundamentals of Logistics, Translation of selected chapters of the book "Logistikens Grunder". Göteborg, August.</p> <p>Pfohl, H.-C.(2004 ): Logistiksysteme - Betriebswirtschaftliche Grundlagen. Berlin, Heidelberg, New York. Springer-Verlag, 7. Auflage.</p> <p>Rodrigue, J. (2020): The geography of transport systems. Fifth edition. London, New York: Routledge.</p> <p>Schulte, C. (1991): Logistik. Wege zur Optimierung des Material- und Informationsflusses. Verlag Franz Vahlen, München.</p> <p>Sjöstedt, L. (1996): A Theoretical Framework - from an Applied Engineering Perspective. In: Eurocase - Mobility, Transport and Traffic: in the perspective of growth, competitiveness, employment. Paris, France.</p> |

| Module M1034: Technology Entrepreneurship              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |        |    |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------|----|
| Courses                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |        |    |
| Title                                                  | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  | Hrs/wk | CP |
| Creation of Business Opportunities (L1280)             | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  | 3      | 3  |
| Entrepreneurship (L1279)                               | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  | 2      | 3  |
| <b>Module Responsible</b>                              | Prof. Christoph Ihl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |        |    |
| <b>Admission Requirements</b>                          | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |        |    |
| <b>Recommended Previous Knowledge</b>                  | Basic knowledge in business economics obtained in the compulsory modules as well as an interest in new technologies and the pursuit of new business opportunities either in corporate or startup contexts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |        |    |
| <b>Educational Objectives</b>                          | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |        |    |
| <b>Professional Competence</b><br><i>Knowledge</i>     | <p>Wissen (subject-related knowledge and understanding):</p> <ul style="list-style-type: none"> <li>• develop a working knowledge and understanding of the entrepreneurial perspective</li> <li>• understand the difference between a good idea and scalable business opportunity</li> <li>• understand the process of taking a technology idea and finding a high-potential commercial opportunity</li> <li>• understand the components of business models</li> <li>• understand the components of business opportunity assessment and business plans</li> </ul>                                                                                                                                                                                    |  |        |    |
| <i>Skills</i>                                          | <ul style="list-style-type: none"> <li>• Fertigkeiten (subject-related skills): <ul style="list-style-type: none"> <li>◦ identify and define business opportunities</li> <li>◦ assess and validate entrepreneurial opportunities</li> <li>◦ create and verify a business model of how to sell and market an entrepreneurial opportunity</li> <li>◦ formulate and test business model assumptions and hypotheses</li> <li>◦ conduct customer and expert interviews regarding business opportunities</li> <li>◦ prepare business opportunity assessment</li> <li>◦ create and verify a plan for gathering resources such as talent and capital</li> <li>◦ pitch a business opportunity to your classmates and the teaching team</li> </ul> </li> </ul> |  |        |    |
| <b>Personal Competence</b><br><i>Social Competence</i> | <p>Sozialkompetenz (Social Competence):</p> <ul style="list-style-type: none"> <li>• team work</li> <li>• communication and presentation</li> <li>• give and take critical comments</li> <li>• engaging in fruitful discussions</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |        |    |
| <i>Autonomy</i>                                        | <p>Selbständigkeit (Autonomy):</p> <ul style="list-style-type: none"> <li>• autonomous work and time management</li> <li>• project management</li> <li>• analytical skills</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |        |    |
| <b>Workload in Hours</b>                               | Independent Study Time 110, Study Time in Lecture 70                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |        |    |
| <b>Credit points</b>                                   | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |        |    |
| <b>Course achievement</b>                              | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |        |    |
| <b>Examination</b>                                     | Subject theoretical and practical work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |        |    |
| <b>Examination duration and scale</b>                  | Three presentations on the respective project status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |        |    |
| <b>Assignment for the Following Curricula</b>          | <p>Global Technology and Innovation Management &amp; Entrepreneurship: Core Qualification: Elective Compulsory</p> <p>International Management and Engineering: Specialisation I. Electives Management: Elective Compulsory</p> <p>Logistics, Infrastructure and Mobility: Core Qualification: Elective Compulsory</p> <p>Mechanical Engineering and Management: Specialisation Management: Elective Compulsory</p>                                                                                                                                                                                                                                                                                                                                  |  |        |    |

| Course L1280: Creation of Business Opportunities |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                       | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Hrs/wk</b>                                    | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>CP</b>                                        | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Workload in Hours</b>                         | Independent Study Time 48, Study Time in Lecture 42                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Lecturer</b>                                  | Prof. Christoph Ihl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Language</b>                                  | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Cycle</b>                                     | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Content</b>                                   | <p>Important note: This course is part of an 6 ECTS module consisting of two courses "Entrepreneurship" &amp; "Creation of Business Opportunities", which have to be taken together in one semester.</p> <p>Startups are temporary, team-based organizations, which can form both within and outside of established companies, to pursue one central objective: taking a new venture idea to market by designing a business model that can be scaled to a full-grown company. In this course, students will form startup teams around self-selected ideas and run through the process just like real startups would do in the first three months of intensive work. Startup Engineering takes an incremental and iterative approach, in that it favors variety and alternatives over one detailed, linear five-year business plan to reach steady state operations. From a problem solving and systems thinking perspective, student teams create different possible versions of a new venture and alternative hypotheses about value creation for customers and value capture vis-à-vis competitors. We will draw on recent scientific findings about international success factors of new venture design. To test critical hypotheses early on, student teams engage in scientific, evidence-based, experimental trial-and-error learning process that measures real progress.</p> <p>Upon completion of this course, students will be able to:</p> <ul style="list-style-type: none"> <li>· Apply a modern innovation toolkit relevant in both the corporate &amp; startup world</li> <li>· Analyze given business opportunities in terms of its constituent elements</li> <li>· Design new business models by gathering and combining relevant ideas, facts and information</li> <li>· Evaluate business opportunities and derive judgment about next steps &amp; decisions</li> </ul> <p>Course language is English, but participants can decide to give their graded presentations in German. Students are invited to apply to this course module already with a startup idea and/ or team, but this is not a requirement! We will form teams and ideas in the beginning of the course. Class meetings have alternate intervals of lecture inputs, teamwork, mentoring, and peer feedback. Attendance is mandatory for at least 80% of class time due to large proportion of teamwork sessions.</p> <p>Student teams give three presentations and submit them with backup analyses. Grading scheme:</p> <ul style="list-style-type: none"> <li>· Startup discovery presentation after 5 weeks: 30%</li> <li>· Startup validation presentation after 10 weeks: 30%</li> <li>· Final startup pitches after 13 weeks: 40%</li> </ul> |
| <b>Literature</b>                                | <ul style="list-style-type: none"> <li>• Blank, S. &amp; Dorf, B. (2012). The startup owner's manual.</li> <li>• Gans, J. &amp; Stern, S. (2016). Entrepreneurial Strategy.</li> <li>• Osterwalder, A. &amp; Yves, P. (2010). Business model generation.</li> <li>• Maurya, A. (2012). Running lean: Iterate from plan A to a plan that works.</li> <li>• Maurya, A. (2016). Scaling lean: Mastering the Key Metrics for Startup Growth.</li> <li>• Wilcox, J. (2016). FOCUS Framework: How to Find Product-Market Fit.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Course L1279: Entrepreneurship |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                     | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Hrs/wk</b>                  | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>CP</b>                      | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Workload in Hours</b>       | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Lecturer</b>                | Prof. Christoph Ihl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Language</b>                | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Cycle</b>                   | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Content</b>                 | <p>Important note: This course is part of an 6 ECTS module consisting of two courses "Entrepreneurship" &amp; "Creation of Business Opportunities", which have to be taken together in one semester.</p> <p>Startups are temporary, team-based organizations, which can form both within and outside of established companies, to pursue one central objective: taking a new venture idea to market by designing a business model that can be scaled to a full-grown company. In this course, students will form startup teams around self-selected ideas and run through the process just like real startups would do in the first three months of intensive work. Startup Engineering takes an incremental and iterative approach, in that it favors variety and alternatives over one detailed, linear five-year business plan to reach steady state operations. From a problem solving and systems thinking perspective, student teams create different possible versions of a new venture and alternative hypotheses about value creation for customers and value capture vis-à-vis competitors. We will draw on recent scientific findings about international success factors of new venture design. To test critical hypotheses early on, student teams engage in scientific, evidence-based, experimental trial-and-error learning process that measures real progress.</p> <p>Upon completion of this course, students will be able to:</p> <ul style="list-style-type: none"> <li>· Apply a modern innovation toolkit relevant in both the corporate &amp; startup world</li> <li>· Analyze given business opportunities in terms of its constituent elements</li> <li>· Design new business models by gathering and combining relevant ideas, facts and information</li> <li>· Evaluate business opportunities and derive judgment about next steps &amp; decisions</li> </ul> <p>Course language is English, but participants can decide to give their graded presentations in German. Students are invited to apply to this course module already with a startup idea and/ or team, but this is not a requirement! We will form teams and ideas in the beginning of the course. Class meetings have alternate intervals of lecture inputs, teamwork, mentoring, and peer feedback. Attendance is mandatory for at least 80% of class time due to large proportion of teamwork sessions.</p> <p>Student teams give three presentations and submit them with backup analyses. Grading scheme:</p> <ul style="list-style-type: none"> <li>· Startup discovery presentation after 5 weeks: 30%</li> <li>· Startup validation presentation after 10 weeks: 30%</li> <li>· Final startup pitches after 13 weeks: 40%</li> </ul> |
| <b>Literature</b>              | <ul style="list-style-type: none"> <li>• Blank, S. &amp; Dorf, B. (2012). The startup owner's manual.</li> <li>• Gans, J. &amp; Stern, S. (2016). Entrepreneurial Strategy.</li> <li>• Osterwalder, A. &amp; Yves, P. (2010). Business model generation.</li> <li>• Maurya, A. (2012). Running lean: Iterate from plan A to a plan that works.</li> <li>• Maurya, A. (2016). Scaling lean: Mastering the Key Metrics for Startup Growth.</li> <li>• Wilcox, J. (2016). FOCUS Framework: How to Find Product-Market Fit.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Module M0750: Economics                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                 |              |             |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|--------------|-------------|
| Courses                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                 |              |             |
| Title                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Typ                             | Hrs/wk       | CP          |
| International Economics (L0700)                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Lecture                         | 2            | 2           |
| Main Theoretical and Political Concepts (L0641) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Lecture                         | 2            | 2           |
| Economics (L2714)                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Project-/problem-based Learning | 1            | 2           |
| Module Responsible                              | Prof. Timo Heinrich                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                 |              |             |
| Admission Requirements                          | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |              |             |
| Recommended Previous Knowledge                  | Basic knowledge of economics is expected.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                 |              |             |
|                                                 | The prior knowledge in the field of economics required for successful completion of this module is imparted as an e-learning offering. Students will receive access and further information on the associated online learning module when they enroll.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                 |              |             |
|                                                 | By taking an associated online test, the student can acquire points that are added to the result of the final examination of the Economics module.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                 |              |             |
| Educational Objectives                          | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |              |             |
| Professional Competence                         | <p><i>Knowledge</i> The students know</p> <ul style="list-style-type: none"> <li>the most important principles of individual decision making in a national and international context,</li> <li>different market structures,</li> <li>types of market failure,</li> <li>the functioning of a single economy (including money market, financial and goods markets, labor market),</li> <li>the difference between and the interdependence of short and long run equilibria,</li> <li>the significance of expectations on the effects of economic policy,</li> <li>the various links between economies and</li> <li>different economic policies and their effects on the economy.</li> </ul> <p><i>Skills</i> The students are able to model analytically or graphically</p> <ul style="list-style-type: none"> <li>the most important principles of individual decision making in a national and international context,</li> <li>the market results of different market structures and market failure,</li> <li>the welfare effects of the market results,</li> <li>the functioning of an economy (including money market, financial and goods markets, labor market),</li> <li>links between economies and</li> <li>the effects of economic policies.</li> </ul> |                                 |              |             |
| Personal Competence                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                 |              |             |
| <i>Social Competence</i>                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                 |              |             |
| <i>Autonomy</i>                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                 |              |             |
| Workload in Hours                               | Independent Study Time 110, Study Time in Lecture 70                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |              |             |
| Credit points                                   | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |              |             |
| Course achievement                              | Compulsory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Bonus                           | Form         | Description |
|                                                 | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 5 %                             | Exercises    |             |
|                                                 | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 15 %                            | Presentation |             |
| Examination                                     | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                 |              |             |
| Examination duration and scale                  | 60 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                 |              |             |
| Assignment for the Following Curricula          | International Management and Engineering: Core Qualification: Compulsory<br>Logistics, Infrastructure and Mobility: Core Qualification: Elective Compulsory<br>Mechanical Engineering and Management: Specialisation Management: Elective Compulsory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |              |             |

| Course L0700: International Economics |                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                            | Lecture                                                                                                                                                                                                                                                                                                                       |
| <b>Hrs/wk</b>                         | 2                                                                                                                                                                                                                                                                                                                             |
| <b>CP</b>                             | 2                                                                                                                                                                                                                                                                                                                             |
| <b>Workload in Hours</b>              | Independent Study Time 32, Study Time in Lecture 28                                                                                                                                                                                                                                                                           |
| <b>Lecturer</b>                       | Prof. Timo Heinrich                                                                                                                                                                                                                                                                                                           |
| <b>Language</b>                       | EN                                                                                                                                                                                                                                                                                                                            |
| <b>Cycle</b>                          | SoSe                                                                                                                                                                                                                                                                                                                          |
| <b>Content</b>                        | <ul style="list-style-type: none"> <li>International Trade Theory and Policy: <ul style="list-style-type: none"> <li>Comparative Advantage - the Ricardian Model</li> <li>The Heckscher-Ohlin Model</li> <li>The Standard Trade Model</li> <li>Intrasectoral Trade</li> <li>International Trade Policy</li> </ul> </li> </ul> |
| <b>Literature</b>                     | <ul style="list-style-type: none"> <li>Mankiw/Taylor: Economics, Cengage, 5<sup>th</sup> ed., 2020</li> <li>Krugman/Obstfeld/Mehltitz: International Economics, Pearson, 11<sup>th</sup> ed. 2018</li> <li>The CORE Team: The Economy: Economics for a Changing World, Oxford University Press, 2017</li> </ul>               |

| Course L0641: Main Theoretical and Political Concepts |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                            | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Hrs/wk</b>                                         | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>CP</b>                                             | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Workload in Hours</b>                              | Independent Study Time 32, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Lecturer</b>                                       | Prof. Timo Heinrich                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Language</b>                                       | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Cycle</b>                                          | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Content</b>                                        | <ul style="list-style-type: none"> <li>Introduction: Ten Principles of Economics</li> <li>Microeconomics: <ul style="list-style-type: none"> <li>Theory of the Household</li> <li>Theory of the Firm</li> <li>Competitive Markets in Equilibrium</li> <li>Market Failure: Monopoly and External Effects</li> <li>Government Policies</li> </ul> </li> <li>Macroeconomics: <ul style="list-style-type: none"> <li>A Nation's Real Income and Production</li> <li>The Real Economy in the Long Run: Capital and Labour Market</li> <li>Money and Prices in the Long Run</li> <li>Aggregate Demand and Supply: Short-Run Economic Fluctuations</li> <li>Monetary and Fiscal Policy in the Short and the Long Run</li> </ul> </li> </ul> |
| <b>Literature</b>                                     | <ul style="list-style-type: none"> <li>Mankiw/Taylor: Economics, Cengage, 5<sup>th</sup> ed., 2020</li> <li>Pindyck/Rubinfeld: Microeconomics, Prentice Hall International, 7th ed. 2010</li> <li>The CORE Team: The Economy: Economics for a Changing World, Oxford University Press, 2017</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Course L2714: Economics  |                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>               | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Hrs/wk</b>            | 1                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>CP</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Workload in Hours</b> | Independent Study Time 46, Study Time in Lecture 14                                                                                                                                                                                                                                                                                                                                                  |
| <b>Lecturer</b>          | Prof. Timo Heinrich                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Language</b>          | EN                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Cycle</b>             | SoSe                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Content</b>           | Students work in teams on in-depth questions related to the contents of the lectures and present the results.                                                                                                                                                                                                                                                                                        |
| <b>Literature</b>        | <ul style="list-style-type: none"> <li>• Mankiw/Taylor: Economics, Cengage, 5<sup>th</sup> ed., 2020</li> <li>• Krugman/Obstfeld/Mehlitz: International Economics, Pearson, 11<sup>th</sup> ed. 2018</li> <li>• Pindyck/Rubinfeld, Microeconomics, Pearson, 9<sup>th</sup> ed., 2018</li> <li>• The CORE Team: The Economy: Economics for a Changing World, Oxford University Press, 2017</li> </ul> |



| Module M0855: Marketing (Sales and Services / Innovation Marketing) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |        |    |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------|----|
| Courses                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |        |    |
| Title                                                               | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  | Hrs/wk | CP |
| Marketing of Innovations (L2009)                                    | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  | 4      | 4  |
| PBL Marketing of Innovations (L0862)                                | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  | 1      | 2  |
| <b>Module Responsible</b>                                           | Prof. Christian Lühje                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |        |    |
| <b>Admission Requirements</b>                                       | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |        |    |
| <b>Recommended Previous Knowledge</b>                               | <ul style="list-style-type: none"> <li>• Module International Business</li> <li>• Basic understanding of business administration principles (strategic planning, decision theory, project management, international business)</li> <li>• Bachelor-level Marketing Knowledge (Marketing Instruments, Market and Competitor Strategies, Basics of Buying Behavior)</li> <li>• Understanding the differences between B2B and B2C marketing</li> <li>• Understanding of the importance of managing innovation in global industrial markets</li> <li>• Good English proficiency; presentation skills</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                              |  |        |    |
| <b>Educational Objectives</b>                                       | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |        |    |
| <b>Professional Competence</b>                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |        |    |
| <i>Knowledge</i>                                                    | <p>Students will have gained a deep understanding of</p> <ul style="list-style-type: none"> <li>• Specific characteristics in the marketing of innovative products and services</li> <li>• Approaches for analyzing the current market situation and the future market development</li> <li>• The gathering of information about future customer needs and requirements</li> <li>• Concepts and approaches to integrate lead users and their needs into product and service development processes</li> <li>• Approaches and tools for ensuring customer-orientation in the development of new products and innovative services</li> <li>• Marketing mix elements that take into consideration the specific requirements and challenges of innovative products and services</li> <li>• Pricing methods for new products and services</li> <li>• The organization of complex sales forces and personal selling</li> <li>• Communication concepts and instruments for new products and services</li> </ul> |  |        |    |
| <i>Skills</i>                                                       | <p>Based on the acquired knowledge students will be able to:</p> <ul style="list-style-type: none"> <li>• Design and to evaluate decisions regarding marketing and innovation strategies</li> <li>• Analyze markets by applying market and technology portfolios</li> <li>• Conduct forecasts and develop compelling scenarios as a basis for strategic planning</li> <li>• Translate customer needs into concepts, prototypes and marketable offers and successfully apply advanced methods for customer-oriented product and service development</li> <li>• Use adequate methods to foster efficient diffusion of innovative products and services</li> <li>• Choose suitable pricing strategies and communication activities for innovations</li> <li>• Make strategic sales decisions for products and services (i.e. selection of sales channels)</li> <li>• Apply methods of sales force management (i.e. customer value analysis)</li> </ul>                                                     |  |        |    |
| <b>Personal Competence</b>                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |        |    |
| <i>Social Competence</i>                                            | <p>The students will be able to</p> <ul style="list-style-type: none"> <li>• have fruitful discussions and exchange arguments</li> <li>• develop original results in a group</li> <li>• present results in a clear and concise way</li> <li>• carry out respectful team work</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |        |    |
| <i>Autonomy</i>                                                     | <p>The students will be able to</p> <ul style="list-style-type: none"> <li>• Acquire knowledge independently in the specific context and to map this knowledge on other new complex problem fields.</li> <li>• Consider proposed business actions in the field of marketing and reflect on them.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |        |    |
| <b>Workload in Hours</b>                                            | Independent Study Time 110, Study Time in Lecture 70                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |        |    |
| <b>Credit points</b>                                                | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |        |    |
| <b>Course achievement</b>                                           | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |        |    |
| <b>Examination</b>                                                  | Subject theoretical and practical work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |        |    |
| <b>Examination duration and scale</b>                               | Written elaboration, exercises, presentation, oral participation, written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |        |    |
| <b>Assignment for the Following Curricula</b>                       | <p>Global Technology and Innovation Management &amp; Entrepreneurship: Core Qualification: Compulsory</p> <p>International Management and Engineering: Specialisation I. Electives Management: Elective Compulsory</p> <p>Mechanical Engineering and Management: Specialisation Management: Elective Compulsory</p> <p>Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory</p> <p>Biomedical Engineering: Specialisation Implants and Endoprostheses: Elective Compulsory</p> <p>Biomedical Engineering: Specialisation Medical Technology and Control Theory: Elective Compulsory</p> <p>Biomedical Engineering: Specialisation Management and Business Administration: Compulsory</p>                                                                                                                                                                                                                                                             |  |        |    |

| Course L2009: Marketing of Innovations |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                             | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Hrs/wk</b>                          | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>CP</b>                              | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Workload in Hours</b>               | Independent Study Time 64, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Lecturer</b>                        | Prof. Christian Lüthje                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Language</b>                        | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Cycle</b>                           | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Content</b>                         | <p>The "Marketing of Innovation" course offers advanced insights into the unique challenges of marketing innovative products and services. Through a blend of research-based knowledge and practical tools, students will learn to navigate the complexities of dynamic market environments, ensuring that innovations meet customer needs and achieve market success.</p> <p>Why This Course is Essential:</p> <p>In a rapidly changing world where innovation drives competitive advantage, understanding how to market and sell new products and services is critical. This course equips students with the essential skills to bridge the gap between technological advancements and market needs, enabling them to contribute to the success of innovative companies and address societal challenges through effective marketing strategies.</p> <p>Content:</p> <ul style="list-style-type: none"> <li>• What are the unique challenges in marketing innovative products and services?</li> <li>• How can companies effectively gather and utilize information about future customer needs?</li> <li>• What strategies ensure that new products and services are customer-oriented from conception to market launch?</li> <li>• How can innovative products be priced, promoted, and sold effectively in competitive markets?</li> </ul> <p>What You Will Learn and Get:</p> <p>In this course, students will develop a comprehensive understanding of the unique aspects of marketing innovative products and services. They will master advanced tools and strategies for market analysis, enabling them to evaluate current market conditions and predict future trends with confidence. Students will learn how to gather and integrate customer needs into the development process, ensuring that new products and services are both innovative and customer-oriented.</p> <p>Furthermore students will enhance their ability to work collaboratively, engaging in group discussions and projects where they will learn to articulate their ideas clearly, exchange constructive feedback, and synthesize diverse perspectives into original solutions. They will also refine their presentation skills, learning to communicate complex concepts in a clear and concise manner.</p> <p>Additionally, the course encourages self-directed learning, equipping students with the ability to independently acquire and apply knowledge to new and complex problem areas. They will develop critical thinking skills, allowing them to thoughtfully consider and reflect on proposed marketing strategies and business decisions. By the end of the course, students will not only be equipped with the theoretical knowledge to succeed in innovation marketing but also with the personal and social skills necessary to thrive in collaborative, fast-paced environments.</p> |
| <b>Literature</b>                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Course L0862: PBL Marketing of Innovations |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                 | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Hrs/wk</b>                              | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>CP</b>                                  | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Workload in Hours</b>                   | Independent Study Time 46, Study Time in Lecture 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Lecturer</b>                            | Prof. Christian Lüthje                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Language</b>                            | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Cycle</b>                               | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Content</b>                             | <p>This PBL course is segregated into two afternoon sessions. This course aims at enhancing the students' practical skills in (1) forecasting the future development of markets and (2) making appropriate market-related decisions (particularly segmentation, managing the marketing mix). The students will be prompted to use the knowledge gathered in the lecture of this module and will be invited to (1) Conduct a scenario analysis for an innovative product category and (2) Engage in decision making within a market simulation game.</p> |
| <b>Literature</b>                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Module M1999: Quantitative Research Methods |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|
| Courses                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| Title                                       | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Hrs/wk | CP |
| Quantitative Research Methods (L1714)       | Project Seminar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3      | 6  |
| Module Responsible                          | Prof. Christian Ringle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |    |
| Admission Requirements                      | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| Recommended Previous Knowledge              | Basic knowledge in business administration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |    |
| Educational Objectives                      | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| Professional Competence                     | <div>The students will be able to...</div> <ul style="list-style-type: none"><li>describe complex and interrelated constructs in the fields of marketing, management of organizations, strategic and human resource management;</li><li>discuss underlying theories of research models;</li><li>explain strategies of research problem analysis;</li><li>describe the functioning and use of quantitative research methods;</li><li>discuss strengths and weaknesses of quantitative research methods.</li></ul> |        |    |
| Knowledge                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| Skills                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| Personal Competence                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| Social Competence                           | <div>The students will be able to...</div> <ul style="list-style-type: none"><li>have fruitful professional discussions;</li><li>present and defend the results of their work;</li><li>communicate and collaborate successfully and respectfully with others in teams.</li></ul>                                                                                                                                                                                                                                 |        |    |
| Autonomy                                    | <div>The students will be able to...</div> <ul style="list-style-type: none"><li>acquire knowledge in a specific context independently and to map this knowledge onto other new complex problem fields,</li><li>read and understand statistical literature.</li></ul>                                                                                                                                                                                                                                            |        |    |
| Workload in Hours                           | Independent Study Time 138, Study Time in Lecture 42                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| Credit points                               | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |    |
| Course achievement                          | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| Examination                                 | Subject theoretical and practical work                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |    |
| Examination duration and scale              | Written elaboration with presentation and semester-accompanying tasks                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |    |
| Assignment for the Following Curricula      | Mechanical Engineering and Management: Specialisation Management: Elective Compulsory                                                                                                                                                                                                                                                                                                                                                                                                                            |        |    |

| Course L1714: Quantitative Research Methods |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                  | Project Seminar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Hrs/wk</b>                               | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>CP</b>                                   | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Workload in Hours</b>                    | Independent Study Time 138, Study Time in Lecture 42                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Lecturer</b>                             | Prof. Christian Ringle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Language</b>                             | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Cycle</b>                                | WiSe/SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Content</b>                              | <p>Participants will understand the use, requirements, advantages and disadvantages of quantitative methods. Examples illustrate the application of quantitative methods and their use to address business related problems.</p> <p>The course involves three parts:</p> <ul style="list-style-type: none"> <li>• The first part of the course focuses on an introduction of quantitative research methods,</li> <li>• The second part of the course involves working on a seminar thesis. Participants are in teams invited to describe selected quantitative research methods and to address simple research questions with the described method. Students are expected to write a short (empirical) paper that applies methods learned in this course to a research question of their choice,</li> <li>• The third part is the final presentations of the results from the group work. Participants will present their own small research projects and discuss the results in the plenum. Participants are invited to join the discussions as a part of the final grade.</li> </ul> |
| <b>Literature</b>                           | <p>Participants will be provided with a course handout in the form of ppt.-slides which can be downloaded in advance. In the course, the participants will obtain a specific list of relevant literature. Some generally recommended are:</p> <ul style="list-style-type: none"> <li>• Dalgaard, P. (2008). Introductory statistics with R. Springer Science &amp; Business Media.</li> <li>• Hair, J. F., Black, W. C., Babin, B. J., Anderson, R. E., &amp; Tatham, R. L. (2006). Multivariate data analysis (Vol. 6). Upper Saddle River, NJ: Pearson Prentice Hall.</li> <li>• Hair Jr, J. F., Hult, G. T. M., Ringle, C., &amp; Sarstedt, M. (2013). A primer on partial least squares structural equation modeling (PLS-SEM). Sage Publications.</li> </ul>                                                                                                                                                                                                                                                                                                                      |

| Module M1733: Foundations in Organizational Design and Human Resource Management     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |        |    |
|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------|----|
| Courses                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |        |    |
| Title                                                                                | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  | Hrs/wk | CP |
| Foundations in Organizational Design and Human Resource Management (Seminar) (L2800) | Seminar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  | 4      | 6  |
| <b>Module Responsible</b>                                                            | Prof. Christian Ringle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |        |    |
| <b>Admission Requirements</b>                                                        | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |        |    |
| <b>Recommended Previous Knowledge</b>                                                | Basic knowledge on academic writing as well as principles and concepts in business administration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |        |    |
| <b>Educational Objectives</b>                                                        | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |        |    |
| <b>Professional Competence</b>                                                       | <p><i>Knowledge</i> Students will be able to...</p> <ul style="list-style-type: none"> <li>• Explain the core elements and practices of an effective organizational design;</li> <li>• Describe key components of human resource management (e.g., personnel planning, employee testing, training &amp; development) throughout national and international organizations;</li> <li>• Comprehend the meaning and importance of managing human resources in multinational companies and its relation to organizational designs and strategies;</li> <li>• Use adequate data and quantitative methods for decision making in organizational design and human resource management;</li> <li>• Identify critical success in organizations and conduct human resource analytics.</li> </ul> <p><i>Skills</i> Students will be able to...</p> <ul style="list-style-type: none"> <li>• Apply theoretical knowledge to practical examples;</li> <li>• Write a scientific seminar thesis;</li> <li>• Appropriately present results of their work to others, both in terms of a thesis and oral presentations.</li> </ul> <p><b>Personal Competence</b></p> <p><i>Social Competence</i> The students will be able to...</p> <ul style="list-style-type: none"> <li>• Respectfully work in teams;</li> <li>• Have fruitful group discussions;</li> <li>• Present their results in written form and oral presentations.</li> </ul> <p><i>Autonomy</i> The students will be able to...</p> <ul style="list-style-type: none"> <li>• Independently gather knowledge on specific topics;</li> <li>• Critically evaluate and discuss this information;</li> <li>• Transfer the acquired knowledge to practical applications.</li> </ul> |  |        |    |
| <b>Workload in Hours</b>                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |        |    |
| <b>Credit points</b>                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |        |    |
| <b>Course achievement</b>                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |        |    |
| <b>Examination</b>                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |        |    |
| <b>Examination duration and scale</b>                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |        |    |
| <b>Assignment for the Following Curricula</b>                                        | International Management and Engineering: Core Qualification: Elective Compulsory<br>Mechanical Engineering and Management: Specialisation Management: Elective Compulsory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |        |    |

| Course L2800: Foundations in Organizational Design and Human Resource Management (Seminar) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                                                                 | Seminar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Hrs/wk</b>                                                                              | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>CP</b>                                                                                  | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Workload in Hours</b>                                                                   | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Lecturer</b>                                                                            | Prof. Christian Ringle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Language</b>                                                                            | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Cycle</b>                                                                               | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Content</b>                                                                             | <p>This course is structured as a lecture and a seminar. The lecture focuses on gaining an understanding of the fundamentals of human resource management and organizational design. The lecture also introduces quantitative and business analytics methods for decision making in the field. In the lecture, the basic theoretical concepts are explained and discussed, whereas they are applied through the preparation of a seminar thesis in the seminar.</p> <p>Organizational Design &amp; Human Resource Management</p> <ul style="list-style-type: none"> <li>• The processes of developing organizational structures for small and mid-sized corporations as well as for large multinational enterprises;</li> <li>• The adaptation of organizations and their structures to the competitive environment, with special focus on international operating organizations and global markets;</li> <li>• Introduction to human resource management from a strategic and international perspective (incl. the typical challenges of international organizations);</li> <li>• Key elements of human resource management (incl. design of work, employee recruitment, development, separation &amp; retention);</li> <li>• Introduction of methods and models for decision making in organizational design and human resource management.</li> </ul> <p>Possible Applications of the Theoretical Concepts</p> <ul style="list-style-type: none"> <li>• Big data in organizations and human resource analytics;</li> <li>• Business analytics and machine learning methods (e.g., factor analysis, regression analysis, and structural equation modeling);</li> <li>• Models for the management of organizations and human resource management (e.g., job satisfaction and turnover intention, motivation and organizational commitment).</li> </ul> |
| <b>Literature</b>                                                                          | <p>This course is structured as a lecture and a seminar. The lecture focuses on gaining an understanding of the fundamentals of human resource management and organizational design. The lecture also introduces quantitative and business analytics methods for decision making in the field. In the lecture, the basic theoretical concepts are explained and discussed, whereas they are applied through the preparation of a seminar thesis in the seminar.</p> <p>Organizational Design &amp; Human Resource Management</p> <ul style="list-style-type: none"> <li>• The processes of developing organizational structures for small and mid-sized corporations as well as for large multinational enterprises;</li> <li>• The adaptation of organizations and their structures to the competitive environment, with special focus on international operating organizations and global markets;</li> <li>• Introduction to human resource management from a strategic and international perspective (incl. the typical challenges of international organizations);</li> <li>• Key elements of human resource management (incl. design of work, employee recruitment, development, separation &amp; retention);</li> <li>• Introduction of methods and models for decision making in organizational design and human resource management.</li> </ul> <p>Possible Applications of the Theoretical Concepts</p> <ul style="list-style-type: none"> <li>• Big data in organizations and human resource analytics;</li> <li>• Business analytics and machine learning methods (e.g., factor analysis, regression analysis, and structural equation modeling);</li> <li>• Models for the management of organizations and human resource management (e.g., job satisfaction and turnover intention, motivation and organizational commitment).</li> </ul> |

| Module M0996: Supply Chain Management              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |             |                                |
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| Courses                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |             |                                |
| Title                                              | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              | Hrs/wk      | CP                             |
| Advanced Topics in Supply Chain Management (L3228) | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              | 2           | 3                              |
| Supply Chain Management (L1218)                    | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |              | 2           | 3                              |
| <b>Module Responsible</b>                          | Prof. Christian Thies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |             |                                |
| <b>Admission Requirements</b>                      | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              |             |                                |
| <b>Recommended Previous Knowledge</b>              | no                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |              |             |                                |
| <b>Educational Objectives</b>                      | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              |             |                                |
| <b>Professional Competence</b>                     | <p><i>Knowledge</i></p> <p>Current developments in international business activities such as outsourcing, offshoring, internationalization and globalization and emerging markets illustrated by examples from practice.</p> <ul style="list-style-type: none"> <li>Theoretical Approaches and methods in logistics and supply chain management and use in practice.</li> <li>to identify fields of decision in SCM .</li> <li>reasons for the formation of networks based on various theories from institutional economics (transaction cost theory, principal-agent theory, property-right theory) and the resource-based view.</li> <li>Selected approaches to explain the development of networks.</li> <li>to illustrate phases of network formation.</li> <li>to understand the functional mechanisms of inter-organizational and international network relationships.</li> <li>to explain and categorize relationships within networks.</li> <li>to categorize sourcing concepts and explain motives/ barriers or advantages and disadvantages.</li> <li>advantages and disadvantages of offshoring and outsourcing and to illustrate the distinction between the two terms .</li> <li>to state criteria/ factors/ parameters that influence production location decisions at the global level (total network costs).</li> <li>to explain methods for location finding/evaluation.</li> <li>to interpret phenotypes of production networks.</li> <li>recognize relationships between R &amp; D and production and their locations and to describe coherent models.</li> <li>to solve sub-problems with the configuration of logistics networks (distribution and spare parts networks ) by the use of appropriate approaches.</li> <li>to categorise special waste logistics including their duties &amp; objectives and to state and describe practical examples of good networking.</li> </ul> <p><i>Skills</i></p> <ul style="list-style-type: none"> <li>to asses trends and challenges in national and international supply chains and logistics networks and their consequences for companies.</li> <li>to evaluate, analyse and systematise networks and network relations based on the lecture.</li> <li>to analyse partners and their suitability for co-operation in collaborations and cooperative relations.</li> <li>to select sourcing concepts for specific products / product components based on the lecture as well as advantages and disadvantages of each approach.</li> <li>to evaluate location decisions for production and R &amp; D based on concepts.</li> <li>to recognize relationships between R &amp; D and production as well as their locations and to evaluate the suitability of specific models for different situations.</li> <li>to transfer the analyzed concepts to international practices.</li> <li>to analyse and evaluate the product development processes.</li> <li>to analyse concepts of Information and communication management in logistics.</li> <li>to design subcontracting, procurement, production and disposal as well as R &amp; D networks to shape,</li> <li>to plan reorganise efficient and flow-oriented enterprise networks.</li> <li>to adopt methods of complexity management and risk management in logistics.</li> </ul> <p><b>Personal Competence</b></p> <p><i>Social Competence</i></p> <ul style="list-style-type: none"> <li>to evaluate intercultural and international relationships based on discussed case studies.</li> <li>advance planning and design of network formation and their objectives based on content discussed in the lecture.</li> <li>definition of procurement strategies for individual parts using the gained knowledge of procurement networks.</li> <li>design of the procurement network (external/internal/modules etc.) based on the sourcing concepts and core competencies, as well as on the findings of the case studies.</li> <li>to make decision of location for production taking into account global contexts, evaluation methods and buying/selling markets, which were also discussed in the case studies and their dependence on R &amp; D.</li> <li>Decision on R &amp; D locations based on the insights gained from case studies / practical examples and the selection of an appropriate model.</li> </ul> <p><i>Autonomy</i></p> <p>After completing the module students are capable to work independently on the subject of Supply Chain Management and transfer the acquired knowledge to new problems.</p> |              |             |                                |
| <b>Workload in Hours</b>                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |             |                                |
| <b>Credit points</b>                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |             |                                |
| <b>Course achievement</b>                          | <b>Compulsory</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Bonus</b> | <b>Form</b> | <b>Description</b>             |
|                                                    | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 20 %         | Subject     | theoretical and practical work |
| <b>Examination</b>                                 | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |             |                                |
| <b>Examination duration and scale</b>              | 120 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |              |             |                                |

|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Assignment for the Following Curricula</b> | Bioprocess Engineering: Specialisation C - Bioeconomic Process Engineering, Focus Management and Controlling: Elective Compulsory<br>International Management and Engineering: Specialisation I. Electives Management: Elective Compulsory<br>Logistics, Infrastructure and Mobility: Specialisation Production and Logistics: Elective Compulsory<br>Mechanical Engineering and Management: Specialisation Management: Elective Compulsory |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Course L3228: Advanced Topics in Supply Chain Management |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                               | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Hrs/wk</b>                                            | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>CP</b>                                                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Workload in Hours</b>                                 | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Lecturer</b>                                          | Prof. Christian Thies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Language</b>                                          | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Cycle</b>                                             | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Content</b>                                           | <p>This problem-based learning (PBL) course complements the lecture Supply Chain Management by exploring selected topics in-depth. The specific topics may vary each semester, but typically comprise:</p> <ul style="list-style-type: none"> <li>• Data analytics and visualization in supply chains (use of software tools to explore large datasets)</li> <li>• Supply chain dynamics (interactive simulation game illustrating the bullwhip effect)</li> <li>• Case studies (in-depth discussion of company cases related to topics from the lecture)</li> <li>• Current research in supply chain management (discussion of recent scientific articles)</li> </ul> |
| <b>Literature</b>                                        | <p>Wird im Kurs bereitgestellt.</p> <p>Will be provided in the course.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |



| Course L1218: Supply Chain Management |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                            | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Hrs/wk</b>                         | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>CP</b>                             | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Workload in Hours</b>              | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Lecturer</b>                       | Prof. Christian Thies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Language</b>                       | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Cycle</b>                          | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Content</b>                        | <ul style="list-style-type: none"> <li>• Vermittlung eines tiefgreifenden Verständnisses von Logistik und Supply Chain Management</li> <li>• Vermittlung umfassender theoretischer Ansätze und Methoden in der Logistik und im Supply Chain Management; Übertragung der analysierten Konzepte auf Praxisbeispiele</li> <li>• Ausarbeitung und kritische Diskussion unterschiedlicher Supply Chain Konfigurationen sowie strategischer Supply Chain Ansätze (z.B. Effizienz vs. Reaktionsfähigkeit)</li> <li>• Einführung in die Managementprozesse des SCOR-Modells; Vermittlung von Konzepten der Bereiche Planung, Beschaffung/Einkauf und Distribution</li> <li>• Vermittlung von Grundlagen des Supply Chain Risikomanagements; Übertragung der Konzepte auf Praxisbeispiele</li> <li>• Einführung in die digitale Transformation; Identifikation von Trends und Strategien in der Logistik und Supply Chain Management; Ableitung von Chancen der digitalen Transformation in der Logistik und Supply Chain Management</li> <li>• Einführung in die Datenanalyse und -visualisierung mithilfe eines Tools; Anwenden der Kenntnisse auf Themengebiete in der Logistik und Supply Chain Management; Aufbereitung der Ergebnisse mit Hilfe moderner Präsentationsmedien</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Literature</b>                     | <p>Bowersox, D. J., Closs, D. J. und Cooper, M. B. (2010): Supply chain logistics management, 3<sup>rd</sup> edition, Boston [u.a.]: McGraw-Hill/Irwin.</p> <p>Chopra, S. und Meindl, P. (2016): Supply chain management: strategy, planning, and operation, 6<sup>th</sup> edition, Boston [u.a.]: Pearson.</p> <p>Corsten, H., Gössinger, R. (2007): Einführung in das Supply Chain Management, 2. Aufl., München/Wien: Oldenbourg.</p> <p>Corsten, H., Gössinger, R., Spengler, Th. (Hrsg., 2018): Handbuch Produktions- und Logistikmanagement in Wertschöpfungsnetzwerken, Berlin/Boston.</p> <p>Heiserich O., Helbig, K. und Ullmann, W. (2011): Logistik, 4. vollständig überarbeitete und erweiterte Auflage, Wiesbaden: Gabler Verlag/ Springer Fachmedien.</p> <p>Heizer, J., Render, B., Munson, Ch. (2020): Principles of Operations Management, 11<sup>th</sup> edition, Boston: Pearson.</p> <p>Hugos, M. (2018): Essentials of Supply Chain Management, Wiley.</p> <p>Fisher, M. (1997): What is the right supply chain for your product?, Harvard Business Review, Vol. 75, No. pp., S. 105-117.</p> <p>Kersten, W. Seiter, M., von See, B. and Hackius, N. und Maurer, T. (2017): Trends und Strategien in Logistik und Supply Chain Management: Chancen der digitalen Transformation, DVV Media Group GmbH: Hamburg.</p> <p>Kuhn, A. und Hellgrath, B. (2002): Supply Chain Management: optimierte Zusammenarbeit in der Wertschöpfungskette, Berlin [u.a.]: Springer.</p> <p>Larson, P., Poist, R. and Halldórsson, Á. (2007): Perspectives on logistics vs. SCM: a survey of SCM professionals, in: Journal of Business Logistics, Vol. 28, No. 1, S. 1-24.</p> <p>Kummer, S., Grün, O. und Jammerneegg, W. (2018): Grundzüge der Beschaffung, Produktion und Logistik, 4. aktualisierte Auflage, München: Pearson Studium.</p> <p>Obermaier, Robert (Hrsg., 2019): Handbuch Industrie 4.0 und Digitale Transformation: Betriebswirtschaftliche, technische und rechtliche Herausforderungen, Wiesbaden.</p> <p>Porter, M. (1986): Changing Patterns of International Competition, California Management Review, Vol. 28, No. 2, S. 9-40.</p> <p>Schröder, M./ Wegner, K., Hrsg. (2019): Logistik im Wandel der Zeit - Von der Produktionssteuerung zu vernetzten Supply Chains, Wiesbaden: Springer Gabler</p> <p>Simchi-Levi, D., Kaminsky, P. und Simchi-Levi, E. (2008): Designing and managing the supply chain: concepts, strategies and case studies, 3<sup>rd</sup> edition, Boston [u.a.]: McGraw-Hill/Irwin.</p> <p>Supply Chain Council (2014): Supply Chain Operations Reference (SCOR) model: Overview - Version 11.0.</p> <p>Swink, M., Melnyk, S. A., Cooper, M. B. und Hartley, J. L. (2011): Managing Operations - Across the Supply Chain. 2<sup>nd</sup> edition, New York, NY: McGraw-Hill/Irwin.</p> <p>Weele, A. J. v. (2005): Purchasing &amp; supply chain management, 4<sup>th</sup> edition, London [u.a.]: Thomson Learning.</p> |

| Module M1035: Entrepreneurial Finance                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |                  |                    |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------------|--------------------|
| Courses                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |                  |                    |
| Title                                                  | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              | Hrs/wk           | CP                 |
| Entrepreneurial Finance: Case Studies (L1282)          | Seminar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |              | 3                | 4                  |
| Entrepreneurial Finance: Lecture (L1281)               | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |              | 2                | 2                  |
| <b>Module Responsible</b>                              | Prof. Christoph Ihl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              |                  |                    |
| <b>Admission Requirements</b>                          | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              |                  |                    |
| <b>Recommended Previous Knowledge</b>                  | Basic knowledge in business economics and finance obtained in the compulsory modules and participation in the module "Technology Entrepreneurship" is highly recommended.                                                                                                                                                                                                                                                                                                                                                                                  |              |                  |                    |
| <b>Educational Objectives</b>                          | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              |                  |                    |
| <b>Professional Competence</b><br><i>Knowledge</i>     | <p>Wissen (subject-related knowledge and understanding):</p> <ul style="list-style-type: none"> <li>• understand the structure of a financial plan for a new venture</li> <li>• understand the procedures, pros and cons of different valuation methods</li> <li>• understand the design of financial contracts and term sheets</li> <li>• understand the interests of venture capital funds</li> <li>• understand the pros and cons of different growth and exit options</li> </ul>                                                                       |              |                  |                    |
| <i>Skills</i>                                          | <p>Fertigkeiten (subject-related skills):</p> <ul style="list-style-type: none"> <li>• prepare a financial plan for a new venture</li> <li>• value a new venture in financial terms</li> <li>• apply different valuation methods</li> <li>• evaluate the attractiveness of financial contracts</li> <li>• design VC term sheets</li> <li>• design employee contracts in terms of financial compensation</li> <li>• design financial contracts and conduct financial negotiations</li> <li>• assess and justify possible growth and exit options</li> </ul> |              |                  |                    |
| <b>Personal Competence</b><br><i>Social Competence</i> | <p>Sozialkompetenz (Social Competence):</p> <ul style="list-style-type: none"> <li>• team work</li> <li>• communication and presentation</li> <li>• give and take critical comments</li> <li>• engaging in fruitful discussions</li> </ul>                                                                                                                                                                                                                                                                                                                 |              |                  |                    |
| <i>Autonomy</i>                                        | <p>Selbständigkeit (Autonomy):</p> <ul style="list-style-type: none"> <li>• autonomous work and time management</li> <li>• project management</li> <li>• analytical skills</li> </ul>                                                                                                                                                                                                                                                                                                                                                                      |              |                  |                    |
| <b>Workload in Hours</b>                               | Independent Study Time 110, Study Time in Lecture 70                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              |                  |                    |
| <b>Credit points</b>                                   | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |              |                  |                    |
| <b>Course achievement</b>                              | <b>Compulsory</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Bonus</b> | <b>Form</b>      | <b>Description</b> |
|                                                        | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 20 %         | Group discussion |                    |
| <b>Examination</b>                                     | Subject theoretical and practical work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |                  |                    |
| <b>Examination duration and scale</b>                  | Presentations and case study work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |              |                  |                    |
| <b>Assignment for the Following Curricula</b>          | <p>Global Innovation Management: Core Qualification: Elective Compulsory</p> <p>Global Technology and Innovation Management &amp; Entrepreneurship: Core Qualification: Elective Compulsory</p> <p>International Management and Engineering: Specialisation I. Electives Management: Elective Compulsory</p> <p>Mechanical Engineering and Management: Specialisation Management: Elective Compulsory</p>                                                                                                                                                  |              |                  |                    |

| Course L1282: Entrepreneurial Finance: Case Studies |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                          | Seminar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Hrs/wk</b>                                       | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>CP</b>                                           | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Workload in Hours</b>                            | Independent Study Time 78, Study Time in Lecture 42                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Lecturer</b>                                     | Prof. Christoph Ihl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Language</b>                                     | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Cycle</b>                                        | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Content</b>                                      | <p>Entrepreneurial finance is at the center of a clash of two very distant worlds: that of entrepreneurship and that of finance. Finance is disciplined, based on numbers and logical thinking and looking for proven track records. Entrepreneurship is messy, based on intuition and experimentation and treading off the beaten track. Entrepreneurial finance is the provision of funding to young, innovative, growth-oriented companies. Entrepreneurial companies are young, typically less than ten years old, and introduce innovative products or business models. The younger are called "startups," and are typically less than five years old.</p> <p>There is a variety of investors who can finance entrepreneurial companies: family and friends, business angels, accelerators and incubators, crowdfunding platforms, venture capital firms, corporate investors, etc. The course provides a thorough understanding of what motivates them, of the way they invest, and of what support they can provide to a company at what stage in the fundraising cycle. The course addresses the following key questions: How much money can and should be raised? When should it be raised and from whom? What is a reasonable valuation of the company? How should funding, employment contracts and exit decisions be structured?</p> <p>Thus, the course provides an understanding of the whole fundraising cycle, from the moment the entrepreneur conceived her idea to the moment investors exit the company and move on. We examine the entrepreneur's signalling to investors of the qualities of the venture, the investors' evaluation of the venture, the various dimensions of contracting (cash flow rights, control rights, compensation, and other clauses), the negotiation of a deal and the provision of corporate governance, the process of staged financing, the financing through debt, and the exit process through liquidity events such as initial public offering, sale or merger.</p> <p>The following topics will be covered with specific case studies:</p> <ol style="list-style-type: none"> <li>1. Introduction: Evaluating Venture Opportunities</li> <li>2. Financial Planning</li> <li>3. Ownership and Returns</li> <li>4. Valuation Methods</li> <li>5. Term Sheets</li> <li>6. Structuring Deals</li> <li>7. Corporate Governance</li> <li>8. Staged Financing</li> <li>9. Debt Financing</li> <li>10. Exits</li> <li>11. Early Stage &amp; Venture Capital Investors</li> <li>12. Ecosystems</li> </ol> |
| <b>Literature</b>                                   | Da Rin, Marco, and Thomas Hellmann. Fundamentals of Entrepreneurial Finance. Oxford University Press, 2020.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Course L1281: Entrepreneurial Finance: Lecture |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                     | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Hrs/wk</b>                                  | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>CP</b>                                      | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Workload in Hours</b>                       | Independent Study Time 32, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Lecturer</b>                                | Prof. Christoph Ihl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Language</b>                                | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Cycle</b>                                   | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Content</b>                                 | <p>Entrepreneurial finance is at the center of a clash of two very distant worlds: that of entrepreneurship and that of finance. Finance is disciplined, based on numbers and logical thinking and looking for proven track records. Entrepreneurship is messy, based on intuition and experimentation and treading off the beaten track. Entrepreneurial finance is the provision of funding to young, innovative, growth-oriented companies. Entrepreneurial companies are young, typically less than ten years old, and introduce innovative products or business models. The younger are called "startups," and are typically less than five years old.</p> <p>There is a variety of investors who can finance entrepreneurial companies: family and friends, business angels, accelerators and incubators, crowdfunding platforms, venture capital firms, corporate investors, etc. The course provides a thorough understanding of what motivates them, of the way they invest, and of what support they can provide to a company at what stage in the fundraising cycle. The course addresses the following key questions: How much money can and should be raised? When should it be raised and from whom? What is a reasonable valuation of the company? How should funding, employment contracts and exit decisions be structured?</p> <p>Thus, the course provides an understanding of the whole fundraising cycle, from the moment the entrepreneur conceived her idea to the moment investors exit the company and move on. We examine the entrepreneur's signalling to investors of the qualities of the venture, the investors' evaluation of the venture, the various dimensions of contracting (cash flow rights, control rights, compensation, and other clauses), the negotiation of a deal and the provision of corporate governance, the process of staged financing, the financing through debt, and the exit process through liquidity events such as initial public offering, sale or merger.</p> <p>The following topics will be covered in lectures:</p> <ol style="list-style-type: none"> <li>1. Introduction: Evaluating Venture Opportunities</li> <li>2. Financial Planning</li> <li>3. Ownership and Returns</li> <li>4. Valuation Methods</li> <li>5. Term Sheets</li> <li>6. Structuring Deals</li> <li>7. Corporate Governance</li> <li>8. Staged Financing</li> <li>9. Debt Financing</li> <li>10. Exits</li> <li>11. Early Stage &amp; Venture Capital Investors</li> <li>12. Ecosystems</li> </ol> |
| <b>Literature</b>                              | Da Rin, Marco, and Thomas Hellmann. Fundamentals of Entrepreneurial Finance. Oxford University Press, 2020.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| Module M1975: Advanced Topics in Management, Organization, and Human Resource Management |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |           |
|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------|
| <b>Courses</b>                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |           |
| <b>Title</b>                                                                             | <b>Typ</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Hrs/wk</b> | <b>CP</b> |
| Advanced Topics in Management, Organization, and Human Resource Management (L0110)       | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 4             | 6         |
| <b>Module Responsible</b>                                                                | Prof. Christian Ringle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |           |
| <b>Admission Requirements</b>                                                            | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |           |
| <b>Recommended Previous Knowledge</b>                                                    | Foundations in Organizational Design and Human Resource Management<br><br>Basic knowledge on academic writing as well as principles and concepts in business administration and foundations in organizational design and human resource management.                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |           |
| <b>Educational Objectives</b>                                                            | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |           |
| <b>Professional Competence</b>                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |           |
| <i>Knowledge</i>                                                                         | The students are able to...<br><br><ul style="list-style-type: none"> <li>Explain the different organizational designs and strategies in an international environment with a focus on selected forms of cooperation (e.g., virtual organizations or strategic alliances) to compete in global business;</li> <li>Map the need of organizational changes in light of new business lines, strategies, altering employees' attitudes, and international competition;</li> <li>Explain the models and approaches for appropriately measuring employee relations (e.g., job satisfaction models), incl. the development and estimation of causal models.</li> </ul>                        |               |           |
| <i>Skills</i>                                                                            | The students are able to...<br><br><ul style="list-style-type: none"> <li>Work with empirical data, apply business process management and multivariate techniques to the data collected using standard software, and critically evaluate and interpret the results;</li> <li>Critically rethink theoretical concepts and gain analytical abilities in organization management and human resource management;</li> <li>Use their practical knowledge of the analytical toolset to successfully tackle the management challenges in organization and human resource management in internationally acting companies;</li> <li>Present their results in written and oral form.</li> </ul> |               |           |
| <b>Personal Competence</b>                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |               |           |
| <i>Social Competence</i>                                                                 | The students are able to...<br><br><ul style="list-style-type: none"> <li>Respectfully work in teams;</li> <li>Have fruitful group discussions;</li> <li>Present their results in written form and oral presentations.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |           |
| <i>Autonomy</i>                                                                          | The students are able to...<br><br><ul style="list-style-type: none"> <li>Acquire further relevant information independently;</li> <li>Critically reflect and evaluate this information;</li> <li>Transfer the acquired knowledge to practical applications.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                               |               |           |
| <b>Workload in Hours</b>                                                                 | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |           |
| <b>Credit points</b>                                                                     | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |               |           |
| <b>Course achievement</b>                                                                | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |           |
| <b>Examination</b>                                                                       | Subject theoretical and practical work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |           |
| <b>Examination duration and scale</b>                                                    | Thesis with presentation and assignments during the semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |               |           |
| <b>Assignment for the Following Curricula</b>                                            | International Management and Engineering: Specialisation I. Electives Management: Elective Compulsory<br>Mechanical Engineering and Management: Specialisation Management: Elective Compulsory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |           |

| Course L0110: Advanced Topics in Management, Organization, and Human Resource Management |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                                                               | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Hrs/wk</b>                                                                            | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>CP</b>                                                                                | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Workload in Hours</b>                                                                 | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Lecturer</b>                                                                          | Prof. Christian Ringle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Language</b>                                                                          | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Cycle</b>                                                                             | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Content</b>                                                                           | <p>This lecture focuses on multinational firms and advanced issues of management, organizations, and human resource management. This course is structured as a lecture and a seminar. In the lecture, the advanced theoretical concepts are explained and discussed, whereas they are applied in the seminar through the preparation of a seminar thesis. The students learn about the process and structure of a scientific article, and further deepen their knowledge, while working in groups.</p> <p>Example topics:</p> <ul style="list-style-type: none"> <li>• Management: change management and corporate social responsibility;</li> <li>• Organization: exploration &amp; exploitation, networks, and organizational identity;</li> <li>• Human Resource Management: human resource metrics &amp; analytics and recruitment &amp; selection.</li> </ul> |
| <b>Literature</b>                                                                        | <p>The students will be provided with selected journal articles.</p> <p>Bernardin, H.J. (2006): Human Resource Management: An Experiential Approach, 4e, New York: McGraw-Hill.</p> <p>Cascio, W. (2015): Managing Human Resources: Productivity, Quality of Work Life, Profits, revised edition, New York: McGraw-Hill.</p> <p>French, W./Bell, C.H./Zawacki, R.A. (2004): Organization Development and Transformation: Managing Effective Change, 6e, Chicago: McGraw-Hill.</p> <p>Hitt, M.A./Ireland, R.D./Hoskisson, R.E. (2014): Strategic Management: Competitiveness and Globalization, 11e, Ohio: Cengage Learning.</p> <p>Lynch, R. (2015): Strategic Management, 7e, Harlow: Prentice Hall.</p>                                                                                                                                                          |

| Module M1986: Applied Statistics              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |             |                                |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------|--------------------------------|
| Courses                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |             |                                |
| Title                                         | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              | Hrs/wk      | CP                             |
| Applied Statistics (L1584)                    | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |              | 2           | 3                              |
| Applied Statistics (L1586)                    | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              | 2           | 2                              |
| Applied Statistics (L1585)                    | Recitation Section (small)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |              | 1           | 1                              |
| <b>Module Responsible</b>                     | Prof. Matthias Schulte                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |             |                                |
| <b>Admission Requirements</b>                 | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |             |                                |
| <b>Recommended Previous Knowledge</b>         | Basic knowledge of statistical methods                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |             |                                |
| <b>Educational Objectives</b>                 | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |             |                                |
| <b>Professional Competence</b>                | <p><i>Knowledge</i> Students can explain the statistical methods and the conditions of their use.</p> <p><i>Skills</i> Students are able to use the statistics program to solve statistics problems and to interpret and depict the results</p> <p><b>Personal Competence</b></p> <p><i>Social Competence</i> Team Work, joined presentation of results</p> <p><i>Autonomy</i> To understand and interpret the question and solve</p>                                                                                |              |             |                                |
|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |             |                                |
|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |             |                                |
|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |             |                                |
| <b>Workload in Hours</b>                      | Independent Study Time 110, Study Time in Lecture 70                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |             |                                |
| <b>Credit points</b>                          | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |              |             |                                |
| <b>Course achievement</b>                     | <b>Compulsory</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Bonus</b> | <b>Form</b> | <b>Description</b>             |
|                                               | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 10 %         | Subject     | theoretical and practical work |
| <b>Examination</b>                            | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |              |             |                                |
| <b>Examination duration and scale</b>         | 90 minutes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |              |             |                                |
| <b>Assignment for the Following Curricula</b> | <p>Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory</p> <p>Mechanical Engineering and Management: Specialisation Management: Elective Compulsory</p> <p>Mechatronics: Core Qualification: Elective Compulsory</p> <p>Biomedical Engineering: Core Qualification: Compulsory</p> <p>Product Development, Materials and Production: Core Qualification: Elective Compulsory</p> <p>Theoretical Mechanical Engineering: Core Qualification: Elective Compulsory</p> |              |             |                                |

| Course L1584: Applied Statistics |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                       | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Hrs/wk</b>                    | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>CP</b>                        | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Workload in Hours</b>         | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Lecturer</b>                  | Prof. Matthias Schulte                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Language</b>                  | DE/EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Cycle</b>                     | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Content</b>                   | <p>The goal is to introduce students to the basic statistical methods and their application to simple problems. The topics include:</p> <ul style="list-style-type: none"> <li>• Chi square test</li> <li>• Simple regression and correlation</li> <li>• Multiple regression and correlation</li> <li>• One way analysis of variance</li> <li>• Two way analysis of variance</li> <li>• Discriminant analysis</li> <li>• Analysis of categorial data</li> <li>• Chossing the appropriate statistical method</li> <li>• Determining critical sample sizes</li> </ul> |
| <b>Literature</b>                | <p>Applied Regression Analysis and Multivariable Methods, 3rd Edition, David G. Kleinbaum Emory University, Lawrence L. Kupper University of North Carolina at Chapel Hill, Keith E. Muller University of North Carolina at Chapel Hill, Azhar Nizam Emory University, Published by Duxbury Press, CB © 1998, ISBN/ISSN: 0-534-20910-6</p>                                                                                                                                                                                                                          |

| Course L1586: Applied Statistics |                                                                                                                                                                                                                                    |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                       | Project-/problem-based Learning                                                                                                                                                                                                    |
| <b>Hrs/wk</b>                    | 2                                                                                                                                                                                                                                  |
| <b>CP</b>                        | 2                                                                                                                                                                                                                                  |
| <b>Workload in Hours</b>         | Independent Study Time 32, Study Time in Lecture 28                                                                                                                                                                                |
| <b>Lecturer</b>                  | Prof. Matthias Schulte                                                                                                                                                                                                             |
| <b>Language</b>                  | DE/EN                                                                                                                                                                                                                              |
| <b>Cycle</b>                     | WiSe                                                                                                                                                                                                                               |
| <b>Content</b>                   | The students receive a problem task, which they have to solve in small groups (n=5). They do have to collect their own data and work with them. The results have to be presented in an executive summary at the end of the course. |
| <b>Literature</b>                | Selbst zu finden                                                                                                                                                                                                                   |

| Course L1585: Applied Statistics |                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                       | Recitation Section (small)                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Hrs/wk</b>                    | 1                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>CP</b>                        | 1                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Workload in Hours</b>         | Independent Study Time 16, Study Time in Lecture 14                                                                                                                                                                                                                                                                                                                                                     |
| <b>Lecturer</b>                  | Prof. Matthias Schulte                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Language</b>                  | DE/EN                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Cycle</b>                     | WiSe                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Content</b>                   | The different statistical tests are applied for the solution of realistic problems using actual data sets and the most common used commercial statistical software package (SPSS).                                                                                                                                                                                                                      |
| <b>Literature</b>                | Student Solutions Manual for Kleinbaum/Kupper/Muller/Nizam's Applied Regression Analysis and Multivariable Methods, 3rd Edition, David G. Kleinbaum Emory University Lawrence L. Kupper University of North Carolina at Chapel Hill, Keith E. Muller University of North Carolina at Chapel Hill, Azhar Nizam Emory University, Published by Duxbury Press, Paperbound © 1998, ISBN/ISSN: 0-534-20913-0 |



## Specialization Mechatronics

Graduates of the Mechatronics specialization are able to solve mechatronic tasks as well as design tasks systematically and methodically. They have knowledge about current methods, automation and simulation, are able to choose between different strategies and to use them independently for the development of new systems.

The Mechatronics specialization is recommended to students who already bring along basic knowledge in measurement technology, control engineering and computer science.

| Module M0751: Vibration Theory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |        |    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------|----|
| Courses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |        |    |
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  | Hrs/wk | CP |
| Vibration Theory (L0701)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Integrated Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  | 4      | 6  |
| <b>Module Responsible</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Prof. Norbert Hoffmann                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |        |    |
| <b>Admission Requirements</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |        |    |
| <b>Recommended Previous Knowledge</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>• Calculus</li> <li>• Linear Algebra</li> <li>• Engineering Mechanics</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |        |    |
| <b>Educational Objectives</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |        |    |
| <b>Professional Competence</b><br><i>Knowledge</i> <ul style="list-style-type: none"> <li>• Students are able to denote terms and concepts of Vibration Theory and develop them further.</li> <li>• Students know methods of modeling and simulation for free, driven, self-excited and parameter driven vibrations.</li> <li>• Students know about concepts of linear and nonlinear vibration problems.</li> <li>• Students know basic tasks of vibration problems of discrete and continuous systems.</li> </ul> <i>Skills</i> <ul style="list-style-type: none"> <li>• Students are able to denote methods of Vibration Theory and develop them further.</li> <li>• Students are able to apply and expand methods of modeling and simulation for free, forced, self-excited and parameter driven vibrations.</li> <li>• Students are able to solve linear and nonlinear vibration problems.</li> </ul> <b>Personal Competence</b><br><i>Social Competence</i> <ul style="list-style-type: none"> <li>• Students can analyze vibration problems, work on them, and reach working results also in teams or groups.</li> <li>• Students are able to document the results of vibration studies also in groups.</li> </ul> <i>Autonomy</i> <ul style="list-style-type: none"> <li>• Students are able to individually analyze and solve vibration problems.</li> <li>• Students are able to approach individually research tasks in Vibration Theory.</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |        |    |
| <b>Workload in Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |        |    |
| <b>Credit points</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |        |    |
| <b>Course achievement</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |        |    |
| <b>Examination</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |        |    |
| <b>Examination duration and scale</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2 Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |        |    |
| <b>Assignment for the Following Curricula</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Energy Systems: Core Qualification: Elective Compulsory<br>International Management and Engineering: Specialisation II. Mechatronics: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Core Qualification: Compulsory<br>Mechanical Engineering and Management: Specialisation Mechatronics: Elective Compulsory<br>Mechatronics: Core Qualification: Compulsory<br>Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory<br>Biomedical Engineering: Specialisation Implants and Endoprostheses: Elective Compulsory<br>Biomedical Engineering: Specialisation Medical Technology and Control Theory: Elective Compulsory<br>Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory<br>Product Development, Materials and Production: Core Qualification: Compulsory<br>Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory<br>Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory<br>Theoretical Mechanical Engineering: Core Qualification: Elective Compulsory |  |        |    |

| Course L0701: Vibration Theory |                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                     | Integrated Lecture                                                                                                                                                                                                                                                                                                                                             |
| <b>Hrs/wk</b>                  | 4                                                                                                                                                                                                                                                                                                                                                              |
| <b>CP</b>                      | 6                                                                                                                                                                                                                                                                                                                                                              |
| <b>Workload in Hours</b>       | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                           |
| <b>Lecturer</b>                | Prof. Norbert Hoffmann                                                                                                                                                                                                                                                                                                                                         |
| <b>Language</b>                | DE/EN                                                                                                                                                                                                                                                                                                                                                          |
| <b>Cycle</b>                   | WiSe                                                                                                                                                                                                                                                                                                                                                           |
| <b>Content</b>                 | <p>Linear and Nonlinear Single and Multiple Degree of Freedom Vibrations</p> <ul style="list-style-type: none"> <li>• Free vibration</li> <li>• Self-excited vibration</li> <li>• Parameter driven vibration</li> <li>• Forced vibration</li> <li>• Multi degree of freedom vibration</li> <li>• Continuum vibration</li> <li>• Irregular vibration</li> </ul> |
| <b>Literature</b>              | <p>German - K. Magnus, K. Popp, W. Sextro: Schwingungen. Physikalische Grundlagen und mathematische Behandlung von Schwingungen.</p> <p>English - K. Magnus: Vibrations.</p>                                                                                                                                                                                   |

| Module M0752: Nonlinear Dynamics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |        |    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------|----|
| Courses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |        |    |
| Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  | Hrs/wk | CP |
| Nonlinear Dynamics (L0702)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Integrated Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  | 4      | 6  |
| <b>Module Responsible</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Prof. Norbert Hoffmann                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |        |    |
| <b>Admission Requirements</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |        |    |
| <b>Recommended Previous Knowledge</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>• Calculus</li> <li>• Linear Algebra</li> <li>• Engineering Mechanics</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |        |    |
| <b>Educational Objectives</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |        |    |
| <b>Professional Competence</b><br><i>Knowledge</i> <ul style="list-style-type: none"> <li>• Students are able to reflect existing terms and concepts in Nonlinear Dynamics and to develop and research new terms and concepts.</li> <li>• Students are able to denote and expand methods of modeling and analysis for nonlinear dynamical systems.</li> </ul> <i>Skills</i> <ul style="list-style-type: none"> <li>• Students are able to apply existing methods and procedures of Nonlinear Dynamics.</li> <li>• Students are able to develop novel methods and procedures for nonlinear dynamical systems.</li> </ul> <b>Personal Competence</b><br><i>Social Competence</i> <ul style="list-style-type: none"> <li>• Students can analyze problems of nonlinear dynamics also in groups.</li> <li>• Students can achieve solution procedures for problems of nonlinear dynamical systems also in groups.</li> </ul> <i>Autonomy</i> <ul style="list-style-type: none"> <li>• Students are able to approach given research tasks on the basis of given methods individually.</li> <li>• Students are able to identify and follow up novel research tasks by themselves.</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |        |    |
| <b>Workload in Hours</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |        |    |
| <b>Credit points</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |        |    |
| <b>Course achievement</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |        |    |
| <b>Examination</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |        |    |
| <b>Examination duration and scale</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2 Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |        |    |
| <b>Assignment for the Following Curricula</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Aircraft Systems Engineering: Core Qualification: Elective Compulsory<br>International Management and Engineering: Specialisation II. Mechatronics: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory<br>Mechanical Engineering and Management: Specialisation Mechatronics: Elective Compulsory<br>Mechatronics: Core Qualification: Elective Compulsory<br>Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory<br>Biomedical Engineering: Specialisation Implants and Endoprotheses: Elective Compulsory<br>Biomedical Engineering: Specialisation Medical Technology and Control Theory: Elective Compulsory<br>Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory<br>Product Development, Materials and Production: Core Qualification: Elective Compulsory<br>Theoretical Mechanical Engineering: Core Qualification: Elective Compulsory |  |        |    |

| Course L0702: Nonlinear Dynamics |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                       | Integrated Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Hrs/wk</b>                    | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>CP</b>                        | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Workload in Hours</b>         | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Lecturer</b>                  | Prof. Norbert Hoffmann                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Language</b>                  | DE/EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Cycle</b>                     | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Content</b>                   | <p>Fundamentals of Nonlinear Dynamics</p> <ul style="list-style-type: none"> <li>• One dimensional problems <ul style="list-style-type: none"> <li>◦ Linear Stability</li> <li>◦ Local Bifurcations</li> <li>◦ Synchronisation</li> </ul> </li> <li>• Two dimensional problems <ul style="list-style-type: none"> <li>◦ Limit Cycles</li> <li>◦ Global Bifurcations</li> </ul> </li> <li>• Chaos <ul style="list-style-type: none"> <li>◦ Lorenz Equations</li> <li>◦ Fractals and Strange Attractors</li> <li>◦ Predictability and Horizons</li> </ul> </li> </ul> |
| <b>Literature</b>                | Steven Strogatz: Nonlinear Dynamics and Chaos.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Module M0846: Control Systems Theory and Design        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |    |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|
| Courses                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |    |
| Title                                                  | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Hrs/wk | CP |
| Control Systems Theory and Design (L0656)              | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2      | 4  |
| Control Systems Theory and Design (L0657)              | Recitation Section (small)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2      | 2  |
| <b>Module Responsible</b>                              | Prof. Timm Faulwasser                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |    |
| <b>Admission Requirements</b>                          | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Recommended Previous Knowledge</b>                  | Introduction to Control Systems or Grundlagen der Regelungstechnik                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |    |
| <b>Educational Objectives</b>                          | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Professional Competence</b><br><i>Knowledge</i>     | <p>The students can explain the system properties controllability and observability, and their relationship to state feedback and state estimation, respectively. They can explain the significance of a minimal realization. They can explain observer-based state feedback and how it can be used to achieve tracking and disturbance rejection.</p> <p>They can extend all of the above to multi-input multi-output systems.</p> <p>They can explain the z-transform and its relationship with the Laplace Transform; they can explain state space models and transfer function models of discrete-time systems. They can explain the experimental identification of ARX models of dynamic systems.</p> <p>They know concepts for two-degree of freedom designs and flatness-based feedforward control.</p> <p>They know concepts for Lyapunov-based stability analysis, passivity and dissipativity analysis of LTI systems and in very basic form also for nonlinear systems. They can compute Lyapunov functions and storage functions for LTI systems.</p> <p>They know how the stability of equilibria of nonlinear systems can be analyzed using the Jacobi linearization.</p> <p>They know the basic of optimal control of linear systems, they know the Bellman equation also for nonlinear systems. They know the concepts of optimal trajectory tracking designs for LTI systems.</p> |        |    |
| <i>Skills</i>                                          | <ul style="list-style-type: none"> <li>• Students can transform transfer function models into state space models and vice versa</li> <li>• They can assess controllability and observability and construct minimal realisations</li> <li>• They can design LQR controllers for multivariable plants, they can perform a 2 DoF design for LTI systems</li> <li>• They can carry out a controller design both in continuous-time and discrete-time domain, and decide which is appropriate for a given sampling rate</li> <li>• They can identify state space models of dynamic systems from experimental data</li> <li>• They can carry out all these tasks using standard software tools (Matlab Control Toolbox, System Identification Toolbox, Simulink)</li> <li>• They can analyze the stability of equilibria of nonlinear systems, they can compute Lyapunov and storage functions for LTI systems.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Personal Competence</b><br><i>Social Competence</i> | Students can work in small groups on specific problems to arrive at joint solutions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <i>Autonomy</i>                                        | <p>Students can obtain information from provided sources (lecture notes, software documentation, experiment guides) and use it when solving given problems.</p> <p>They can assess their knowledge in weekly on-line tests and thereby control their learning progress.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |    |
| <b>Workload in Hours</b>                               | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Credit points</b>                                   | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| <b>Course achievement</b>                              | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Examination</b>                                     | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |    |
| <b>Examination duration and scale</b>                  | 120 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |    |
| <b>Assignment for the Following Curricula</b>          | <p>Electrical Engineering and Information Technology: Core Qualification: Compulsory</p> <p>Electrical Engineering: Core Qualification: Compulsory</p> <p>Energy Systems: Core Qualification: Elective Compulsory</p> <p>Energy Systems: Core Qualification: Elective Compulsory</p> <p>Energy Systems: Specialisation Energy Systems: Elective Compulsory</p> <p>Aircraft Systems Engineering: Core Qualification: Elective Compulsory</p> <p>Aeronautics: Core Qualification: Elective Compulsory</p> <p>Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory</p> <p>Mechanical Engineering and Management: Specialisation Mechatronics: Elective Compulsory</p> <p>Mechatronics: Core Qualification: Compulsory</p> <p>Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory</p> <p>Biomedical Engineering: Specialisation Implants and Endoprostheses: Elective Compulsory</p> <p>Biomedical Engineering: Specialisation Medical Technology and Control Theory: Compulsory</p>                                                                                                                                                                                                                                                                                                               |        |    |

|  |                                                                                                                                                                                                                                                                    |
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|  | Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory<br>Product Development, Materials and Production: Core Qualification: Elective Compulsory<br>Theoretical Mechanical Engineering: Core Qualification: Compulsory |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Course L0656: Control Systems Theory and Design |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                      | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Hrs/wk</b>                                   | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>CP</b>                                       | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Workload in Hours</b>                        | Independent Study Time 92, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Lecturer</b>                                 | Prof. Timm Faulwasser                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Language</b>                                 | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Cycle</b>                                    | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Content</b>                                  | <p>State space methods (single-input single-output)</p> <ul style="list-style-type: none"> <li>• State space models and transfer functions, state feedback</li> <li>• Coordinate basis, similarity transformations</li> <li>• Solutions of state equations, matrix exponentials, Caley-Hamilton Theorem</li> <li>• Controllability and pole placement</li> <li>• State estimation, observability, Kalman decomposition</li> <li>• Observer-based state feedback control, reference tracking</li> <li>• Transmission zeros</li> <li>• Optimal pole placement, symmetric root locus</li> </ul> <p>Multi-input multi-output systems</p> <ul style="list-style-type: none"> <li>• Transfer function matrices, state space models of multivariable systems, Gilbert realization</li> <li>• Poles and zeros of multivariable systems, minimal realization</li> <li>• Closed-loop stability</li> <li>• Pole placement for multivariable systems, LQR design, Kalman filter</li> </ul> <p>Digital Control</p> <ul style="list-style-type: none"> <li>• Discrete-time systems: difference equations and z-transform</li> <li>• Discrete-time state space models, sampled data systems, poles and zeros</li> <li>• Frequency response of sampled data systems, choice of sampling rate</li> </ul> <p>System identification and model order reduction</p> <ul style="list-style-type: none"> <li>• Least squares estimation, ARX models, persistent excitation</li> <li>• Identification of state space models, subspace identification</li> <li>• Balanced realization and model order reduction</li> </ul> <p>Case study</p> <ul style="list-style-type: none"> <li>• Modelling and multivariable control of a process evaporator using Matlab and Simulink</li> </ul> <p>Software tools</p> <ul style="list-style-type: none"> <li>• Matlab/Simulink</li> </ul> |
| <b>Literature</b>                               | <ul style="list-style-type: none"> <li>• Werner, H., Lecture Notes „Control Systems Theory and Design“</li> <li>• T. Kailath "Linear Systems", Prentice Hall, 1980</li> <li>• K.J. Astrom, B. Wittenmark "Computer Controlled Systems" Prentice Hall, 1997</li> <li>• L. Ljung "System Identification - Theory for the User", Prentice Hall, 1999</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Course L0657: Control Systems Theory and Design |                                                     |
|-------------------------------------------------|-----------------------------------------------------|
| <b>Typ</b>                                      | Recitation Section (small)                          |
| <b>Hrs/wk</b>                                   | 2                                                   |
| <b>CP</b>                                       | 2                                                   |
| <b>Workload in Hours</b>                        | Independent Study Time 32, Study Time in Lecture 28 |
| <b>Lecturer</b>                                 | Prof. Timm Faulwasser                               |
| <b>Language</b>                                 | EN                                                  |
| <b>Cycle</b>                                    | WiSe                                                |
| <b>Content</b>                                  | See interlocking course                             |
| <b>Literature</b>                               | See interlocking course                             |

| Module M0746: Microsystem Engineering         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              |             |                    |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------|--------------------|
| Courses                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              |             |                    |
| Title                                         | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |              | Hrs/wk      | CP                 |
| Microsystem Engineering (L0680)               | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              | 2           | 4                  |
| Microsystem Engineering (L0682)               | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              | 2           | 2                  |
| <b>Module Responsible</b>                     | Dr. Timo Lipka                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              |             |                    |
| <b>Admission Requirements</b>                 | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              |             |                    |
| <b>Recommended Previous Knowledge</b>         | Basic courses in physics, mathematics and electric engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              |             |                    |
| <b>Educational Objectives</b>                 | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              |             |                    |
| <b>Professional Competence</b>                | <p><i>Knowledge</i> The students know about the most important technologies and materials of MEMS as well as their applications in sensors and actuators.</p> <p><i>Skills</i> Students are able to analyze and describe the functional behaviour of MEMS components and to evaluate the potential of microsystems.</p> <p><i>Personal Competence</i></p> <p><i>Social Competence</i> Students are able to solve specific problems alone or in a group and to present the results accordingly.</p> <p><i>Autonomy</i> Students are able to acquire particular knowledge using specialized literature and to integrate and associate this knowledge with other fields.</p>                                                             |              |             |                    |
|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              |             |                    |
|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              |             |                    |
|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              |             |                    |
| <b>Workload in Hours</b>                      | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              |             |                    |
| <b>Credit points</b>                          | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |             |                    |
| <b>Course achievement</b>                     | <b>Compulsory</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Bonus</b> | <b>Form</b> | <b>Description</b> |
|                                               | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | None         | Exercises   |                    |
| <b>Examination</b>                            | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |              |             |                    |
| <b>Examination duration and scale</b>         | 2h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |              |             |                    |
| <b>Assignment for the Following Curricula</b> | <p>Electrical Engineering and Information Technology: Core Qualification: Compulsory</p> <p>Electrical Engineering: Core Qualification: Compulsory</p> <p>International Management and Engineering: Specialisation II. Electrical Engineering: Elective Compulsory</p> <p>International Management and Engineering: Specialisation II. Mechatronics: Elective Compulsory</p> <p>Mechanical Engineering and Management: Specialisation Mechatronics: Elective Compulsory</p> <p>Mechatronics: Core Qualification: Elective Compulsory</p> <p>Microelectronics and Microsystems: Core Qualification: Elective Compulsory</p> <p>Theoretical Mechanical Engineering: Specialisation Bio- and Medical Technology: Elective Compulsory</p> |              |             |                    |

| Course L0680: Microsystem Engineering |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                            | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Hrs/wk</b>                         | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>CP</b>                             | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Workload in Hours</b>              | Independent Study Time 92, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Lecturer</b>                       | Dr. Timo Lipka                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Language</b>                       | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Cycle</b>                          | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Content</b>                        | <p>Object and goal of MEMS</p> <p>Scaling Rules</p> <p>Wafers and thin film deposition</p> <p>Lithography</p> <p>Structuring and etching</p> <p>Laser processing and bonding</p> <p>Energy conversion and force generation</p> <p>Electromagnetic actuators</p> <p>Electrostatic actuators, comb drives</p> <p>Reluctance motors</p> <p>Piezoelectric actuators, bi-metal actuator</p> <p>Transducer principles</p> <p>Signal detection and signal processing</p> <p>Mechanical and physical sensors based on beams and diaphragms</p> <p>Optical sensors and sensor arrays</p> <p>System integration aspects</p> <p>Yield, test and reliability</p> |
| <b>Literature</b>                     | <p>M. Kasper: Mikrosystementwurf, Springer (2000)</p> <p>M. Madou: Fundamentals of Microfabrication, CRC Press (1997)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Course L0682: Microsystem Engineering |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                            | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Hrs/wk</b>                         | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>CP</b>                             | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Workload in Hours</b>              | Independent Study Time 32, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Lecturer</b>                       | Dr. Timo Lipka                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Language</b>                       | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Cycle</b>                          | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Content</b>                        | <p>Scaling effects in microsystems</p> <p>Process flow charts</p> <p>Design aspects and layout consideration</p> <p>Examples of MEMS components</p> <p>Electric, thermal and mechanical behaviour</p> <p>Electrostatic actuators, comb drives</p> <p>Piezoelectric actuators, bi-metal actuator</p> <p>Signal detection and signal processing</p> <p>Mechanical and physical sensors based on beams and diaphragms</p> <p>Yield estimation and reliability issues</p> |
| <b>Literature</b>                     | Wird in der Veranstaltung bekannt gegeben                                                                                                                                                                                                                                                                                                                                                                                                                             |



## Module M0677: Digital Signal Processing and Digital Filters

### Courses

| Title                                                 | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Hrs/wk | CP |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|
| Digital Signal Processing and Digital Filters (L0446) | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3      | 4  |
| Digital Signal Processing and Digital Filters (L0447) | Recitation Section (large)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2      | 2  |
| Module Responsible                                    | Prof. Gerhard Bauch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |    |
| Admission Requirements                                | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| Recommended Previous Knowledge                        | <ul style="list-style-type: none"><li>Mathematics 1-3</li><li>Signals and Systems</li><li>Fundamentals of signal and system theory as well as random processes.</li><li>Fundamentals of spectral transforms (Fourier series, Fourier transform, Laplace transform)</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |    |
| Educational Objectives                                | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| Professional Competence                               | <p><i>Knowledge</i> The students know and understand basic algorithms of digital signal processing. They are familiar with the spectral transforms of discrete-time signals and are able to describe and analyse signals and systems in time and image domain. They know basic structures of digital filters and can identify and assess important properties including stability. They are aware of the effects caused by quantization of filter coefficients and signals. They are familiar with the basics of adaptive filters. They can perform traditional and parametric methods of spectrum estimation, also taking a limited observation window into account.</p> <p>The students are familiar with the contents of lecture and tutorials. They can explain and apply them to new problems.</p> <p><i>Skills</i> The students are able to apply methods of digital signal processing to new problems. They can choose and parameterize suitable filter structures. In particular, they can design adaptive filters according to the minimum mean squared error (MMSE) criterion and develop an efficient implementation, e.g. based on the LMS or RLS algorithm. Furthermore, the students are able to apply methods of spectrum estimation and to take the effects of a limited observation window into account.</p> <p><b>Personal Competence</b></p> <p><i>Social Competence</i> The students can jointly solve specific problems.</p> <p><i>Autonomy</i> The students are able to acquire relevant information from appropriate literature sources. They can control their level of knowledge during the lecture period by solving tutorial problems, software tools, clicker system.</p> |        |    |
|                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |    |
|                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |    |
|                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |    |
|                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |    |
| Workload in Hours                                     | Independent Study Time 110, Study Time in Lecture 70                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| Credit points                                         | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |    |
| Course achievement                                    | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| Examination                                           | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |    |
| Examination duration and scale                        | 90 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |    |
| Assignment for the Following Curricula                | <p>Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory</p> <p>Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory</p> <p>Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory</p> <p>Information and Communication Systems: Specialisation Communication Systems, Focus Signal Processing: Elective Compulsory</p> <p>Mechanical Engineering and Management: Specialisation Mechatronics: Elective Compulsory</p> <p>Mechatronics: Core Qualification: Elective Compulsory</p> <p>Microelectronics and Microsystems: Specialisation Communication and Signal Processing: Elective Compulsory</p> <p>Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |    |

| Course L0446: Digital Signal Processing and Digital Filters |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                                  | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Hrs/wk</b>                                               | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>CP</b>                                                   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Workload in Hours</b>                                    | Independent Study Time 78, Study Time in Lecture 42                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Lecturer</b>                                             | Prof. Gerhard Bauch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Language</b>                                             | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Cycle</b>                                                | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Content</b>                                              | <ul style="list-style-type: none"> <li>Transforms of discrete-time signals: <ul style="list-style-type: none"> <li>Discrete-time Fourier Transform (DTFT)</li> <li>Discrete Fourier-Transform (DFT), Fast Fourier Transform (FFT)</li> <li>Z-Transform</li> </ul> </li> <li>Correspondence of continuous-time and discrete-time signals, sampling, sampling theorem</li> <li>Fast convolution, Overlap-Add-Method, Overlap-Save-Method</li> <li>Fundamental structures and basic types of digital filters</li> <li>Characterization of digital filters using pole-zero plots, important properties of digital filters</li> <li>Quantization effects</li> <li>Design of linear-phase filters</li> <li>Fundamentals of stochastic signal processing and adaptive filters <ul style="list-style-type: none"> <li>MMSE criterion</li> <li>Wiener Filter</li> <li>LMS- and RLS-algorithm</li> </ul> </li> <li>Traditional and parametric methods of spectrum estimation</li> </ul> |
| <b>Literature</b>                                           | <p>K.-D. Kammeyer, K. Kroschel: Digitale Signalverarbeitung. Vieweg Teubner.</p> <p>V. Oppenheim, R. W. Schaffer, J. R. Buck: Zeitdiskrete Signalverarbeitung. Pearson StudiumA. V.</p> <p>W. Hesse: Digitale Filter. Teubner.</p> <p>Oppenheim, R. W. Schaffer: Digital signal processing. Prentice Hall.</p> <p>S. Haykin: Adaptive filter theory.</p> <p>L. B. Jackson: Digital filters and signal processing. Kluwer.</p> <p>T.W. Parks, C.S. Burrus: Digital filter design. Wiley.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Course L0447: Digital Signal Processing and Digital Filters |                                                     |
|-------------------------------------------------------------|-----------------------------------------------------|
| <b>Typ</b>                                                  | Recitation Section (large)                          |
| <b>Hrs/wk</b>                                               | 2                                                   |
| <b>CP</b>                                                   | 2                                                   |
| <b>Workload in Hours</b>                                    | Independent Study Time 32, Study Time in Lecture 28 |
| <b>Lecturer</b>                                             | Prof. Gerhard Bauch                                 |
| <b>Language</b>                                             | EN                                                  |
| <b>Cycle</b>                                                | WiSe                                                |
| <b>Content</b>                                              | See interlocking course                             |
| <b>Literature</b>                                           | See interlocking course                             |

| Module M0633: Industrial Process Automation            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |             |                    |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------|--------------------|
| Courses                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |             |                    |
| Title                                                  | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              | Hrs/wk      | CP                 |
| Industrial Process Automation (L0344)                  | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              | 2           | 3                  |
| Industrial Process Automation (L0345)                  | Recitation Section (small)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |              | 2           | 3                  |
| <b>Module Responsible</b>                              | Prof. Alexander Schlaefer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |              |             |                    |
| <b>Admission Requirements</b>                          | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              |             |                    |
| <b>Recommended Previous Knowledge</b>                  | mathematics and optimization methods<br>principles of automata<br>principles of algorithms and data structures<br>programming skills                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              |             |                    |
| <b>Educational Objectives</b>                          | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              |             |                    |
| <b>Professional Competence</b><br><i>Knowledge</i>     | The students can evaluate and assess discrete event systems. They can evaluate properties of processes and explain methods for process analysis. The students can compare methods for process modelling and select an appropriate method for actual problems. They can discuss scheduling methods in the context of actual problems and give a detailed explanation of advantages and disadvantages of different programming methods. The students can relate process automation to methods from robotics and sensor systems as well as to recent topics like 'cyberphysical systems' and 'industry 4.0'.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |              |             |                    |
| <b>Personal Competence</b><br><i>Skills</i>            | The students are able to develop and model processes and evaluate them accordingly. This involves taking into account optimal scheduling, understanding algorithmic complexity, and implementation using PLCs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |             |                    |
| <b>Personal Competence</b><br><i>Social Competence</i> | The students can independently define work processes within their groups, distribute tasks within the group and develop solutions collaboratively.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              |             |                    |
| <b>Personal Competence</b><br><i>Autonomy</i>          | The students are able to assess their level of knowledge and to document their work results adequately.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |             |                    |
| <b>Workload in Hours</b>                               | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              |             |                    |
| <b>Credit points</b>                                   | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |              |             |                    |
| <b>Course achievement</b>                              | <b>Compulsory</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Bonus</b> | <b>Form</b> | <b>Description</b> |
|                                                        | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 10 %         | Exercices   |                    |
| <b>Examination</b>                                     | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              |             |                    |
| <b>Examination duration and scale</b>                  | 90 minutes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |              |             |                    |
| <b>Assignment for the Following Curricula</b>          | Bioprocess Engineering: Specialisation A - General Bioprocess Engineering: Elective Compulsory<br>Chemical and Bioprocess Engineering: Specialisation General Process Engineering: Elective Compulsory<br>Chemical and Bioprocess Engineering: Specialisation Chemical Process Engineering: Elective Compulsory<br>Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory<br>Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory<br>Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory<br>Aircraft Systems Engineering: Core Qualification: Elective Compulsory<br>International Management and Engineering: Specialisation II. Mechatronics: Elective Compulsory<br>International Management and Engineering: Specialisation II. Product Development and Production: Elective Compulsory<br>Mechanical Engineering and Management: Specialisation Mechatronics: Elective Compulsory<br>Mechatronics: Core Qualification: Elective Compulsory<br>Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory<br>Process Engineering: Specialisation Chemical Process Engineering: Elective Compulsory<br>Process Engineering: Specialisation Process Engineering: Elective Compulsory |              |             |                    |

| Course L0344: Industrial Process Automation |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                  | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Hrs/wk</b>                               | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>CP</b>                                   | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Workload in Hours</b>                    | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Lecturer</b>                             | Prof. Alexander Schlaefer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Language</b>                             | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Cycle</b>                                | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Content</b>                              | <ul style="list-style-type: none"> <li>- foundations of problem solving and system modeling, discrete event systems</li> <li>- properties of processes, modeling using automata and Petri-nets</li> <li>- design considerations for processes (mutex, deadlock avoidance, liveness)</li> <li>- optimal scheduling for processes</li> <li>- optimal decisions when planning manufacturing systems, decisions under uncertainty</li> <li>- software design and software architectures for automation, PLCs</li> </ul> |
| <b>Literature</b>                           | <p>J. Lunze: „Automatisierungstechnik“, Oldenbourg Verlag, 2012</p> <p>Reisig: Petrinetze: Modellierungstechnik, Analysemethoden, Fallstudien; Vieweg+Teubner 2010</p> <p>Hrúz, Zhou: Modeling and Control of Discrete-event Dynamic Systems; Springer 2007</p> <p>Li, Zhou: Deadlock Resolution in Automated Manufacturing Systems, Springer 2009</p> <p>Pinedo: Planning and Scheduling in Manufacturing and Services, Springer 2009</p>                                                                          |

| Course L0345: Industrial Process Automation |                                                     |
|---------------------------------------------|-----------------------------------------------------|
| <b>Typ</b>                                  | Recitation Section (small)                          |
| <b>Hrs/wk</b>                               | 2                                                   |
| <b>CP</b>                                   | 3                                                   |
| <b>Workload in Hours</b>                    | Independent Study Time 62, Study Time in Lecture 28 |
| <b>Lecturer</b>                             | Prof. Alexander Schlaefer                           |
| <b>Language</b>                             | EN                                                  |
| <b>Cycle</b>                                | WiSe                                                |
| <b>Content</b>                              | See interlocking course                             |
| <b>Literature</b>                           | See interlocking course                             |

| Module M2162: Digital IC Design                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |        |    |
|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------|----|
| Courses                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |        |    |
| Title                                                                                                                                            | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  | Hrs/wk | CP |
| Digital IC Design (L3463)                                                                                                                        | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  | 3      | 4  |
| Digital IC Design (L3464)                                                                                                                        | Recitation Section (small)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  | 1      | 2  |
| <b>Module Responsible</b>                                                                                                                        | Prof. Qiang Li                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |        |    |
| <b>Admission Requirements</b>                                                                                                                    | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |        |    |
| <b>Recommended Previous Knowledge</b>                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |        |    |
| <b>Educational Objectives</b>                                                                                                                    | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |        |    |
| <b>Professional Competence</b><br><i>Knowledge</i><br><i>Skills</i><br><b>Personal Competence</b><br><i>Social Competence</i><br><i>Autonomy</i> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |        |    |
| <b>Workload in Hours</b>                                                                                                                         | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |        |    |
| <b>Credit points</b>                                                                                                                             | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |        |    |
| <b>Course achievement</b>                                                                                                                        | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |        |    |
| <b>Examination</b>                                                                                                                               | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |        |    |
| <b>Examination duration and scale</b>                                                                                                            | 90 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |        |    |
| <b>Assignment for the Following Curricula</b>                                                                                                    | Electrical Engineering and Information Technology: Specialisation Nanoelectronics and Microsystems Technology: Elective Compulsory<br>International Management and Engineering: Specialisation II. Electrical Engineering: Elective Compulsory<br>Mechanical Engineering and Management: Specialisation Mechatronics: Elective Compulsory<br>Microelectronics and Microsystems: Specialisation Embedded Systems: Elective Compulsory<br>Microelectronics and Microsystems: Specialisation Microelectronics Complements: Elective Compulsory |  |        |    |

| Course L3463: Digital IC Design |                                                     |
|---------------------------------|-----------------------------------------------------|
| <b>Typ</b>                      | Lecture                                             |
| <b>Hrs/wk</b>                   | 3                                                   |
| <b>CP</b>                       | 4                                                   |
| <b>Workload in Hours</b>        | Independent Study Time 78, Study Time in Lecture 42 |
| <b>Lecturer</b>                 | Prof. Qiang Li                                      |
| <b>Language</b>                 | EN                                                  |
| <b>Cycle</b>                    | WiSe                                                |
| <b>Content</b>                  |                                                     |
| <b>Literature</b>               |                                                     |

| Course L3464: Digital IC Design |                                                     |
|---------------------------------|-----------------------------------------------------|
| <b>Typ</b>                      | Recitation Section (small)                          |
| <b>Hrs/wk</b>                   | 1                                                   |
| <b>CP</b>                       | 2                                                   |
| <b>Workload in Hours</b>        | Independent Study Time 46, Study Time in Lecture 14 |
| <b>Lecturer</b>                 | Prof. Qiang Li                                      |
| <b>Language</b>                 | EN                                                  |
| <b>Cycle</b>                    | WiSe                                                |
| <b>Content</b>                  | See interlocking course                             |
| <b>Literature</b>               | See interlocking course                             |

| Module M2161: Analog IC Design         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |        |    |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------|----|
| Courses                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |        |    |
| Title                                  | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  | Hrs/wk | CP |
| Analog IC Design (L0691)               | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  | 3      | 4  |
| Analog IC Design (L0998)               | Recitation Section (small)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  | 1      | 2  |
| Module Responsible                     | Prof. Qiang Li                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |        |    |
| Admission Requirements                 | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |        |    |
| Recommended Previous Knowledge         | <p>Basic knowledge of (solid-state) physics and mathematics.</p> <p>Knowledge in fundamentals of electrical engineering and electrical networks.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |        |    |
| Educational Objectives                 | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |        |    |
| Professional Competence                | <ul style="list-style-type: none"> <li>Students can explain basic concepts of electron transport in semiconductor devices (energy bands, generation/recombination, carrier concentrations, drift and diffusion current densities, semiconductor device equations).</li> <li>Students are able to explain functional principles of pn-diodes, MOS capacitors, and MOSFETs using energy band diagrams.</li> <li>Students can present and discuss current-voltage relationships and small-signal equivalent circuits of these devices.</li> <li>Students can explain the physics and current-voltage behavior transistors based on charged carrier flow.</li> <li>Students are able to explain the basic concepts for static and dynamic logic gates for integrated circuits</li> <li>Students can exemplify approaches for low power consumption on the device and circuit level</li> <li>Students can describe the potential and limitations of analytical expression for device and circuit analysis.</li> <li>Students can explain characterization techniques for MOS devices.</li> </ul> |  |        |    |
| <i>Knowledge</i>                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |        |    |
| <i>Skills</i>                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |        |    |
| <i>Personal Competence</i>             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |        |    |
| <i>Social Competence</i>               | <ul style="list-style-type: none"> <li>Students can team up with other experts in the field to work out innovative solutions.</li> <li>Students are able to work by their own or in small groups for solving problems and answer scientific questions.</li> <li>Students have the ability to critically question the value of their contributions to working groups.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |        |    |
| <i>Autonomy</i>                        | <ul style="list-style-type: none"> <li>Students are able to assess their knowledge in a realistic manner.</li> <li>Students are able to define their personal approaches to solve challenging problems</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |        |    |
| Workload in Hours                      | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |        |    |
| Credit points                          | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |        |    |
| Course achievement                     | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |        |    |
| Examination                            | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |        |    |
| Examination duration and scale         | 90 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |        |    |
| Assignment for the Following Curricula | <p>Electrical Engineering and Information Technology: Specialisation Nanoelectronics and Microsystems Technology: Elective Compulsory</p> <p>International Management and Engineering: Specialisation II. Electrical Engineering: Elective Compulsory</p> <p>Mechanical Engineering and Management: Specialisation Mechatronics: Elective Compulsory</p> <p>Mechatronics: Core Qualification: Elective Compulsory</p> <p>Microelectronics and Microsystems: Core Qualification: Elective Compulsory</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |        |    |

| Course L0691: Analog IC Design |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                     | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Hrs/wk</b>                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>CP</b>                      | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Workload in Hours</b>       | Independent Study Time 78, Study Time in Lecture 42                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Lecturer</b>                | Prof. Qiang Li                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Language</b>                | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Cycle</b>                   | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Content</b>                 | <ul style="list-style-type: none"> <li>• Electron transport in semiconductors</li> <li>• Electronic operating principles of diodes, MOS capacitors, and MOS field-effect transistors</li> <li>• MOS transistor as four terminal device</li> <li>• Performance degradation due to short channel effects</li> <li>• Scaling-down of MOS technology</li> <li>• Digital logic circuits</li> <li>• Basic analog circuits</li> <li>• Operational amplifiers</li> <li>• Bipolar and BiCMOS circuits</li> </ul>                                                                                                            |
| <b>Literature</b>              | <ul style="list-style-type: none"> <li>• Yuan Taur, Tak H. Ning: Fundamentals of Modern VLSI Devices, Cambridge University Press 1998</li> <li>• R. Jacob Baker: CMOS, Circuit Design, Layout and Simulation, IEEE Press, Wiley Interscience, 3rd Edition, 2010</li> <li>• Neil H.E. Weste and David Money Harris, Integrated Circuit Design, Pearson, 4th International Edition, 2013</li> <li>• John E. Ayers, Digital Integrated Circuits: Analysis and Design, CRC Press, 2009</li> <li>• Richard C. Jaeger and Travis N. Blalock: Microelectronic Circuit Design, Mc Graw-Hill, 4rd. Edition, 2010</li> </ul> |

| Course L0998: Analog IC Design |                                                     |
|--------------------------------|-----------------------------------------------------|
| <b>Typ</b>                     | Recitation Section (small)                          |
| <b>Hrs/wk</b>                  | 1                                                   |
| <b>CP</b>                      | 2                                                   |
| <b>Workload in Hours</b>       | Independent Study Time 46, Study Time in Lecture 14 |
| <b>Lecturer</b>                | Prof. Qiang Li                                      |
| <b>Language</b>                | EN                                                  |
| <b>Cycle</b>                   | WiSe                                                |
| <b>Content</b>                 | See interlocking course                             |
| <b>Literature</b>              | See interlocking course                             |

## Specialization Product Engineering

In the specialization *Product Engineering*, graduates acquire in-depth knowledge of various production processes and materials. They are able to evaluate and select these processes and materials considering aspects such as geometry generation, defect control, and cost-effectiveness. Graduates are capable of developing products at the cutting edge of technology.

The specialization *Product Engineering* is recommended for students who, during their bachelor's studies, have gained foundational knowledge in the methodological design, calculation, and manufacturing of products.

### Module M1150: Continuum Mechanics

#### Courses

| Title                                                  | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Hrs/wk | CP |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|
| Continuum Mechanics (L1533)                            | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2      | 3  |
| Continuum Mechanics (Exercise) (L1534)                 | Recitation Section (small)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2      | 3  |
| <b>Module Responsible</b>                              | Prof. Christian Cyron                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |    |
| <b>Admission Requirements</b>                          | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Recommended Previous Knowledge</b>                  | Basics of mechanics as taught, e.g., in the modules Engineering Mechanics I and Engineering Mechanics II at TUHH (forces and moments, stress, linear strain, free-body principle, linear-elastic constitutive laws, strain energy); basics of mathematics as taught, e.g., in the modules Mathematics I and Mathematics II at TUHH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |    |
| <b>Educational Objectives</b>                          | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Professional Competence</b><br><i>Knowledge</i>     | In this module, students learn the fundamental concepts of nonlinear continuum mechanics. This theory enables students to describe arbitrary deformations of continuous bodies (solid, liquid or gaseous) under arbitrary loads. The module is a continuation of the basic module Engineering Mechanics II (elastostatics), the limiting assumptions (isotropic, linear-elastic material behavior, small deformations, simple geometries) of which are successively eliminated.<br><br>First, the students learn the necessary fundamentals of tensor calculus. Based on this, the description of the deformations / strains of arbitrarily deformable bodies is dealt with. The students learn the mathematical formalism for characterizing the stress state of a body and for formulating the balance equations for mass, momentum, energy and entropy in various forms. Furthermore, the students know which constitutive assumptions have to be made for modeling the material behavior of a mechanical body. |        |    |
| <i>Skills</i>                                          | The students can set up balance laws and apply basics of deformation theory to specific aspects, both in applied contexts as in research contexts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |    |
| <b>Personal Competence</b><br><i>Social Competence</i> | The students are able to develop solutions also for complex problems of solid mechanics, to present them to specialists in written form and to develop ideas further.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |    |
| <i>Autonomy</i>                                        | The students are able to assess their own strengths and weaknesses. They can independently and on their own identify and solve problems in the area of continuum mechanics and acquire the knowledge required to this end.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |    |
| <b>Workload in Hours</b>                               | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Credit points</b>                                   | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| <b>Course achievement</b>                              | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Examination</b>                                     | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |    |
| <b>Examination duration and scale</b>                  | 60 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| <b>Assignment for the Following Curricula</b>          | Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory<br>Materials Science and Engineering: Specialisation Modeling: Elective Compulsory<br>Materials Science: Specialisation Modeling: Elective Compulsory<br>Mechanical Engineering and Management: Specialisation Product Engineering: Elective Compulsory<br>Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory<br>Biomedical Engineering: Specialisation Implants and Endoprostheses: Elective Compulsory<br>Biomedical Engineering: Specialisation Medical Technology and Control Theory: Elective Compulsory<br>Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory<br>Product Development, Materials and Production: Core Qualification: Elective Compulsory<br>Theoretical Mechanical Engineering: Core Qualification: Elective Compulsory                                                             |        |    |



| Course L1533: Continuum Mechanics |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                        | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Hrs/wk</b>                     | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>CP</b>                         | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Workload in Hours</b>          | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Lecturer</b>                   | Prof. Christian Cyron                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Language</b>                   | DE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Cycle</b>                      | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Content</b>                    | <p>Continuum mechanics is a general theory to describe the effect of mechanical forces on continuous mechanical (both solid and fluid) bodies. An important part of continuum mechanics is the mathematical description of strains and stresses as well as the stress-strain response of continuous mechanical bodies. The lecture continuum mechanics builds on the foundations taught in the lecture Engineering Mechanics II (Elastostatics) but extends them significantly. While in the lecture Engineering Mechanics II (Elastostatics) the focus was by and large limited to small deformations of simple bodies under simple loading, the lecture continuum mechanics introduces a general mathematical framework to deal with arbitrarily shaped bodies under arbitrary loading undergoing very general kinds of deformations. This lecture focuses primarily on theoretical aspects of continuum mechanics but its content is key to numerous applications in modern engineering, for example, in production, automotive, and biomedical engineering. The lecture covers:</p> <ul style="list-style-type: none"> <li>• Fundamentals of tensor calculus <ul style="list-style-type: none"> <li>◦ Transformation invariance</li> <li>◦ Tensor algebra</li> <li>◦ Tensor analysis</li> </ul> </li> <li>• Kinematics <ul style="list-style-type: none"> <li>◦ Motion of continuum</li> <li>◦ Deformation of infinitesimal line, area and volume elements</li> <li>◦ Material and spatial description</li> <li>◦ Polar decomposition</li> <li>◦ Spectral decomposition</li> <li>◦ Objectivity</li> <li>◦ Strain measures</li> <li>◦ Time derivatives <ul style="list-style-type: none"> <li>▪ Partial / material time derivatives</li> <li>▪ Objective time rates</li> <li>▪ Strain and deformation rates</li> </ul> </li> <li>◦ Transport theorems</li> </ul> </li> <li>• Balance equations (global and local form) <ul style="list-style-type: none"> <li>◦ Balance of mass</li> <li>◦ The stress state <ul style="list-style-type: none"> <li>▪ Surface traction vectors</li> <li>▪ Cauchy's fundamental theorem</li> <li>▪ Stress tensors (Cauchy, 1. and 2. Piola-Kirchhoff, Kirchhoff stress tensor)</li> </ul> </li> <li>◦ Balance of linear momentum</li> <li>◦ Balance of angular momentum</li> <li>◦ Balance of energy</li> <li>◦ Balance of entropy</li> <li>◦ Clausius-Duhem inequality</li> </ul> </li> <li>• Constitutive laws <ul style="list-style-type: none"> <li>◦ Constitutive assumptions</li> <li>◦ Fluids</li> <li>◦ Elastic solids <ul style="list-style-type: none"> <li>▪ Hyperelasticity</li> <li>▪ Material symmetry</li> </ul> </li> <li>◦ Elasto-plastic solids</li> </ul> </li> <li>• Analysis <ul style="list-style-type: none"> <li>◦ Initial-boundary value problems and their numerical solution</li> </ul> </li> </ul> |
| <b>Literature</b>                 | <p>R. Greve: Kontinuumsmechanik: Ein Grundkurs für Ingenieure und Physiker</p> <p>I-S. Liu: Continuum Mechanics, Springer</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Course L1534: Continuum Mechanics (Exercise) |                                                                                                                                                             |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                   | Recitation Section (small)                                                                                                                                  |
| <b>Hrs/wk</b>                                | 2                                                                                                                                                           |
| <b>CP</b>                                    | 3                                                                                                                                                           |
| <b>Workload in Hours</b>                     | Independent Study Time 62, Study Time in Lecture 28                                                                                                         |
| <b>Lecturer</b>                              | Prof. Christian Cyron, Kian Philipp Abdolazizi                                                                                                              |
| <b>Language</b>                              | DE                                                                                                                                                          |
| <b>Cycle</b>                                 | WiSe                                                                                                                                                        |
| <b>Content</b>                               | The exercise on Continuum Mechanics explains the theoretical content of the lecture on Continuum Mechanics by way of a series of specific example problems. |
| <b>Literature</b>                            | R. Greve: Kontinuumsmechanik: Ein Grundkurs für Ingenieure und Physiker<br><br>I-S. Liu: Continuum Mechanics, Springer                                      |

| Module M1950: Advanced Functional Materials   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |    |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|
| Courses                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |    |
| Title                                         | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Hrs/wk | CP |
| Advanced Functional Materials (L1625)         | Seminar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2      | 6  |
| <b>Module Responsible</b>                     | Prof. Patrick Huber                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |    |
| <b>Admission Requirements</b>                 | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |    |
| <b>Recommended Previous Knowledge</b>         | Basic knowledge in Materials Science, e.g. Materials Science I/II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |    |
| <b>Educational Objectives</b>                 | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |    |
| <b>Professional Competence</b>                | <p><i>Knowledge</i> The students will be able to explain the properties of advanced materials along with their applications in technology, in particular metallic, ceramic, polymeric, semiconductor, modern composite materials (biomaterials) and nanomaterials.</p> <p><i>Skills</i> The students will be able to select material configurations according to the technical needs and, if necessary, to design new materials considering architectural principles from the micro- to the macroscale. The students will also gain an overview on modern materials science, which enables them to select optimum materials combinations depending on the technical applications.</p> <p><b>Personal Competence</b></p> <p><i>Social Competence</i> The students are able to present solutions to specialists and to develop ideas further.</p> <p><i>Autonomy</i> The students are able to ...</p> <ul style="list-style-type: none"> <li>• assess their own strengths and weaknesses.</li> <li>• gather new necessary expertise by their own.</li> </ul> |        |    |
|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |    |
|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |    |
|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |    |
| <b>Workload in Hours</b>                      | Independent Study Time 152, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |    |
| <b>Credit points</b>                          | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |    |
| <b>Course achievement</b>                     | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |    |
| <b>Examination</b>                            | Presentation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Examination duration and scale</b>         | 30 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |    |
| <b>Assignment for the Following Curricula</b> | Materials Science and Engineering: Core Qualification: Compulsory<br>Mechanical Engineering and Management: Specialisation Product Engineering: Elective Compulsory<br>Biomedical Engineering: Specialisation Implants and Endoprostheses: Elective Compulsory<br>Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory<br>Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory<br>Biomedical Engineering: Specialisation Medical Technology and Control Theory: Elective Compulsory<br>Theoretical Mechanical Engineering: Specialisation Materials Science: Elective Compulsory                                                                                                                                                                                                                                                                                                                                                                          |        |    |

| Course L1625: Advanced Functional Materials |                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                  | Seminar                                                                                                                                                                                                                                                                                                               |
| <b>Hrs/wk</b>                               | 2                                                                                                                                                                                                                                                                                                                     |
| <b>CP</b>                                   | 6                                                                                                                                                                                                                                                                                                                     |
| <b>Workload in Hours</b>                    | Independent Study Time 152, Study Time in Lecture 28                                                                                                                                                                                                                                                                  |
| <b>Lecturer</b>                             | Prof. Patrick Huber, Prof. Bodo Fiedler, Prof. Gerold Schneider, Prof. Jörg Weißmüller, Prof. Kaline Pagnan Furlan, Prof. Robert Meißner                                                                                                                                                                              |
| <b>Language</b>                             | DE/EN                                                                                                                                                                                                                                                                                                                 |
| <b>Cycle</b>                                | WiSe                                                                                                                                                                                                                                                                                                                  |
| <b>Content</b>                              | 1. Porous Solids - Preparation, Characterization and Functionalities<br>2. Fluidics with nanoporous membranes<br>3. Thermoplastic elastomers<br>4. Optimization of polymer properties by nanoparticles<br>5. Fiber composites in automotive<br>6. Modeling of materials based on quantum mechanics<br>7. Biomaterials |
| <b>Literature</b>                           | Aktuelle Publikationen aus der Fachliteratur werden während der Veranstaltung bekanntgegeben.                                                                                                                                                                                                                         |

## Module M0604: High-Order FEM

### Courses

| Title                                                                                                                                                        | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Hrs/wk       | CP           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|
| High-Order FEM (L0280)                                                                                                                                       | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3            | 4            |
| High-Order FEM (L0281)                                                                                                                                       | Recitation Section (large)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1            | 2            |
| <b>Module Responsible</b>                                                                                                                                    | Prof. Alexander Düster                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |              |
| <b>Admission Requirements</b>                                                                                                                                | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |              |
| <b>Recommended Previous Knowledge</b>                                                                                                                        | Knowledge of partial differential equations is recommended.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |              |              |
| <b>Educational Objectives</b>                                                                                                                                | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |              |
| <b>Professional Competence</b><br><i>Knowledge</i><br><br><i>Skills</i><br><br><b>Personal Competence</b><br><i>Social Competence</i><br><br><i>Autonomy</i> | <p>Students are able to</p> <ul style="list-style-type: none"> <li>+ give an overview of the different (h, p, hp) finite element procedures.</li> <li>+ explain high-order finite element procedures.</li> <li>+ specify problems of finite element procedures, to identify them in a given situation and to explain their mathematical and mechanical background.</li> </ul> <p>Students are able to</p> <ul style="list-style-type: none"> <li>+ apply high-order finite elements to problems of structural mechanics.</li> <li>+ select for a given problem of structural mechanics a suitable finite element procedure.</li> <li>+ critically judge results of high-order finite elements.</li> <li>+ transfer their knowledge of high-order finite elements to new problems.</li> </ul> <p>Students are able to</p> <ul style="list-style-type: none"> <li>+ solve problems in heterogeneous groups.</li> <li>+ present and discuss their results in front of others.</li> <li>+ give and accept professional constructive criticism.</li> </ul> <p>Students are able to</p> <ul style="list-style-type: none"> <li>+ assess their knowledge by means of exercises and E-Learning.</li> <li>+ acquaint themselves with the necessary knowledge to solve research oriented tasks.</li> <li>+ to transform the acquired knowledge to similar problems.</li> </ul> |              |              |
| <b>Workload in Hours</b>                                                                                                                                     | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |              |
| <b>Credit points</b>                                                                                                                                         | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |              |              |
| <b>Course achievement</b>                                                                                                                                    | <b>Compulsory</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Bonus</b> | <b>Form</b>  |
|                                                                                                                                                              | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 10 %         | Presentation |
| <b>Examination</b>                                                                                                                                           | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |              |              |
| <b>Examination duration and scale</b>                                                                                                                        | 120 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |              |              |
| <b>Assignment for the Following Curricula</b>                                                                                                                | Civil Engineering: Specialisation Computational Engineering: Elective Compulsory<br>Computational Engineering: Core Qualification: Elective Compulsory<br>International Management and Engineering: Specialisation II. Product Development and Production: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory<br>Materials Science and Engineering: Specialisation Modeling: Elective Compulsory<br>Mechanical Engineering and Management: Specialisation Product Engineering: Elective Compulsory<br>Mechatronics: Technical Complementary Course: Elective Compulsory<br>Product Development, Materials and Production: Core Qualification: Elective Compulsory<br>Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory<br>Naval Architecture and Ocean Engineering: Core Qualification: Elective Compulsory<br>Technomathematics: Specialisation III. Engineering Science: Elective Compulsory<br>Theoretical Mechanical Engineering: Core Qualification: Elective Compulsory                                                                                                                                                                                                                                                                               |              |              |

| Course L0280: High-Order FEM |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                   | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Hrs/wk</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>CP</b>                    | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Workload in Hours</b>     | Independent Study Time 78, Study Time in Lecture 42                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Lecturer</b>              | Prof. Alexander Düster                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Language</b>              | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Cycle</b>                 | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Content</b>               | <ol style="list-style-type: none"> <li>1. Introduction</li> <li>2. Motivation</li> <li>3. Hierarchic shape functions</li> <li>4. Mapping functions</li> <li>5. Computation of element matrices, assembly, constraint enforcement and solution</li> <li>6. Convergence characteristics</li> <li>7. Mechanical models and finite elements for thin-walled structures</li> <li>8. Computation of thin-walled structures</li> <li>9. Error estimation and hp-adaptivity</li> <li>10. High-order fictitious domain methods</li> </ol> |
| <b>Literature</b>            | <p>[1] Alexander Düster, High-Order FEM, Lecture Notes, Technische Universität Hamburg-Harburg, 164 pages, 2014</p> <p>[2] Barna Szabo, Ivo Babuska, Introduction to Finite Element Analysis – Formulation, Verification and Validation, John Wiley &amp; Sons, 2011</p>                                                                                                                                                                                                                                                         |

| Course L0281: High-Order FEM |                                                     |
|------------------------------|-----------------------------------------------------|
| <b>Typ</b>                   | Recitation Section (large)                          |
| <b>Hrs/wk</b>                | 1                                                   |
| <b>CP</b>                    | 2                                                   |
| <b>Workload in Hours</b>     | Independent Study Time 46, Study Time in Lecture 14 |
| <b>Lecturer</b>              | Prof. Alexander Düster                              |
| <b>Language</b>              | EN                                                  |
| <b>Cycle</b>                 | SoSe                                                |
| <b>Content</b>               | See interlocking course                             |
| <b>Literature</b>            | See interlocking course                             |

## Module M1256: AdditiveProduction

### Courses

| Title                                                  | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Hrs/wk | CP |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|
| Additive Production (L1128)                            | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2      | 3  |
| Additive Production (L1129)                            | Seminar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2      | 3  |
| <b>Module Responsible</b>                              | Prof. Claus Emmelmann                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |    |
| <b>Admission Requirements</b>                          | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| <b>Recommended Previous Knowledge</b>                  | <ul style="list-style-type: none"> <li>• Production Engineering</li> <li>• Fundamental of Material Science</li> <li>• Fundamentals of Mechanical Engineering Design</li> </ul>                                                                                                                                                                                                                                                                                                                                                   |        |    |
| <b>Educational Objectives</b>                          | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| <b>Professional Competence</b><br><i>Knowledge</i>     | Students will be able to: <ul style="list-style-type: none"> <li>• give an overview of Additive Manufacturing Technologies, namely</li> <li>• describe basics of Laser Technologies</li> <li>• discuss laser Additive Manufacturing, specifically</li> <li>• design Guidelines for Additive Manufacturing</li> <li>• describe the Digital Process Chain for Additive Manufacturing</li> <li>• discuss Quality Assurance for Additive Manufacturing</li> <li>• describe Product Development for Additive Manufacturing</li> </ul> |        |    |
| <i>Skills</i>                                          | The students will be able to: <ul style="list-style-type: none"> <li>• give an overview of Potential and Challenges of Additive Manufacturing Technologies</li> <li>• show that Additive Manufacturing offers new possibilities for product development</li> <li>• show major differences between Additive Manufacturing and conventional manufacturing technologies</li> <li>• apply basic skills to develop and design Additive Manufacturing parts</li> <li>• design and build own Additive Manufacturing parts</li> </ul>    |        |    |
| <b>Personal Competence</b><br><i>Social Competence</i> | Students are able to <ul style="list-style-type: none"> <li>• interact within a team</li> <li>• organize workload in a team</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                           |        |    |
| <i>Autonomy</i>                                        | Students are able to <ul style="list-style-type: none"> <li>• develop and optimize a product with limited resources, based on defined requirements</li> <li>• present results skillfully</li> </ul>                                                                                                                                                                                                                                                                                                                              |        |    |
| <b>Workload in Hours</b>                               | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| <b>Credit points</b>                                   | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |    |
| <b>Course achievement</b>                              | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| <b>Examination</b>                                     | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |    |
| <b>Examination duration and scale</b>                  | 75 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |    |
| <b>Assignment for the Following Curricula</b>          | Mechanical Engineering and Management: Specialisation Product Engineering: Elective Compulsory                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |    |

### Course L1128: Additive Production

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>               | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Hrs/wk</b>            | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>CP</b>                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Workload in Hours</b> | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Lecturer</b>          | Prof. Claus Emmelmann, Dr. Dirk Herzog                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Language</b>          | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Cycle</b>             | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Content</b>           | Learn the Basics of Additive Manufacturing, with focus on the Selective Laser Melting and Selective Laser Sintering. Understand the advantages the technologies offer for product development and what current challenges Additive Manufacturing faces. Get to know the design restrictions as well as basic knowledge about material characteristics, post processing and quality assurance.<br><br><b>This lecture is part of the Module Rapid Production and cannot be chosen separately</b> |
| <b>Literature</b>        | Will be announced during the course                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Course L1129: Additive Production |                                                                                                                                                                                                                                                                 |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                        | Seminar                                                                                                                                                                                                                                                         |
| <b>Hrs/wk</b>                     | 2                                                                                                                                                                                                                                                               |
| <b>CP</b>                         | 3                                                                                                                                                                                                                                                               |
| <b>Workload in Hours</b>          | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                             |
| <b>Lecturer</b>                   | Prof. Claus Emmelmann, Dr. Dirk Herzog                                                                                                                                                                                                                          |
| <b>Language</b>                   | EN                                                                                                                                                                                                                                                              |
| <b>Cycle</b>                      | SoSe                                                                                                                                                                                                                                                            |
| <b>Content</b>                    | <p>Intensify learning from the lecture, especially regarding design principles and product development by design of own Selective Laser Sintering parts.</p> <p><b>This seminar is part of the Module Rapid Production and cannot be chosen separately.</b></p> |
| <b>Literature</b>                 | Will be announced during the course                                                                                                                                                                                                                             |

## Module M1143: Applied Design Methodology in Mechatronics

### Courses

| Title                                              | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Hrs/wk | CP |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|
| Applied Design Methodology in Mechatronics (L1523) | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2      | 2  |
| Applied Design Methodology in Mechatronics (L1524) | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3      | 4  |
| <b>Module Responsible</b>                          | Prof. Thorsten Kern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| <b>Admission Requirements</b>                      | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |    |
| <b>Recommended Previous Knowledge</b>              | Basics of mechanical design, electrical design or computer-sciences                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| <b>Educational Objectives</b>                      | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |    |
| <b>Professional Competence</b>                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |    |
| <i>Knowledge</i>                                   | Science-based working on interdisciplinary product design considering targeted application of specific product design techniques                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |    |
| <i>Skills</i>                                      | Creative handling of processes used for scientific preparation and formulation of complex product design problems / Application of various product design techniques following theoretical aspects.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| <b>Personal Competence</b>                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |    |
| <i>Social Competence</i>                           | Students will solve and execute technical-scientific tasks from an industrial context in small design-teams with application of common, creative methodologies.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |    |
| <i>Autonomy</i>                                    | Students are enabled to optimize the design and development process according to the target and topic of the design                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
|                                                    | Students are educated to operate in a development team                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |    |
|                                                    | Students learn about the right application of creative methods in engineering.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| <b>Workload in Hours</b>                           | Independent Study Time 110, Study Time in Lecture 70                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |    |
| <b>Credit points</b>                               | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Course achievement</b>                          | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |    |
| <b>Examination</b>                                 | Subject theoretical and practical work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |    |
| <b>Examination duration and scale</b>              | 30 min Presentation for a group design-work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |    |
| <b>Assignment for the Following Curricula</b>      | International Management and Engineering: Specialisation II. Product Development and Production: Elective Compulsory<br>International Management and Engineering: Specialisation II. Mechatronics: Elective Compulsory<br>Mechanical Engineering and Management: Specialisation Product Engineering: Elective Compulsory<br>Mechatronics: Core Qualification: Elective Compulsory<br>Biomedical Engineering: Specialisation Implants and Endoprotheses: Elective Compulsory<br>Biomedical Engineering: Specialisation Medical Technology and Control Theory: Elective Compulsory<br>Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory<br>Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory<br>Theoretical Mechanical Engineering: Specialisation Product Development and Production: Elective Compulsory |        |    |



| Course L1523: Applied Design Methodology in Mechatronics |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                               | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Hrs/wk</b>                                            | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>CP</b>                                                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Workload in Hours</b>                                 | Independent Study Time 32, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Lecturer</b>                                          | Prof. Thorsten Kern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Language</b>                                          | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Cycle</b>                                             | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Content</b>                                           | <ul style="list-style-type: none"> <li>• Systematic analysis and planning of the design process for products combining a multitude of disciplines</li> <li>• Structure of the engineering process with focus on engineering steps (task-definition, functional decomposition, physical principles, elements for solution, combination to systems and products, execution of design, component-tests, system-tests, product-testing and qualification/validation)</li> <li>• Creative methods (Basics, methods like lead-user-method, 6-3-5, BrainStorming, Intergalactic Thinking, ... - Applications in examples all around mechatronics topics)</li> <li>• Several design-supporting methods and tools (functional structures, GALFMOS, AEIOU-method, GAMPFT, simulation and its application, TRIZ, design for SixSigma, continuous integration and testing, ...)</li> <li>• Evaluation and final selection of solution (technical and business-considerations, preference-matrix, pair-comparison), dealing with uncertainties, decision-making</li> <li>• Value-analysis</li> <li>• Derivation of architectures and architectural management</li> <li>• Project-tracking and -guidance (project-lead, guiding of employees, organization of multidisciplinary R&amp;D departments, idea-identification, responsibilities and communication)</li> <li>• Project-execution methods (Scrum, Kanban, ...)</li> <li>• Presentation-skills</li> <li>• Questions of aesthetic product design and design for subjective requirements (industrial design, color, haptic/optic/acoustic interfaces)</li> <li>• Evaluation of selected methods at practical examples in small teams</li> </ul> |
| <b>Literature</b>                                        | <ul style="list-style-type: none"> <li>• Definition folgt...</li> <li>• Pahl, G.; Beitz, W.; Feldhusen, J.; Grote, K.-H.: Konstruktionslehre: Grundlage erfolgreicher Produktentwicklung, Methoden und Anwendung, 7. Auflage, Springer Verlag, Berlin 2007</li> <li>• VDI-Richtlinien: 2206; 2221ff</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Course L1524: Applied Design Methodology in Mechatronics |                                                     |
|----------------------------------------------------------|-----------------------------------------------------|
| <b>Typ</b>                                               | Project-/problem-based Learning                     |
| <b>Hrs/wk</b>                                            | 3                                                   |
| <b>CP</b>                                                | 4                                                   |
| <b>Workload in Hours</b>                                 | Independent Study Time 78, Study Time in Lecture 42 |
| <b>Lecturer</b>                                          | Prof. Thorsten Kern                                 |
| <b>Language</b>                                          | EN                                                  |
| <b>Cycle</b>                                             | SoSe                                                |
| <b>Content</b>                                           | See interlocking course                             |
| <b>Literature</b>                                        | See interlocking course                             |

| Module M1220: Interfaces and interface-dominated Materials |                                                                                                                                                                                                                                                                                                                                                                                                                |               |           |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------|
| <b>Courses</b>                                             |                                                                                                                                                                                                                                                                                                                                                                                                                |               |           |
| <b>Title</b>                                               | <b>Typ</b>                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Hrs/wk</b> | <b>CP</b> |
| Nature's Hierarchical Materials (L1663)                    | Seminar                                                                                                                                                                                                                                                                                                                                                                                                        | 2             | 3         |
| Interfaces (L1654)                                         | Lecture                                                                                                                                                                                                                                                                                                                                                                                                        | 2             | 3         |
| <b>Module Responsible</b>                                  | Prof. Patrick Huber                                                                                                                                                                                                                                                                                                                                                                                            |               |           |
| <b>Admission Requirements</b>                              | None                                                                                                                                                                                                                                                                                                                                                                                                           |               |           |
| <b>Recommended Previous Knowledge</b>                      | Basic knowledge in Materials Science, e.g. Materials Science I/II, and physical chemistry                                                                                                                                                                                                                                                                                                                      |               |           |
| <b>Educational Objectives</b>                              | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                           |               |           |
| <b>Professional Competence</b>                             |                                                                                                                                                                                                                                                                                                                                                                                                                |               |           |
| <i>Knowledge</i>                                           | The students will be able to explain the structural and thermodynamic properties of interfaces in comparison to the bulk systems. They will be able to describe the relevance of interfaces and physico-chemical modifications of interfaces. Moreover, they are able to outline the characteristics of biomaterials and to relate them to classical materials systems, such as metals, ceramics and polymers. |               |           |
| <i>Skills</i>                                              | The students are able to rationalize the impact of interfaces on material properties and functionalities. Moreover, they are able to trace the peculiar properties of biomaterials to their hierarchical hybrid structure.                                                                                                                                                                                     |               |           |
| <b>Personal Competence</b>                                 |                                                                                                                                                                                                                                                                                                                                                                                                                |               |           |
| <i>Social Competence</i>                                   | The students are able to present solutions to specialists and to develop ideas further.                                                                                                                                                                                                                                                                                                                        |               |           |
| <i>Autonomy</i>                                            | The students are able to ... <ul style="list-style-type: none"> <li>• assess their own strengths and weaknesses.</li> <li>• define tasks independently.</li> </ul>                                                                                                                                                                                                                                             |               |           |
| <b>Workload in Hours</b>                                   | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                           |               |           |
| <b>Credit points</b>                                       | 6                                                                                                                                                                                                                                                                                                                                                                                                              |               |           |
| <b>Course achievement</b>                                  | None                                                                                                                                                                                                                                                                                                                                                                                                           |               |           |
| <b>Examination</b>                                         | Written exam                                                                                                                                                                                                                                                                                                                                                                                                   |               |           |
| <b>Examination duration and scale</b>                      | 90 min                                                                                                                                                                                                                                                                                                                                                                                                         |               |           |
| <b>Assignment for the Following Curricula</b>              | Materials Science and Engineering: Specialisation Nano and Hybrid Materials: Elective Compulsory<br>Mechanical Engineering and Management: Specialisation Product Engineering: Elective Compulsory                                                                                                                                                                                                             |               |           |

| Course L1663: Nature's Hierarchical Materials |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                    | Seminar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Hrs/wk</b>                                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>CP</b>                                     | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Workload in Hours</b>                      | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Lecturer</b>                               | NN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Language</b>                               | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Cycle</b>                                  | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Content</b>                                | <p>Nature hierarchical materials are omnipresent in the world around us, being fundamental for both the plants and animal kingdoms. Nature engineering is quite impressive since nature uses only a small variety of building blocks (minerals, proteins and sugar) and yet, is capable of producing an incredible large number of structures with different functions, i.e. multifunctional materials. This is one of the reasons why materials science and engineering research on bioinspiration or biomimicry has been increasing significantly over the past 20 years. Moreover, the scientists and engineers have one major advantage over nature: they can combine bioinspiration with a wide variety of other building blocks (metals, ceramics, polymers and derived composites). The main goal of this seminar series is to provide an introduction about the state of the art on bioinspired materials from an engineering point of view, while providing students opportunities to develop skills relevant to their master thesis work such as proper literature search, systematic literature review, presentation preparation and presentation - all connected to the main topic of Nature hierarchical materials.</p> <p>.</p> |
| <b>Literature</b>                             | <p>Peter Fratzl and Richard Weinkamer. Nature's hierarchical materials. Progress in Materials Science 52 (2007) 1263-1334</p> <p>Journal publications</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Course L1654: Interfaces |                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>               | Lecture                                                                                                                                                                                                                                                                                                                                                              |
| <b>Hrs/wk</b>            | 2                                                                                                                                                                                                                                                                                                                                                                    |
| <b>CP</b>                | 3                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Workload in Hours</b> | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                  |
| <b>Lecturer</b>          | Prof. Patrick Huber                                                                                                                                                                                                                                                                                                                                                  |
| <b>Language</b>          | DE                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Cycle</b>             | SoSe                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Content</b>           | <ul style="list-style-type: none"> <li>• Microscopic structure and thermodynamics of interfaces (gas/solid, gas/liquid, liquid/liquid, liquid/solid)</li> <li>• Experimental methods for the study of interfaces</li> <li>• Interfacial forces</li> <li>• wetting</li> <li>• surfactants, foams, bio-membranes</li> <li>• chemical grafting of interfaces</li> </ul> |
| <b>Literature</b>        | <p>"Physics and Chemistry of Interfaces", K.H. Butt, K. Graf, M. Kappl, Wiley-VCH Weinheim (2006)</p> <p>"Interfacial Science", G.T. Barnes, I.R. Gentle, Oxford University Press (2005)</p>                                                                                                                                                                         |

| Module M1344: Processing of Fibre-Polymer-Composites |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |    |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|
| Courses                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |    |
| Title                                                | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Hrs/wk | CP |
| Processing of fibre-polymer-composites (L1895)       | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2      | 3  |
| From Molecule to Composites Part (L1516)             | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2      | 3  |
| Module Responsible                                   | Prof. Bodo Fiedler                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| Admission Requirements                               | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |    |
| Recommended Previous Knowledge                       | Knowledge in the basics of chemistry / physics / materials science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| Educational Objectives                               | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |    |
| Professional Competence                              | <p><i>Knowledge</i> Students are able to give a summary of the technical details of the manufacturing processes composites and illustrate respective relationships. They are capable of describing and communicating relevant problems and questions using appropriate technical language. They can explain the typical process of solving practical problems and present related results.</p> <p><i>Skills</i> Students can use the knowledge of fiber-reinforced composites (FRP) and its constituents (fiber / matrix) and define the necessary testing and analysis.</p> <p>They can explain the complex structure-property relationship and</p> <p>the interactions of chemical structure of the polymers, their processing with the different fiber types, including to explain neighboring contexts (e.g. sustainability, environmental protection).</p>                                                                                                                                                                                                                                                |        |    |
| Personal Competence                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |    |
| <i>Social Competence</i>                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |    |
| <i>Autonomy</i>                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |    |
| Workload in Hours                                    | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |    |
| Credit points                                        | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |    |
| Course achievement                                   | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |    |
| Examination                                          | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |    |
| Examination duration and scale                       | 90 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |    |
| Assignment for the Following Curricula               | Aeronautics: Core Qualification: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory<br>Materials Science and Engineering: Specialisation Engineering Materials: Elective Compulsory<br>Biomedical Engineering and Management: Specialisation Product Engineering: Elective Compulsory<br>Biomedical Engineering: Specialisation Implants and Endoprostheses: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Production: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Materials: Elective Compulsory<br>Theoretical Mechanical Engineering: Specialisation Materials Science: Elective Compulsory |        |    |

| Course L1895: Processing of fibre-polymer-composites |                                                                                                      |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                           | Lecture                                                                                              |
| <b>Hrs/wk</b>                                        | 2                                                                                                    |
| <b>CP</b>                                            | 3                                                                                                    |
| <b>Workload in Hours</b>                             | Independent Study Time 62, Study Time in Lecture 28                                                  |
| <b>Lecturer</b>                                      | Prof. Bodo Fiedler                                                                                   |
| <b>Language</b>                                      | DE/EN                                                                                                |
| <b>Cycle</b>                                         | SoSe                                                                                                 |
| <b>Content</b>                                       | Manufacturing of Composites: Hand Lay-Up; Pre-Preg; GMT, BMC; SMC, RIM; Pultrusion; Filament Winding |
| <b>Literature</b>                                    | Åström: Manufacturing of Polymer Composites, Chapman and Hall                                        |

| Course L1516: From Molecule to Composites Part |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                     | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Hrs/wk</b>                                  | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>CP</b>                                      | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Workload in Hours</b>                       | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Lecturer</b>                                | Prof. Bodo Fiedler                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Language</b>                                | DE/EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Cycle</b>                                   | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Content</b>                                 | <p>Students get the task in the form of a customer request for the development and production of a MTB handlebar made of fiber composites. In the task technical and normative requirements (standards) are given, all other required information come from the lectures and tutorials, and the respective documents (electronically and in conversation).</p> <p>The procedure is to specify in a milestone schedule and allows students to plan tasks and to work continuously. At project end, each group has a made handlebar with approved quality.</p> <p>In each project meeting the design (discussion of the requirements and risks) are discussed. The calculations are analyzed, evaluated and established manufacturing methods are selected. Materials are selected bar will be produced. The quality and the mechanical properties are checked. At the end of the final report created (compilation of the results for the "customers").</p> <p>After the test during the "customer / supplier conversation" there is a mutual feedback-talk ("lessons learned") in order to ensure the continuous improvement.</p> |
| <b>Literature</b>                              | Åström: Manufacturing of Polymer Composites, Chapman and Hall                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Module M1948: Mechanical Properties               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |    |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|
| Courses                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |    |
| Title                                             | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Hrs/wk | CP |
| Mechanical Behaviour of Brittle Materials (L1661) | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2      | 2  |
| Dislocation Theory of Plasticity (L1662)          | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2      | 2  |
| Exercise on mechanical properties (HÜ) (L3352)    | Recitation Section (large)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2      | 2  |
| <b>Module Responsible</b>                         | Prof. Shan Shi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |    |
| <b>Admission Requirements</b>                     | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| <b>Recommended Previous Knowledge</b>             | Basics in Materials Science I/II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |    |
| <b>Educational Objectives</b>                     | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| <b>Professional Competence</b>                    | <p><i>Knowledge</i> Students can explain basic principles of crystallography, statics (free body diagrams, tractions) and thermodynamics (energy minimization, energy barriers, entropy)</p> <p><i>Skills</i> Students are capable of using standardized calculation methods: tensor calculations, derivatives, integrals, tensor transformations</p> <p><i>Social Competence</i> Students can provide appropriate feedback and handle feedback on their own performance constructively.</p> <p><i>Autonomy</i> Students are able to</p> <ul style="list-style-type: none"> <li>- assess their own strengths and weaknesses</li> <li>- assess their own state of learning in specific terms and to define further work steps on this basis guided by teachers.</li> <li>- work independently based on lectures and notes to solve problems, and to ask for help or clarifications when needed</li> </ul>                                              |        |    |
| <i>Knowledge</i>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |    |
| <i>Skills</i>                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |    |
| <b>Personal Competence</b>                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |    |
| <i>Social Competence</i>                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |    |
| <i>Autonomy</i>                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |    |
| <b>Workload in Hours</b>                          | Independent Study Time 96, Study Time in Lecture 84                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |    |
| <b>Credit points</b>                              | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |    |
| <b>Course achievement</b>                         | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| <b>Examination</b>                                | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |    |
| <b>Examination duration and scale</b>             | 180 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Assignment for the Following Curricula</b>     | <p>Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory</p> <p>Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory</p> <p>Mechanical Engineering - Product Development and Production: Specialisation Materials: Compulsory</p> <p>Materials Science and Engineering: Core Qualification: Compulsory</p> <p>Mechanical Engineering and Management: Specialisation Product Engineering: Elective Compulsory</p> <p>Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory</p> <p>Product Development, Materials and Production: Specialisation Production: Elective Compulsory</p> <p>Product Development, Materials and Production: Specialisation Materials: Compulsory</p> <p>Theoretical Mechanical Engineering: Specialisation Materials Science: Elective Compulsory</p> |        |    |

| Course L1661: Mechanical Behaviour of Brittle Materials |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                              | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Hrs/wk</b>                                           | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>CP</b>                                               | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Workload in Hours</b>                                | Independent Study Time 32, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Lecturer</b>                                         | Dr. Tim Fischer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Language</b>                                         | DE/EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Cycle</b>                                            | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Content</b>                                          | <p><b>Theoretical Strength</b><br/>Of a perfect crystalline material, theoretical critical shear stress</p> <p><b>Real strength of brittle materials</b><br/>Energy release reate, stress intensity factor, fracture criterion</p> <p><b>Scattering of strength of brittle materials</b><br/>Defect distribution, strength distribution, Weibull distribution</p> <p><b>Heterogeneous materials I</b><br/>Internal stresses, micro cracks, weight function,</p> <p><b>Heterogeneous materials II</b><br/>Toughening mechanisms: crack bridging, fibres</p> <p><b>Heterogeneous materials III</b><br/>Toughening mechanisms. Process zone</p> <p><b>Testing methods to determine the fracture toughness of brittle materials</b></p> <p><b>R-curve, stable/unstable crack growth, fractography</b></p> <p><b>Thermal shock</b></p> <p><b>Subcritical crack growth)</b><br/>v-K-curve, life time prediction</p> <p><b>Kriechen</b></p> <p><b>Mechanical properties of biological materials</b></p> <p><b>Examples of use for a mechanically reliable design of ceramic components</b></p> |
| <b>Literature</b>                                       | <p>D R H Jones, Michael F. Ashby, Engineering Materials 1, An Introduction to Properties, Applications and Design, Elsevier</p> <p>D.J. Green, An introduction to the mechanical properties of ceramics", Cambridge University Press, 1998</p> <p>B.R. Lawn, Fracture of Brittle Solids", Cambridge University Press, 1993</p> <p>D. Munz, T. Fett, Ceramics, Springer, 2001</p> <p>D.W. Richerson, Modern Ceramic Engineering, Marcel Decker, New York, 1992</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Course L1662: Dislocation Theory of Plasticity |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                     | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Hrs/wk</b>                                  | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>CP</b>                                      | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Workload in Hours</b>                       | Independent Study Time 32, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Lecturer</b>                                | Prof. Shan Shi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Language</b>                                | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Cycle</b>                                   | SoSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Content</b>                                 | <p>This class will cover the principles of dislocation theory from a physical metallurgy perspective, providing a fundamental understanding of the relations between the strength and of crystalline solids and distributions of defects.</p> <p>We will review the concept of dislocations, defining terminology used, and providing an overview of important concepts (e.g. linear elasticity, stress-strain relations, and stress transformations) for theory development. We will develop the theory of dislocation plasticity through derived stress-strain fields, associated self-energies, and the induced forces on dislocations due to internal and externally applied stresses. Dislocation structure will be discussed, including core models, stacking faults, and dislocation arrays (including grain boundary descriptions). Mechanisms of dislocation multiplication and strengthening will be covered along with general principles of creep and strain rate sensitivity. Final topics will include non-FCC dislocations, emphasizing the differences in structure and corresponding implications on dislocation mobility and macroscopic mechanical behavior; and dislocations in finite volumes.</p> |
| <b>Literature</b>                              | <p>Vorlesungsskript</p> <p>Aktuelle Publikationen</p> <p>Bücher:</p> <p>Introduction to Dislocations, by D. Hull and D.J. Bacon</p> <p>Theory of Dislocations, by J.P. Hirth and J. Lothe</p> <p>Physical Metallurgy, by Peter Hassen</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Course L3352: Exercise on mechanical properties (HÜ) |                                                     |
|------------------------------------------------------|-----------------------------------------------------|
| <b>Typ</b>                                           | Recitation Section (large)                          |
| <b>Hrs/wk</b>                                        | 2                                                   |
| <b>CP</b>                                            | 2                                                   |
| <b>Workload in Hours</b>                             | Independent Study Time 32, Study Time in Lecture 28 |
| <b>Lecturer</b>                                      | Prof. Shan Shi                                      |
| <b>Language</b>                                      | EN                                                  |
| <b>Cycle</b>                                         | SoSe                                                |
| <b>Content</b>                                       |                                                     |
| <b>Literature</b>                                    |                                                     |



## Module M1257: 3D Printing Laboratory

### Courses

| Title                                                  | Typ                                                                                                                                                                                                                                                                                                                   | Hrs/wk | CP |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|
| 3D Printing Laboratory (L1701)                         | Practical Course                                                                                                                                                                                                                                                                                                      | 3      | 6  |
| <b>Module Responsible</b>                              | Prof. Claus Emmelmann                                                                                                                                                                                                                                                                                                 |        |    |
| <b>Admission Requirements</b>                          | None                                                                                                                                                                                                                                                                                                                  |        |    |
| <b>Recommended Previous Knowledge</b>                  | Rapid Production<br>Computer Aided Design and Computation                                                                                                                                                                                                                                                             |        |    |
| <b>Educational Objectives</b>                          | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                  |        |    |
| <b>Professional Competence</b><br><i>Knowledge</i>     | Students will be able to give an overview over <ul style="list-style-type: none"> <li>• 3D printing based on fused deposition modeling,</li> <li>• printer setup and hardware components,</li> <li>• software and CAD data preparation,</li> <li>• and process parameters and quality aspects.</li> </ul>             |        |    |
| <i>Skills</i>                                          | The students will be able to <ul style="list-style-type: none"> <li>• prepare CAD models for 3D printing,</li> <li>• calibrate and operate a 3D printer,</li> <li>• conduct designed experiments,</li> <li>• and find optimal printing parameters.</li> </ul>                                                         |        |    |
| <b>Personal Competence</b><br><i>Social Competence</i> | The students will be able to <ul style="list-style-type: none"> <li>• coordinate work in a team,</li> <li>• set up, monitor and adapt a project plan,</li> <li>• share information with team members,</li> <li>• deal with different personal knowledge backgrounds,</li> <li>• and handle team conflicts.</li> </ul> |        |    |
| <i>Autonomy</i>                                        | Without external support the students will be able to <ul style="list-style-type: none"> <li>• do literature research,</li> <li>• organize work according to a schedule,</li> <li>• conduct experiments,</li> <li>• and operate and troubleshoot a production machine.</li> </ul>                                     |        |    |
| <b>Workload in Hours</b>                               | Independent Study Time 138, Study Time in Lecture 42                                                                                                                                                                                                                                                                  |        |    |
| <b>Credit points</b>                                   | 6                                                                                                                                                                                                                                                                                                                     |        |    |
| <b>Course achievement</b>                              | None                                                                                                                                                                                                                                                                                                                  |        |    |
| <b>Examination</b>                                     | Written elaboration                                                                                                                                                                                                                                                                                                   |        |    |
| <b>Examination duration and scale</b>                  | ca. 30 pages, approximately eight hours of preparation                                                                                                                                                                                                                                                                |        |    |
| <b>Assignment for the Following Curricula</b>          | Mechanical Engineering and Management: Specialisation Product Engineering: Elective Compulsory                                                                                                                                                                                                                        |        |    |

### Course L1701: 3D Printing Laboratory

|                          |                                                                                                                                                                                                                                                                                                                 |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>               | Practical Course                                                                                                                                                                                                                                                                                                |
| <b>Hrs/wk</b>            | 3                                                                                                                                                                                                                                                                                                               |
| <b>CP</b>                | 6                                                                                                                                                                                                                                                                                                               |
| <b>Workload in Hours</b> | Independent Study Time 138, Study Time in Lecture 42                                                                                                                                                                                                                                                            |
| <b>Lecturer</b>          | Prof. Claus Emmelmann, Dr. Dirk Herzog                                                                                                                                                                                                                                                                          |
| <b>Language</b>          | EN                                                                                                                                                                                                                                                                                                              |
| <b>Cycle</b>             | WiSe                                                                                                                                                                                                                                                                                                            |
| <b>Content</b>           | The 3D Printing lab consists of: <ul style="list-style-type: none"> <li>· Preparation of CAD models for 3D printing,</li> <li>· Design of Experiments for 3D-printing</li> <li>· Hands-on operation of 3D printer</li> <li>· Printing parameter variation and detection of influences on the process</li> </ul> |
| <b>Literature</b>        | wird in der Veranstaltung bekannt gegeben                                                                                                                                                                                                                                                                       |

## Module M1258: Laser Systems and Metallic Materials

### Courses

| Title                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Type    | Hrs/wk | CP |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------|----|
| Laser Systems and Process Technologies (L1612)         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Lecture | 2      | 3  |
| Structural Metallic Materials (L1702)                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Lecture | 2      | 3  |
| <b>Module Responsible</b>                              | Prof. Claus Emmelmann                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |        |    |
| <b>Admission Requirements</b>                          | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |        |    |
| <b>Recommended Previous Knowledge</b>                  | Fundamentals of Materials Science I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |        |    |
| <b>Educational Objectives</b>                          | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |        |    |
| <b>Professional Competence</b><br><i>Knowledge</i>     | <p>Students can give an overview over laser systems for material processing, specifically:</p> <ul style="list-style-type: none"> <li>• beam sources,</li> <li>• transport and manipulation of Laser beams,</li> <li>• and laser Safety.</li> </ul> <p>They can also describe applications of laser systems in material processing, namely:</p> <ul style="list-style-type: none"> <li>• primary forming,</li> <li>• marking,</li> <li>• cutting,</li> <li>• joining,</li> <li>• and surface treatment.</li> </ul> <p>They can also explain the material science of technically relevant metals as for example</p> <ul style="list-style-type: none"> <li>• carbon steels,</li> <li>• micro alloyed steels</li> <li>• low- and high-alloyed steels,</li> <li>• stainless steels,</li> <li>• aluminium alloys,</li> <li>• and magnesium alloys.</li> </ul> |         |        |    |
| <b>Skills</b>                                          | <p>After successful completion of this course, students should be able to</p> <ul style="list-style-type: none"> <li>• give an overview on current laser technology,</li> <li>• classify its applications in today's manufacturing processes,</li> <li>• evaluate economical and quality aspects,</li> <li>• find suitable laser systems for given tasks.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |        |    |
| <b>Personal Competence</b><br><i>Social Competence</i> | <ul style="list-style-type: none"> <li>• Students are able to discuss their solutions to problems with others. They communicate in English.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |        |    |
| <i>Autonomy</i>                                        | <ul style="list-style-type: none"> <li>• Students are able of checking their understanding of complex concepts by solving variants of concrete problems</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |         |        |    |
| <b>Workload in Hours</b>                               | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |        |    |
| <b>Credit points</b>                                   | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |         |        |    |
| <b>Course achievement</b>                              | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |        |    |
| <b>Examination</b>                                     | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |        |    |
| <b>Examination duration and scale</b>                  | approx. 20 pages                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |         |        |    |
| <b>Assignment for the Following Curricula</b>          | Mechanical Engineering and Management: Specialisation Product Engineering: Elective Compulsory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |        |    |

| Course L1612: Laser Systems and Process Technologies |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                           | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Hrs/wk</b>                                        | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>CP</b>                                            | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Workload in Hours</b>                             | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Lecturer</b>                                      | Prof. Claus Emmelmann, Dr. Dirk Herzog                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Language</b>                                      | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Cycle</b>                                         | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Content</b>                                       | <ul style="list-style-type: none"> <li>• Fundamentals of laser technology</li> <li>• Laser beam sources: CO<sub>2</sub>-, Nd:YAG-, Fiber- and Diodelasers</li> <li>• Laser system technology: beam forming, beam guidance systems, beam motion and beam control</li> <li>• Laser-based manufacturing technologies: generation, marking, cutting, joining, surface treatment</li> <li>• Quality assurance and economical aspects of laser material processing</li> <li>• Markets and Applications of laser technology</li> <li>• Student group exercises</li> </ul>                                                                                                                            |
| <b>Literature</b>                                    | <ul style="list-style-type: none"> <li>• Hügel, H. , T. Graf: Laser in der Fertigung : Strahlquellen, Systeme, Fertigungsverfahren, 3. Aufl., Vieweg + Teubner Wiesbaden 2014.</li> <li>• Eichler, J., Eichler, H. J.: Laser: Bauformen, Strahlführung, Anwendungen, 7. Aufl., Springer-Verlag Berlin Heidelberg 2010.</li> <li>• Steen W. M.; Mazumder J.: Laser material processing, 4th Edition, Springer-Verlag London 2010.</li> <li>• J.C. Ion: Laser processing of engineering materials: principles, procedure and industrial applications, Elsevier Butterworth-Heinemann 2005.</li> <li>• Gebhardt, A.: Understanding additive manufacturing, München [u.a.] Hanser 2011</li> </ul> |

| Course L1702: Structural Metallic Materials |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                  | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Hrs/wk</b>                               | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>CP</b>                                   | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Workload in Hours</b>                    | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Lecturer</b>                             | Dr. Nikolai Kashaev                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Language</b>                             | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Cycle</b>                                | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Content</b>                              | <p>The course enfold the relationships between metallic materials, their properties, processing technologies as well as fields of application. Because of the ever-increasing loads and demands for resource efficiency, the optimization of material properties through the tailored processing as well as the targeted sequence of processing steps for the manufacturing of the final part are becoming more important than ever. In terms of selecting of an appropriate material for a targeted application, the necessary and appropriate manufacturing technologies have to be taking into consideration. In order to reflect the effects of manufacturing methods, students are imparted knowledge about metallic materials combined with processing technologies. Particular attention is also paid to loading cases as well as damage mechanisms of the materials used in industrial applications. Furthermore, the possible methods for life extension are analysed and discussed. The aim of the course is to make students aware to perform a correct selection of appropriate materials with technological processes for potential applications taking into consideration the different kinds of stress (fatigue, creep, corrosion etc.).</p> <p>Lecture 1: Introduction. Requirements to structural metallic materials depending on their application. Typical examples for material usage in automotive, airplane and wind energy structures, power plants structures as well as in automotive components including transmissions, bearings, engines etc. Classification of the used materials into groups depending on their application requirements.</p> <p>Lecture 2: Fundamental aspects of Fe-C-alloys. Mechanical properties, material classes (austenitic and ferritic steels, cast iron etc.), Fe-C phase diagram. Fundamental aspects of heat treatment for Fe-base materials. Discussion of specific alloys and their typical applications.</p> <p>Lecture 3: Fundamentals of Fe-base materials processing for fabrication of components. From raw material to the component. Typical fabrication routes: casting, forging, machining. Fundamentals of common manufacturing technologies. Cold forming and forging of steels. Fundamentals of formability and materials strengthening mechanisms, typical alloys and applications (e.g. TRIP steels).</p> <p>Lecture 4: Fundamental aspects of Al-alloys and their base processing technologies for fabrication of components. Fundamental aspects of Mg-alloys and their base processing technologies for fabrication of components.</p> <p>Lecture 5: Fundamental aspects of Ti-alloys and their base processing technologies for fabrication of components. Intermetallic alloys and metallic glasses: properties, applications and fundamental aspects of production and processing.</p> <p>Lecture 6: Cu-base alloys: classes of alloys, their typical applications and fundamental aspects of processing; examples for components. Ni- und Co-base alloys: classes of alloys, their properties and typical applications. Fundamental aspects of processing and manufacturing of components.</p> <p>Lecture 7: Fatigue and fracture of metallic materials. Fundamental aspects of fatigue loading (stress amplitudes, mean stress, high- and low cycle fatigue). Notch effects, crack initiation and propagation. Damage tolerance assessment.</p> <p>Lecture 8: Degradation and failure of materials and components in service. Stress corrosion cracking and corrosion fatigue of metallic materials.</p> <p>Lecture 9: Surface engineering: coatings. Functional coatings for wear and corrosion protection, as well as decorative purposes. Electrochemical and physical coating deposition, deposit welding and thermal spraying.</p> <p>Lecture 10: Surface engineering: modifications. Metallurgical surface modifications (nitriding, surface hardening ect.) and (thermo-)mechanical methods (shot peening, laser shock peening, rolling, friction stir processing ect.).</p> |
| <b>Literature</b>                           | <ol style="list-style-type: none"> <li>1. George Krauss, Steels: Processing, Structure, and Performance, 978-0-87170-817-5, 2006.</li> <li>2. Hans Berns, Werner Theisen, Ferrous Materials: Steel and Cast Iron, 2008. <a href="http://dx.doi.org/10.1007/978-3-540-71848-2">http://dx.doi.org/10.1007/978-3-540-71848-2</a></li> <li>3. Bruno C., De Cooman / John G. Speer: Fundamentals of Steel Product Physical Metallurgy, 2011, 642 S.</li> <li>4. Harry Chandler, Steel Metallurgy for the Non-Metallurgist 0-87170-652-0, 2006, 84 S.</li> <li>5. Catrin Kammer, Aluminium Taschenbuch 1, Grundlagen und Werkstoffe, Beuth,16. Auflage 2009. 784 S., ISBN 978-3-410-22028-2</li> <li>6. Günter Drossel, Susanne Friedrich, Catrin Kammer und Wolfgang Lehnert, Aluminium Taschenbuch 2, Umformung von Aluminium-Werkstoffen, Gießen von Aluminiumteilen, Oberflächenbehandlung von Aluminium, Recycling und Ökologie, Beuth, 16. Auflage 2009. 768 S., ISBN 978-3-410-22029-9</li> <li>7. Catrin Kammer, Aluminium Taschenbuch 3, Weiterverarbeitung und Anwendung, Beuth,17. Auflage 2014. 892 S., ISBN 978-3-410-22311-5</li> <li>8. Lütjering, J.C. Williams: Titanium, 2nd ed., Springer, Berlin, Heidelberg, 2007, ISBN 978-3-540-71397</li> <li>9. Magnesium - Alloys and Technologies, K. U. Kainer (Hrsg.), Wiley-VCH, Weinheim 2003, ISBN 3-527-30570-x</li> <li>10. Mihriban O. Pekguleryuz, Karl U. Kainer and Ali Kaya "Fundamentals of Magnesium Alloy Metallurgy", Woodhead Publishing Ltd, 2013,ISBN 10: 0857090887</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Module M1151: Materials Modeling                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |    |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|
| Courses                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |    |
| Title                                                  | Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Hrs/wk | CP |
| Material Modeling (L1535)                              | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2      | 3  |
| Material Modeling (Exercise) (L1536)                   | Recitation Section (small)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2      | 3  |
| <b>Module Responsible</b>                              | Prof. Christian Cyron                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |    |
| <b>Admission Requirements</b>                          | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |    |
| <b>Recommended Previous Knowledge</b>                  | Basics of mechanics as taught, e.g., in the modules Engineering Mechanics I and Engineering Mechanics II at TUHH (forces and moments, stress, linear strain, free-body principle, linear-elastic constitutive laws, strain energy); basics of mathematics as taught, e.g., in the modules Mathematics I and Mathematics II at TUHH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |    |
| <b>Educational Objectives</b>                          | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |    |
| <b>Professional Competence</b><br><i>Knowledge</i>     | The students understand the theoretical foundations of anisotropic elasticity, viscoelasticity and elasto-plasticity in the realm of three-dimensional (linear) continuum mechanics. In the area of anisotropic elasticity, they know the concept of material symmetry and its application in orthotropic, transversely isotropic and isotropic materials. They understand the concept of stiffness and compliance and how both can be characterized by appropriate parameters. Moreover, the students understand viscoelasticity both in the time and frequency domain using the concepts of relaxation modulus, creep modulus, storage modulus and loss modulus. In the area of elasto-plasticity, the students know the concept of yield stress or (in higher dimensions) yield surface and of plastic potential. Additionally, they know the concepts of ideal plasticity, hardening and weakening. Moreover, they know von-Mises plasticity as a specific model of elasto-plasticity.                                                                                                                                                                                                                              |        |    |
| <i>Skills</i>                                          | The students can independently identify and solve problems in the area of materials modeling and acquire the knowledge to do so. This holds in particular for the area of anisotropically elastic, viscoelastic and elasto-plastic material behavior. In these areas, the students can independently develop models for complex material behavior. To this end, they have the ability to read and understand relevant literature and identify the relevant results reported there. Moreover, they can implement models which they developed or found in the literature in computational software (e.g., based on the finite element method) and use it for practical calculations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |    |
| <b>Personal Competence</b><br><i>Social Competence</i> | The students are able to develop constitutive models for materials and present them to specialists. Moreover, they have the ability to discuss challenging problems of materials modeling with experts using the proper terminology, to identify and ask critical questions in such discussions and to identify and discuss potential caveats in models presented to them.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |    |
| <i>Autonomy</i>                                        | The students have the ability to independently develop abstract models that allow them to classify observed phenomena within an more general abstract framework and to predict their further evolution. Moreover, the students understand the advantages but also limitations of mathematical models and can thus independently decide when and to which extent they make sense as a basis for decisions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Workload in Hours</b>                               | Independent Study Time 124, Study Time in Lecture 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |    |
| <b>Credit points</b>                                   | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |    |
| <b>Course achievement</b>                              | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |    |
| <b>Examination</b>                                     | Written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |    |
| <b>Examination duration and scale</b>                  | 60 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| <b>Assignment for the Following Curricula</b>          | Mechanical Engineering - Product Development and Production: Core Qualification: Elective Compulsory<br>Materials Science and Engineering: Specialisation Modeling: Elective Compulsory<br>Mechanical Engineering and Management: Specialisation Product Engineering: Elective Compulsory<br>Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory<br>Biomedical Engineering: Specialisation Implants and Endoprostheses: Elective Compulsory<br>Biomedical Engineering: Specialisation Medical Technology and Control Theory: Elective Compulsory<br>Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory<br>Product Development, Materials and Production: Core Qualification: Elective Compulsory<br>Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory<br>Theoretical Mechanical Engineering: Specialisation Materials Science: Elective Compulsory<br>Theoretical Mechanical Engineering: Specialisation Product Development and Production: Elective Compulsory<br>Theoretical Mechanical Engineering: Specialisation Bio- and Medical Technology: Elective Compulsory |        |    |

| Course L1535: Material Modeling |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                      | Lecture                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Hrs/wk</b>                   | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>CP</b>                       | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Workload in Hours</b>        | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Lecturer</b>                 | Prof. Christian Cyron                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Language</b>                 | DE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Cycle</b>                    | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Content</b>                  | <p>One of the most important questions when modeling mechanical systems in practice is how to model the behavior of the materials of their different components. In addition to simple isotropic elasticity in particular the following phenomena play key roles</p> <ul style="list-style-type: none"> <li>- anisotropy (material behavior depending on direction, e.g., in fiber-reinforced materials)</li> <li>- plasticity (permanent deformation due to one-time overload, e.g., in metal forming)</li> <li>- viscoelasticity (absorption of energy, e.g., in dampers)</li> <li>- creep (slow deformation under permanent load, e.g., in pipes)</li> </ul> <p>This lecture briefly introduces the theoretical foundations and mathematical modeling of the above phenomena. It is complemented by exercises where simple examples problems are solved by calculations and where the implementation of the content of the lecture in computer simulations is explained. It will also briefly discussed how important material parameters can be determined from experimental data.</p> |
| <b>Literature</b>               | <p>Empfohlene Literatur / Recommended literature:</p> <p>1) Dietmar Gross, Werner Hauger, Peter Wriggers, Technische Mechanik 4, Springer 2018, DOI: 10.1007/978-3-662-55694-8</p> <p>2) Peter Haupt, Continuum Mechanics and Theory of Materials, Springer 2002, DOI: 10.1007/978-3-662-04775-0</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Course L1536: Material Modeling (Exercise) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                                 | Recitation Section (small)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Hrs/wk</b>                              | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>CP</b>                                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Workload in Hours</b>                   | Independent Study Time 62, Study Time in Lecture 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Lecturer</b>                            | Prof. Christian Cyron, Daniel Paukner                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Language</b>                            | DE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Cycle</b>                               | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Content</b>                             | <p>One of the most important questions when modeling mechanical systems in practice is how to model the behavior of the materials of their different components. In addition to simple isotropic elasticity in particular the following phenomena play key roles</p> <ul style="list-style-type: none"> <li>- anisotropy (material behavior depending on direction, e.g., in fiber-reinforced materials)</li> <li>- plasticity (permanent deformation due to one-time overload, e.g., in metal forming)</li> <li>- viscoelasticity (absorption of energy, e.g., in dampers)</li> <li>- creep (slow deformation under permanent load, e.g., in pipes)</li> </ul> <p>This lecture briefly introduces the theoretical foundations and mathematical modeling of the above phenomena. It is complemented by exercises where simple examples problems are solved by calculations and where the implementation of the content of the lecture in computer simulations is explained. It will also briefly discussed how important material parameters can be determined from experimental data.</p> |
| <b>Literature</b>                          | <p>Empfohlene Literatur / Recommended literature:</p> <p>1) Dietmar Gross, Werner Hauger, Peter Wriggers, Technische Mechanik 4, Springer 2018, DOI: 10.1007/978-3-662-55694-8</p> <p>2) Peter Haupt, Continuum Mechanics and Theory of Materials, Springer 2002, DOI: 10.1007/978-3-662-04775-0</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Module M1984: Leading Agile Teams                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              |                                    |                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------------------------------|----------------------------------------------------------------------------------------------|
| Courses                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              |                                    |                                                                                              |
| Title                                                                                                                                            | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              | Hrs/wk                             | CP                                                                                           |
| Leading Agile Teams (L3245)                                                                                                                      | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              | 6                                  | 6                                                                                            |
| <b>Module Responsible</b>                                                                                                                        | Prof. Nikola Bursac                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |                                    |                                                                                              |
| <b>Admission Requirements</b>                                                                                                                    | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |              |                                    |                                                                                              |
| <b>Recommended Previous Knowledge</b>                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              |                                    |                                                                                              |
| <b>Educational Objectives</b>                                                                                                                    | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |              |                                    |                                                                                              |
| <b>Professional Competence</b><br><i>Knowledge</i><br><i>Skills</i><br><b>Personal Competence</b><br><i>Social Competence</i><br><i>Autonomy</i> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              |                                    |                                                                                              |
| <b>Workload in Hours</b>                                                                                                                         | Independent Study Time 96, Study Time in Lecture 84                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |                                    |                                                                                              |
| <b>Credit points</b>                                                                                                                             | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |              |                                    |                                                                                              |
| <b>Course achievement</b>                                                                                                                        | <b>Compulsory</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Bonus</b> | <b>Form</b>                        | <b>Description</b>                                                                           |
|                                                                                                                                                  | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 20 %         | Subject theoretical practical work | andThere is the opportunity to receive a bonus through the successful completion of the PBL. |
| <b>Examination</b>                                                                                                                               | Oral exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |                                    |                                                                                              |
| <b>Examination duration and scale</b>                                                                                                            | 30 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |              |                                    |                                                                                              |
| <b>Assignment for the Following Curricula</b>                                                                                                    | Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory<br>Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory<br>Mechanical Engineering and Management: Specialisation Product Engineering: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Production: Elective Compulsory<br>Product Development, Materials and Production: Specialisation Materials: Elective Compulsory |              |                                    |                                                                                              |

| Course L3245: Leading Agile Teams |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Typ</b>                        | Project-/problem-based Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Hrs/wk</b>                     | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>CP</b>                         | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Workload in Hours</b>          | Independent Study Time 96, Study Time in Lecture 84                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Lecturer</b>                   | Prof. Nikola Bursac                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Language</b>                   | EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Cycle</b>                      | WiSe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Content</b>                    | <p>Within the scope of the "Leading Agile Teams" module, you will receive a comprehensive introduction to leading agile teams in modern work environments. The module focuses on essential concepts and practices necessary for effectively managing agile teams.</p> <p>Module Contents:</p> <ol style="list-style-type: none"> <li><b>Fundamentals of Agile Methods:</b> <ul style="list-style-type: none"> <li>Agile Manifesto and Principles</li> <li>Scrum, Kanban, and other agile frameworks</li> </ul> </li> <li><b>Team Dynamics and Communication:</b> <ul style="list-style-type: none"> <li>Fostering communication and collaboration</li> <li>Conflict management in agile teams</li> </ul> </li> <li><b>Leadership in Agile Contexts:</b> <ul style="list-style-type: none"> <li>Servant Leadership</li> <li>Empowering team members</li> </ul> </li> <li><b>Agile Planning and Implementation:</b> <ul style="list-style-type: none"> <li>Sprint planning and backlog management</li> <li>Effective implementation of agile practices</li> </ul> </li> <li><b>Measurement and Improvement:</b> <ul style="list-style-type: none"> <li>Metrics for agile teams</li> <li>Continuous improvement of processes</li> </ul> </li> </ol> <p>The module combines theoretical concepts with practical case studies and interactive exercises to provide you with a holistic insight into leading agile teams.</p> |
| <b>Literature</b>                 | <ol style="list-style-type: none"> <li><b>"Scrum: The Art of Doing Twice the Work in Half the Time" by Jeff Sutherland</b> <ul style="list-style-type: none"> <li>This book, written by one of the co-creators of Scrum, provides insights into the Scrum framework and how it can enhance team productivity.</li> </ul> </li> <li><b>"Spiral Dynamics: Mastering Values, Leadership, and Change" by Don Edward Beck and Christopher C. Cowan</b> <ul style="list-style-type: none"> <li>"Spiral Dynamics" offers a deep dive into the theory of Spiral Dynamics, providing a framework for understanding human values and behavior, which is essential for effective leadership.</li> </ul> </li> <li><b>"The Five Dysfunctions of a Team: A Leadership Fable" by Patrick Lencioni</b> <ul style="list-style-type: none"> <li>Lencioni's book delves into the common challenges that teams face and offers practical advice on overcoming dysfunctions to build a cohesive and high-performing team.</li> </ul> </li> <li><b>"Leaders Eat Last: Why Some Teams Pull Together and Others Don't" by Simon Sinek</b> <ul style="list-style-type: none"> <li>Simon Sinek explores the dynamics of effective leadership, emphasizing the importance of creating a trusting and supportive environment for teams to thrive.</li> </ul> </li> </ol>                                                                           |



## Thesis

### Module M-002: Master Thesis

#### Courses

| Title                                                  | Typ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Hrs/wk | CP |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|
| <b>Module Responsible</b>                              | Professoren der TUHH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| <b>Admission Requirements</b>                          | <ul style="list-style-type: none"> <li>According to General Regulations §21 (1):</li> </ul> <p>At least 60 credit points have to be achieved in study programme. The examinations board decides on exceptions.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Recommended Previous Knowledge</b>                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |    |
| <b>Educational Objectives</b>                          | After taking part successfully, students have reached the following learning results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| <b>Professional Competence</b><br><i>Knowledge</i>     | <ul style="list-style-type: none"> <li>The students can use specialized knowledge (facts, theories, and methods) of their subject competently on specialized issues.</li> <li>The students can explain in depth the relevant approaches and terminologies in one or more areas of their subject, describing current developments and taking up a critical position on them.</li> <li>The students can place a research task in their subject area in its context and describe and critically assess the state of research.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                            |        |    |
| <i>Skills</i>                                          | <p>The students are able:</p> <ul style="list-style-type: none"> <li>To select, apply and, if necessary, develop further methods that are suitable for solving the specialized problem in question.</li> <li>To apply knowledge they have acquired and methods they have learnt in the course of their studies to complex and/or incompletely defined problems in a solution-oriented way.</li> <li>To develop new scientific findings in their subject area and subject them to a critical assessment.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Personal Competence</b><br><i>Social Competence</i> | <p>Students can</p> <ul style="list-style-type: none"> <li>Both in writing and orally outline a scientific issue for an expert audience accurately, understandably and in a structured way.</li> <li>Deal with issues competently in an expert discussion and answer them in a manner that is appropriate to the addressees while upholding their own assessments and viewpoints convincingly.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |    |
| <i>Autonomy</i>                                        | <p>Students are able:</p> <ul style="list-style-type: none"> <li>To structure a project of their own in work packages and to work them off accordingly.</li> <li>To work their way in depth into a largely unknown subject and to access the information required for them to do so.</li> <li>To apply the techniques of scientific work comprehensively in research of their own.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |    |
| <b>Workload in Hours</b>                               | Independent Study Time 900, Study Time in Lecture 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |    |
| <b>Credit points</b>                                   | 30                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |    |
| <b>Course achievement</b>                              | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |    |
| <b>Examination</b>                                     | Thesis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |    |
| <b>Examination duration and scale</b>                  | According to General Regulations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |    |
| <b>Assignment for the Following Curricula</b>          | Civil Engineering: Thesis: Compulsory<br>Bioprocess Engineering: Thesis: Compulsory<br>Chemical and Bioprocess Engineering: Thesis: Compulsory<br>Chemical and Bioprocess Engineering: Thesis: Compulsory<br>Computational Engineering: Thesis: Compulsory<br>Computer Science: Thesis: Compulsory<br>Data Science: Thesis: Compulsory<br>Electrical Engineering and Information Technology: Thesis: Compulsory<br>Electrical Engineering: Thesis: Compulsory<br>Energy Systems: Thesis: Compulsory<br>Environmental Engineering: Thesis: Compulsory<br>Aircraft Systems Engineering: Thesis: Compulsory<br>Global Innovation Management: Thesis: Compulsory<br>Computer Science in Engineering: Thesis: Compulsory<br>Information and Communication Systems: Thesis: Compulsory<br>Interdisciplinary Mathematics: Thesis: Compulsory<br>International Production Management: Thesis: Compulsory<br>International Management and Engineering: Thesis: Compulsory |        |    |

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|  | Joint European Master in Environmental Studies - Cities and Sustainability: Thesis: Compulsory |
|  | Logistics, Infrastructure and Mobility: Thesis: Compulsory                                     |
|  | Aeronautics: Thesis: Compulsory                                                                |
|  | Mechanical Engineering - Product Development and Production: Thesis: Compulsory                |
|  | Materials Science and Engineering: Thesis: Compulsory                                          |
|  | Materials Science: Thesis: Compulsory                                                          |
|  | Mechanical Engineering and Management: Thesis: Compulsory                                      |
|  | Mechatronics: Thesis: Compulsory                                                               |
|  | Biomedical Engineering: Thesis: Compulsory                                                     |
|  | Microelectronics and Microsystems: Thesis: Compulsory                                          |
|  | Product Development, Materials and Production: Thesis: Compulsory                              |
|  | Renewable Energies: Thesis: Compulsory                                                         |
|  | Naval Architecture and Ocean Engineering: Thesis: Compulsory                                   |
|  | Naval Architecture and Ocean Engineering: Thesis: Compulsory                                   |
|  | Ship and Offshore Technology: Thesis: Compulsory                                               |
|  | Theoretical Mechanical Engineering: Thesis: Compulsory                                         |
|  | Process Engineering: Thesis: Compulsory                                                        |
|  | Water and Environmental Engineering: Thesis: Compulsory                                        |
|  | Certification in Engineering & Advisory in Aviation: Thesis: Compulsory                        |